



Review of the Papua New Guinea Tariff Reduction Programme

Technical cooperation outcome



© 2026, United Nations Conference on Trade and Development

The findings, interpretations and conclusions expressed herein are those of the authors and do not necessarily reflect the views of the United Nations or its officials or Member States.

The designations employed and the presentation of material on any map in this work do not imply the expression of any opinion whatsoever on the part of the United Nations concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries.

Mention of any firm or licensed process does not imply the endorsement of the United Nations.

This publication has not been formally edited.

UNCTAD/TCS/DITC/ INF/2026/1

Acknowledgements

This report was prepared under the overall guidance of Luz Maria de la Mora, Director of the Division on International Trade and Commodities of the United Nations Conference on Trade and Development (UNCTAD). Julien Bliesener and Ralf Peters led the report team and work. Invaluable inputs and comments were provided by Luz Maria de la Mora, Graham Mott and Alessandro Nicita.

UNCTAD gratefully acknowledges the substantive contributions of Paul R. Baker (Chief Executive of International Economics Consulting), as well as the data collection, research and analytics contributed by Neetish Hurry, Akshay Bhoolee and Loan Le.

UNCTAD further gratefully acknowledges the substantial inputs from the Department of Treasury of the Government of Papua New Guinea from which the report benefited. Inputs and extensive data from different departments and technical agencies of the Government of Papua New Guinea are also greatly appreciated, particularly the National Trade Office and Papua New Guinea Customs Service.

UNCTAD extends its appreciation to respondents for the information provided by 50 businesses in Papua New Guinea, as well as by many government officials who responded to the survey supporting this report.

The study on which this report is based was funded by the European Union under the UNCTAD Improving Pacific Islands Customs and Trade (IMPACT) project, which is a component of the Pacific Regional Integration Support Programme under the 11th cycle of the European Development Fund.

Gilles Maury undertook desktop publishing and prepared the cover for the report.



**Funded by
the European Union**



PAPUA NEW GUINEA
DEPARTMENT OF TREASURY

Abbreviations

APEC	Asia-Pacific Economic Cooperation
ASEAN	Association of Southeast Asian Nations
ASEAN+1	Association of Southeast Asian Nations Plus One
AVE	Ad valorem equivalent
CPTA	Framework Agreement on the Facilitation of Cross-border Paperless Trade in Asia and the Pacific
ECI	Economic Complexity Index
FTA	Free trade agreement
GDP	Gross domestic product
GST	Government sales tax
GVA	Gross value added
HS	Harmonized System
iEPA	Interim Economic Partnership Agreement
IMF	International Monetary Fund
LNG	Liquefied natural gas
MFN	Most-favoured-nation
MSG	Melanesian Spearhead Group
MSGTA	Melanesian Spearhead Group Trade Agreement
MTDP	Medium-Term Development Plan
MTRS	Medium Term Revenue Strategy
NSO	National Statistical Office
NTP	National trade policy
OECD	Organisation of Economic Cooperation and Development
PACER Plus	Pacific Agreement on Closer Economic Relations Plus
PATCRA	Australia - Papua New Guinea Agreement on Trade and Commercial Relations
PE	Partial equilibrium
PICTA	Pacific Island Countries Trade Agreement
PNGCS	Papua New Guinea Customs Service
PGK	Papua New Guinea Kina
PPML	Poisson pseudo-maximum likelihood
PPP	Purchasing power parity
R&D	Research and development
RCEP	Regional Comprehensive Economic Partnership
RQI	Regulatory quality index
SME	Small and medium enterprise
SPARTECA	South Pacific Regional Trade, Economic and Commercial Agreement
TRP	Tariff Reduction Programme
UNCTAD	United Nations Trade and Development
UNDP	United Nations Development Programme
UNECA	United Nations Economic Commission for Asia and the Pacific
VAT	Value added tax
WITS	World Integrated Trade Solution
WTO	World Trade Organisation

Table of Content

Acknowledgements.....	iii
Abbreviations.....	iv
Executive summary	ix
Chapter 1 Introduction	1
Chapter 2 Context.....	5
A. Macroeconomic overview	7
B. Papua New Guinea’s tariff reduction programme: a chronology	11
1. Initiation of the tariff reduction programme (late 1990s – early 2000s).....	11
2. Phased tariff reductions (1999–2006)	11
3. Continued tariff reductions (2007–2017)	12
4. Reversal and shift towards protectionism (2017–present).....	14
Chapter 3 Government policies on trade and industry.....	17
A. National Trade Policy (2017–2032).....	19
B. Medium Term Development Plan IV (2023–2027).....	20
C. Medium Term Revenue Strategy (MTRS) (2023–2027)	20
D. Sectoral plans (fisheries, forestry, agriculture, industry)	21
E. Multilateral and regional trade agreements	21
Chapter 4 Assessment of TRP outcomes	25
A. Tariff outcomes	28
B. Economic outcomes and sector performance	32
Overall observations.....	32
Impacts by product category	35
Impact by sector.....	37
Impact by partner.....	38
C. Tariff revenue	39
D. Industry perspectives and cross cutting observations	44
E. Innovation	47
F. Consumer welfare.....	49

Chapter 5 Conclusions and Recommendations	53
1. Restore a path of gradual tariff rationalization	55
2. Focus on fostering a competitive business environment	57
3. Use targeted support for infant industries	58
4. Safeguard consumer welfare and food security	58
5. Enhance regional integration efforts	58
6. Institutionalize evidence-based review mechanisms	59
7. Invest in the capacity to monitor, investigate and deploy trade defense mechanisms	59
References	61
Annexes	63
Annex 1: Project terms of reference	63
Review of the Papua New Guinea tariff reduction programme	63
Annex 2: Methodology	67
Background	67
Protection measures	68
Categories of products	69
Impact analysis	71
Methodology for measures of protection and trade flow analysis	72
Economic modelling strategy	73
Stakeholder engagement	75
Annex 3: Results from the Gravity Model	76
Methodology and data	76
Results from the Gravity model	78
Appendix to Annex 3. Summary statistics by year	79
Annex 4: Births of products	81

Figures

Figure 1.	Papua New Guinea has experienced steady economic growth	7
Figure 2.	The kina has remained stable over the past 10 years following a strong increase at the turn of the millennium.....	8
Figure 3.	Papua New Guinea ranks low on economic complexity	8
Figure 4.	Papua New Guinea's product space has evolved little over time	10
Figure 5.	Tariff levels vary across products in Papua New Guinea's	15
Figure 6.	Average applied MFN tariff rates differ across products in Papua New Guinea.....	22
Figure 7.	MFN AVE tariffs have changed since 2010	29
Figure 8.	MFN AVE tariffs in Papua New Guinea changed over the 2018–2023 period	29
Figure 9.	Simple average AVE tariffs of Papua New Guinea	30
Figure 10.	Impact of tariff policy on investment decisions.....	31
Figure 11.	Tariff levels by product category changed during and after the TRP	31
Figure 12.	Tariff dispersions across HS 2 sectors.....	32
Figure 13.	Exports in Papua New Guinea have expanded.....	33
Figure 14.	Growth has slowed after the TRP suspension	34
Figure 15.	Overall, growth in Papua New Guinea has been strong	34
Figure 16.	Consumer goods as a share of imports have grown steadily	36
Figure 17.	Intermediate, consumer and capital goods returned to modest positive growth ...	36
Figure 18.	Manufacturing has stagnated	37
Figure 19.	Tax-to-GDP ratio has fallen in Papua New Guinea	40
Figure 20.	Total tax revenue has increased in Papua New Guinea.....	40
Figure 21.	Tariff revenue increased since the TRP suspension.....	42
Figure 22.	Results from tariff revenue simulations	43
Figure 23.	Perceived impacts of the TRP 2012-2017.....	44
Figure 24.	Perceived impacts of the tariff increases post-2018.....	45
Figure 25.	Perceived level of competition linked to the TRP	45
Figure 26.	Factors affecting competitiveness	46
Figure 27.	Main constraints to business operations	47
Figure 28.	Innovation and decay in the export portfolio	48
Figure 29.	Reported innovation attempts: introducing new products	49
Figure 30.	Welfare implications of the TRP	50
Figure 31.	Perceived price sensitivity of Papua New Guinea consumers	51
Figure 32.	Recommended areas of support for competitiveness	57
Figure 33.	Tariffs and transmission of firm decisions	67
Figure 34.	Impact of a tariff on consumers, producers, government and welfare.....	68
Figure 35.	PNGCS and UNDS data for Papua New Guinea imports.....	70

Tableaux

Table 1.	Tariff measures in support of sectoral strategies (2018–2020)	21
Table 2.	Papua New Guinea’s network of FTAs.....	22
Table 3.	Trade effects by partner	38
Table 4.	Number of products affected by tariff change during and post TRP for China	39
Table 5.	Number of products affected by tariff change during and post TRP for Australia ...	39
Table 6.	Mauritius tariff profile in 2025	56
Table 7.	Papua New Guinea tariff profile in 2025.....	56
Table 8.	UNCTAD and TRP classification of goods.....	70
Table 9.	PNGCS and UNDS import data for Papua New Guinea	71
Table 10.	Tariff rates estimated for Papua New Guinea.....	71
Table 11.	Import values in \$ thousands, current prices.....	77
Table 12.	Summary statistics.....	77
Table 13.	Estimation results.....	78
Table 14.	List of products that were exported in 2018 but not exported in 2010	81

Boxes

Box 1.	TRP summary (2007-2017).....	13
Box 2.	Tariffs in the context of free trade agreements.....	13
Box 3.	TRP suspension summary (2017-2020)	16
Box 4.	Analytical approach employed for the analysis.....	27

Executive summary

Papua New Guinea embarked on the Tariff Reduction Programme (TRP) as part of a broader economic recovery and reform agenda that aimed to improve economic efficiency and integrate with global markets. By 2015, Papua New Guinea's tariffs were at historic lows, around 3.2 per cent on average, making it one of the most open economies in the region. After further tariff reductions, the government suspended the final stage of tariff cuts in 2018. It progressively raised tariffs on some products, citing the need to protect infant industries and boost revenue. As Papua New Guinea implements new national development frameworks, including the Papua New Guinea National Trade Policy 2017–2032, the Medium-Term Development Plan IV (2023–2027), and the Medium-Term Revenue Strategy (2023–2027), as well as other sector-specific plans in fisheries, forestry, agriculture, and manufacturing, there is a need for a structured and evidence-based assessment of the TRP.

This report provides a review of Papua New Guinea's TRP from 2010 to the present, examining its chronology and phases, the policy rationale for both liberalisation and reversal, and the impacts. To evaluate the outcomes and impacts of the TRP, the study relies on two complementary methodologies: an *ex-post* econometric analysis of trade flows, which assesses the results of the implementation of the TRP on specific economic variables, and an *ex-ante* partial equilibrium simulation, which evaluates the expected changes to occur as a result of tariff changes on trade flows, tariff revenue, sectoral impacts, consumer welfare, and other broader business impacts, holding all other variables constant. The analysis focuses on two periods for comparisons: TRP phase 2 (2012–2017) and the TRP suspension (2018–2023). To supplement the quantitative analysis of the TRP impact assessment, stakeholder consultations, in the form of a survey, were conducted with national government agencies, business associations, and companies to identify the perceived benefits, challenges, and recommendations for future measures to support Papua New Guinea firms in building competitiveness.

On tariff outcomes, the review highlights a policy shift from liberalization to selective protectionism. During the TRP, tariffs declined across all product categories, with consumer goods falling from 6.4 to 5.2 per cent, raw materials from 11.4 to 10.8 per cent, intermediates from 2 to 1.5 per cent, and capital goods near zero. This period also saw reduced dispersion of rates, leading to a simpler tariff schedule. However, after the TRP's suspension in 2018, tariffs moderately increased, with consumer goods rising to 5.3 per cent and raw materials to 11.3 per cent (both in simple average terms), reflecting renewed protection for targeted industries. By 2023, tariffs signalled support for infant industries, but overall dispersion was more rationalized than in 2010. This policy volatility created uncertainty: 61 per cent of surveyed firms reported that tariff revisions between 2018–2020 disrupted investment decisions. While the overall trend of moving from broad liberalization to selective protectionism aimed to foster domestic industries and reduce import dependence, it may come at the cost of higher consumer prices, complex trade-offs for firms, and weaker competitiveness if uncertainty persists.

On economic outcomes and sectoral impacts, the TRP coincided with a period of reduced import demand. Therefore, Papua New Guinea did not experience an import surge, linked to the lowering of tariffs. Instead, it benefited from strong export earnings coming from mining and petroleum. When Papua New Guinea reversed course and raised tariffs in 2018, the tariff hikes had a limited additional dampening effect on import levels, which had already been low and stable. Overall, Papua New Guinea's economy overall exhibited remarkable gross domestic product (GDP) growth averaging 6.3 per cent annually during the active TRP period, compared to the post-suspension period growth of two per cent. GDP per capita at Purchasing

Power Parity (PPP) showed a similar trend in growth of 4.4 per cent for the 2010-2017 period and 3.8 per cent for the 2018-2024 period. It is, however, important to consider multiple factors that contributed to the differential trends, including global commodity cycles and major resource project developments (for the TRP), as well as the COVID-19 outbreak (for the TRP suspension phase). The gravity model's findings support the conventional wisdom that tariff increases yield a reduction in imports: model runs on data during 2012-2023 showed that a 1 per cent increase in tariffs (ad valorem) is associated with roughly a 2.1 per cent decrease in import value on average. The partial equilibrium findings showed single-digit millions of dollars change in imports for major partners (e.g. a \$6.8 million drop in imports from China due to post-2018 tariff hikes), reinforcing the notion that other factors overshadowed the tariff policy in determining Papua New Guinea's trade volumes.

On tariff revenue, tariff policy plays an important but diminishing role in Papua New Guinea's fiscal revenue. The tax-to-GDP ratio declined to 15.8 per cent in 2023, even as tax revenue grew in absolute terms. Import duties rose from \$71.5 million in 2010 to \$112 million in 2023,¹ yet their share in total revenue fell as the Global Sales Tax and other taxes became more prominent. Consumer goods remain the largest source of tariff revenue because of high import volumes, especially for fuel, vehicles, alcohol, and tobacco. In contrast, raw materials and intermediates generate little revenue despite some high nominal rates, due to low import volumes. Simulations indicate tariff cuts during the TRP (2012–2017) reduced revenue by only about \$7.8 million annually, or just one per cent of tariff revenue, a negligible effect. After 2018, tariff hikes were expected to boost revenue, but collections were about \$15.6 million lower than they would have been under 2018 baseline tariffs. This outcome was driven largely by the sharp cut in tobacco duties, one of Papua New Guinea's top revenue sources, alongside reduced import volumes and likely evasion incentives. The experience showed tariff liberalization cost Papua New Guinea little in revenue terms.

Industry views, as expressed through the business survey, paint a complex picture of competitive challenges that are informative for trade policy considerations beyond direct tariff impacts on trade flows. During the TRP, 36 per cent of firms reported benefits from tariff reductions through lower input costs, enabling price cuts, expansion, and productivity gains, though 55 per cent saw no impact. After tariff increases from 2018, 53 per cent of firms reported negative effects, suggesting policy volatility created winners and losers, with losses more visible than gains. Import competition was reportedly intensified, with 53 per cent of firms reporting impacts on sales and business models due to consumer preferences for cheaper imports. Price is perceived as the main problem: Papua New Guinea businesses were concerned that they cannot match import prices because their costs (transport, electricity, inputs) are too high. Next to price consideration, quality perception emerged as the second most important factor affecting competitiveness, per Papua New Guinea businesses' perception. Businesses further identified transport and logistics costs among the most severe constraints, alongside expensive inputs, policy unpredictability, and regulatory burdens. These findings show Papua New Guinea's competitive challenges are structural and require supportive infrastructure and regulatory reforms alongside trade policy.

Innovation under this report is considered broadly via the ability to introduce new products to the market. During the TRP, Papua New Guinea experienced a notable surge in export diversification, followed by a decline once tariffs were raised. Between 2010 and 2018, the number of distinct export products expanded significantly, with 591 new products entering Papua New Guinea's export portfolio, representing 45 per cent of the total basket. However, the majority of these exported items were of low value, with only 44 new products exceeding \$100,000 in value in 2018. This latest export drive was supported by

¹ 189 million and 403 million Papua New Guinea Kina.

cheaper imported inputs and machinery, which lowered production costs and enabled firms to test new product lines in international markets. However, sustainability was limited, with only 465 products (35 per cent) surviving through the period, underscoring the importance of addressing structural constraints beyond tariff reductions.

Consumer welfare was affected by the radical change in tariffs of a few product groups.

During 2010–2018, simulations show a net welfare loss of around \$5.9 million for Papua New Guinea consumers. While tariffs fell on many consumer goods, Papua New Guinea retained or raised very high tariffs on beverages and tobacco, which alone caused losses of about \$6.5 million. Beyond beverages and tobacco products, Papua New Guinea increased tariffs on about \$390 million of other goods, lifting average rates from 5 to 14 per cent which contributes a welfare loss of \$1 million. The total consumer welfare loss, therefore, outweighed the \$1.6 million gain consumers enjoyed from all the other tariff reductions.

The evidence reveals that tariff policy alone cannot determine economic development outcomes in contexts where structural constraints fundamentally limit competitiveness.

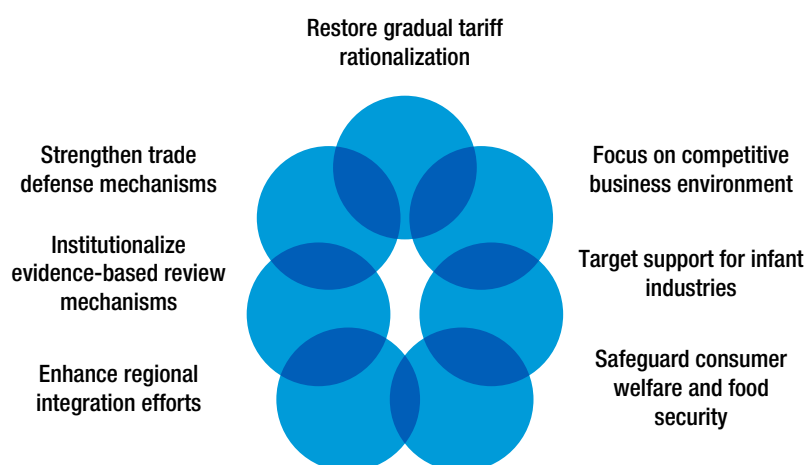
Papua New Guinea’s businesses face transport costs, energy expenses, and regulatory complexity that often dwarf the effects of tariff modifications, suggesting that successful development strategies must also address these underlying constraints to deliver more results from trade policy instruments like tariffs.

Yet trade policy remains an important component of the government policy toolbox for a comprehensive development strategy.

The TRP period’s correlation with stronger economic growth, export diversification, and increased business dynamism suggests that trade openness can contribute to economic vitality when implemented within supportive policy frameworks. Conversely, contraction in exports and GDP following TRP suspension appears to show some association between the two.

The study of the TRP reveals both potential benefits and limitations of trade policy as a development tool, which provide valuable lessons for future policy design and implementation. It is suggested that to yield results, trade policy requires consistent efforts to address fundamental constraints while maintaining openness to international opportunities that can accelerate development progress. Moving forward, Papua New Guinea should aim to formulate a tariff policy that balances development objectives with fiscal revenue needs and a realistic dependence on outside sourcing for its industry.

The following **recommendations** are suggested **for future tariff policy formulation**:



1. Restore gradual tariff rationalization

Papua New Guinea should adopt a predictable, phased tariff reduction path to allow industries time to adjust. A simpler, more uniform structure with low rates would reduce distortions, improve certainty, and avoid counterproductive high tariffs that harm consumers, encourage smuggling, and undermine industrial growth.

2. Focus on competitive business environment

Policy should prioritize reducing the high cost of doing business rather than shielding firms with tariffs. Investments in infrastructure, power supply, and streamlined customs can lower costs, while reducing tariffs on key inputs will directly enhance competitiveness and help firms produce goods that can compete globally.

3. Use targeted support for infant industries

Support to priority sectors should be time-bound, specific, and performance-based rather than relying on broad tariff protection. Targeted measures like subsidies, tax relief, or technical assistance will boost productivity and encourage industries to mature into competitive sectors, while public-private dialogue can ensure support meets actual business needs.

4. Safeguard consumer welfare and food security

Tariff policy must protect low-income households by keeping tariffs on essential goods like food, fuel, and medicine at low or zero rates. Excise taxes, rather than tariffs, should be used to regulate socially harmful goods such as alcohol and tobacco without distorting trade or raising consumer costs unfairly.

5. Enhance regional integration efforts

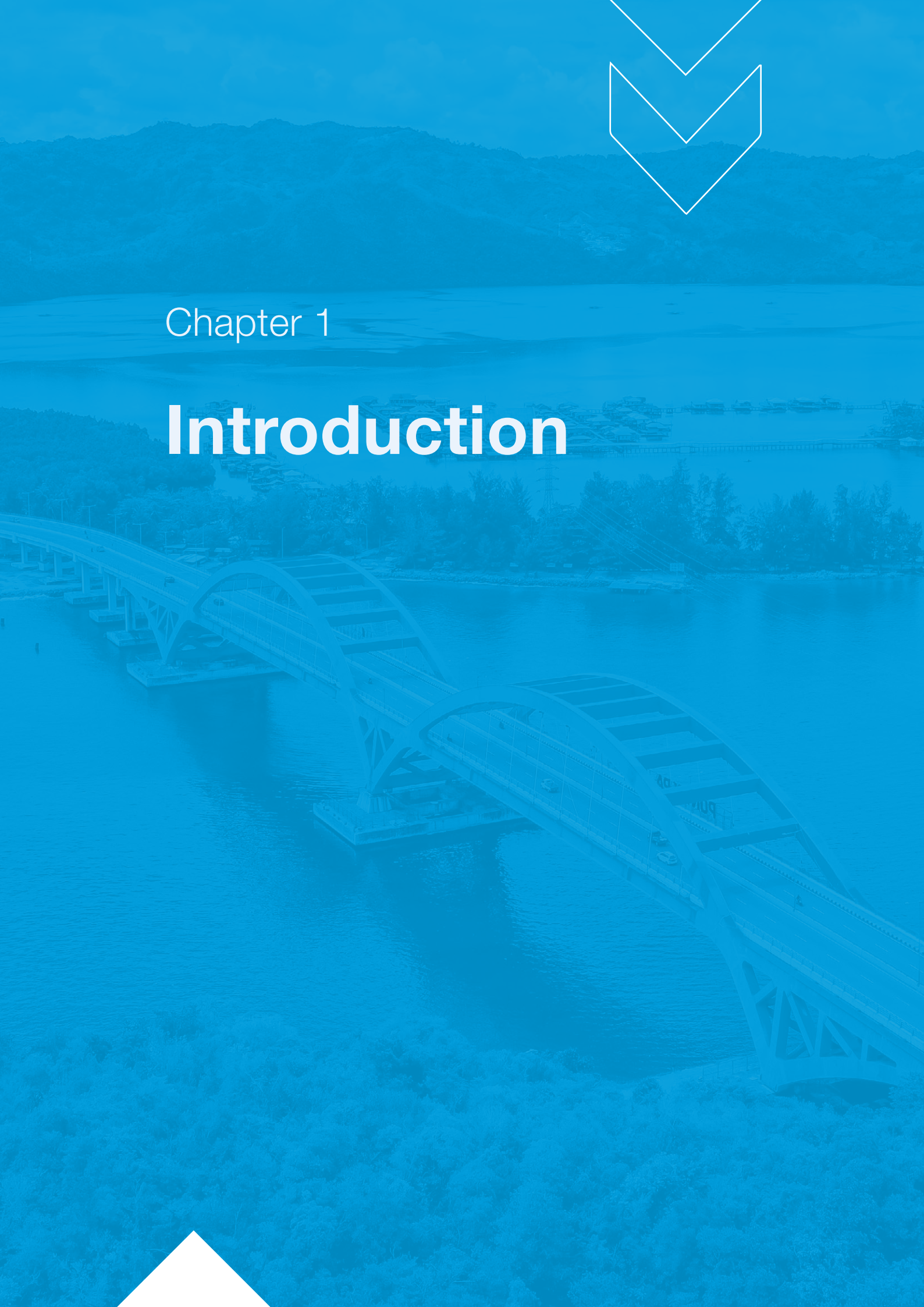
Beyond lowering tariffs, Papua New Guinea should reduce non-tariff barriers and strengthen customs efficiency to lower trade costs. Greater engagement in regional and multilateral agreements, such as PACER Plus, MSG, ASEAN+1, and WTO initiatives, can expand market access, improve input sourcing, and support diversification of export opportunities.

6. Institutionalize evidence-based review mechanisms

Tariff policy should follow a transparent, institutionalized review process rather than ad hoc budget announcements. Establishing a Tariff Advisory Board and conducting regular impact assessments would help ensure tariff measures are evidence-based, predictable, and phased out when no longer needed, boosting investor confidence and policy credibility.

7. Strengthen trade defence mechanisms

Papua New Guinea should build institutional capacity to monitor, investigate, and respond to unfair trade practices such as dumping or subsidised exports. Enacting anti-dumping and safeguard laws, supported by a specialised trade remedies unit, would help protect domestic industries, maintain a level playing field, and align with WTO rules.



Chapter 1

Introduction



Papua New Guinea rolled out the Tariff Reduction Programme (TRP) in 1999 as part of a broader economic recovery and reform agenda. After a phase of deep tariff cuts, which started in 2012, the TRP was suspended in 2018 with the reversing of rates for some tariff lines. As Papua New Guinea implement new national development frameworks, including the Papua New Guinea National Trade Policy 2017–2032, the Medium-Term Development Plan IV (2023–2027), and the Medium-Term Revenue Strategy (2023–2027), as well as other sector-specific plans in fisheries, forestry, agriculture, and manufacturing, there is a need for a structured and evidence-based assessment of the TRP.

This TRP review is intended to conduct a comprehensive analysis of the cost and benefits of the TRP to date, assess the economic implications of its suspension, and provide well-informed policy options for the future. This includes determining the TRP's impact on domestic industries, consumers, trade competitiveness, and alignment with international agreements such as the Interim Economic Partnership Agreement (iEPA) with the European Union and commitments under regional and multilateral frameworks.

This study meets several specific objectives as follows:

- To evaluate the impact of tariff changes on the private sector, including determining effects on competitiveness, efficiency and productivity by surveying companies to understand their response to the policy.
- To analyse the benefits and costs of tariff changes on different stakeholders,

including consumers (through choice/variety of products and price reductions), businesses (competitiveness), and trade participants (importers/exporters).

- To determine the impact of tariff changes supporting downstream processing and value addition in sectors such as manufacturing, fisheries, and agriculture.
- To evaluate whether certain exogenous factors explain any unintended outcomes arising from the tariff changes.

The review provides a set of conclusions and recommendations in order to guide the development of a future coherent, transparent, and development-friendly import tariff policy that supports the government's overarching objectives of promoting value addition, import substitution, and economic resilience in a globally competitive context.



Chapter 2

Context





A. Macroeconomic overview

Papua New Guinea is a lower-middle-income economy with a Gross Domestic Product (GDP) of \$32.8 billion and GDP per capita of \$2.6 thousand in 2025 (IMF, 2025). This income level is modest by global standards and also below the East Asia and Pacific averages of \$3.6 thousand. The Papua New Guinea economy has been on a positive growth trajectory since

2000, with episodes of negative growth due to low commodity prices (early 2000s), financial crisis (2008), the end of the liquefied natural gas (LNG)-led construction boom (2018), and the COVID-19 pandemic (2020-2021) (ADB, 2012; IMF, 2025). Post-pandemic, the country managed to achieve stable growth of around 4-5 per cent annually (see Figure 1).

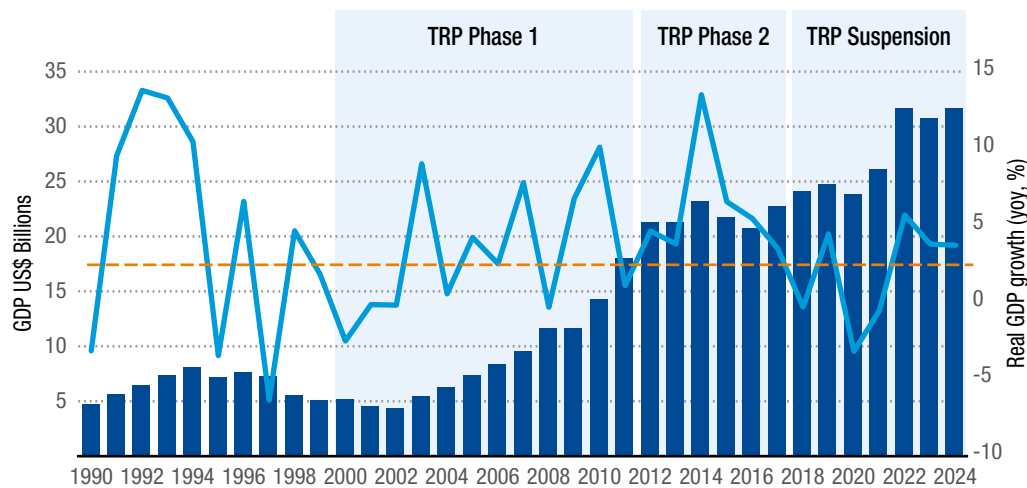
The Papua New Guinea economy has been on a positive growth trajectory since 2000



Figure 1.

Papua New Guinea has experienced steady economic growth

GDP in current prices (US\$ billion) and growth (per cent)



Source: IMF (2025)

Extractive industries (minerals, petroleum) and agriculture form the backbone of Papua New Guinea's economy. According to data from the Papua New Guinea National Statistical Office (2025), Mining and Quarrying contributed to 27 per cent of the national gross output in 2022, followed by Agriculture, Forestry and Fishing (15 per cent), and Construction (9 per cent). Meanwhile, manufacturing and broader industrial sectors remain small, accounting for 5 per cent of the national gross output in 2022. This narrow production base means growth surges during commodity booms (or project construction phases) but also leaves Papua New Guinea vulnerable to global price swings and incidental shocks.

Mining, after an initial decline from 2010 to 2012, saw a significant surge from 2014 to 2019, followed by a further surge in 2021-22, in line with the global commodity

boom and Papua New Guinea's increased mineral exports. Manufacturing, on the other hand, did not grow as quickly as agriculture and mining until 2020 but then expanded strongly between 2021 and 2023.

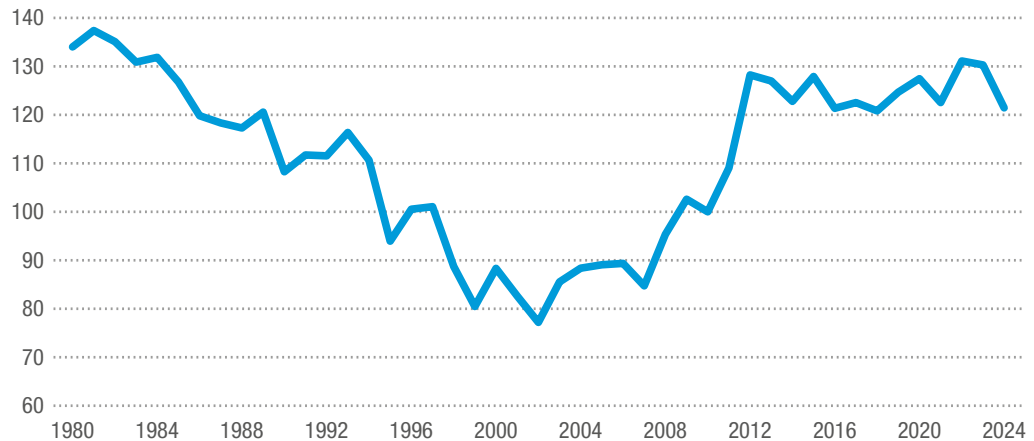
Exchange rate developments during this period were influenced by capital inflows associated with the construction phase of a large liquefied natural gas project, which led to a marked appreciation of the kina in real terms up to around 2012 (IMF, 2015). Following the completion of the construction phase, the real effective exchange rate remained at a higher level and has since fluctuated within a relatively narrow range (Figure 2). This sustained high real effective exchange rate has implications for international competitiveness, as it makes exports relatively more expensive and imports relatively cheaper than domestically produced goods, all else being equal.



Figure 2.

The kina has remained stable over the past 10 years following a strong increase at the turn of the millennium

Index of Papua New Guinea kina's real effective exchange rate against major trading partners, 2010 = 100



Note: Real effective exchange rate is the nominal effective exchange rate (a measure of the value of a currency against a weighted average of several foreign currencies) divided by a price deflator or index of costs.

Source: World Bank WDI. Accessed on 20 March 2025

Based on the level of diversification of exports over this period, Papua New Guinea had a relatively low complexity

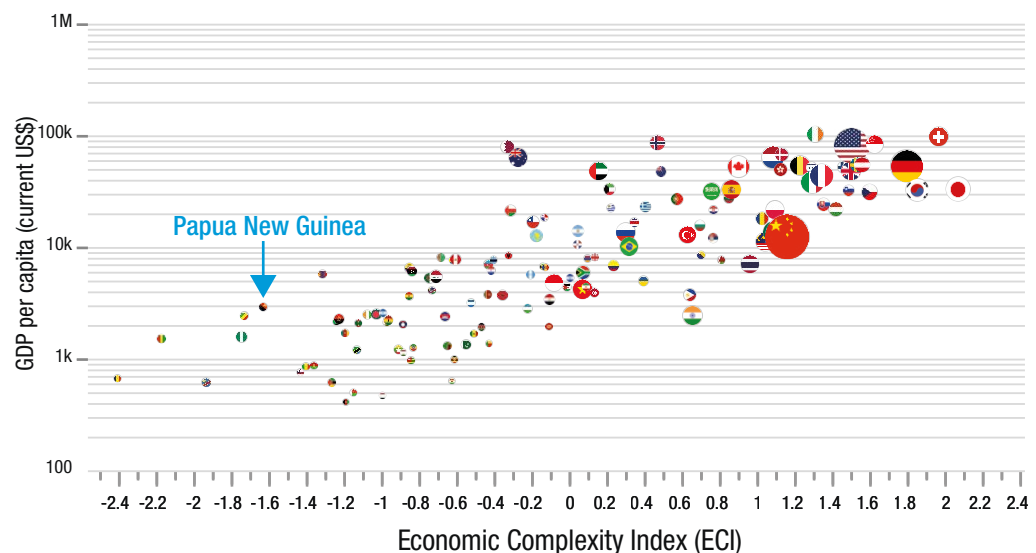
of its economy compared to other countries of similar GDP (Figure 3).



Figure 3.

Papua New Guinea ranks low on economic complexity

Comparison of Papua New Guinea's economic complexity index and GDP per capita (US\$ PPP) with other economies, 2023



Source: Harvard Growth Lab. Accessed August 2025

The country ranks near the bottom globally: in 2023, its estimates of economic complexity index (ECI) based on trade data

(ECI trade), patent data (ECI technology), and research publication data (ECI research), reached -1.5, placing Papua



New Guinea 127th out of 132 economies. Over 20 years, Papua New Guinea's ranking in economic complexity has drifted downward, indicating that it is not moving much into new, higher-capability products. Furthermore, product specialization remained limited. During 2010-2023, Papua New Guinea's product space has remained concentrated on some products in the Minerals and Agriculture clusters (Figure 4).

The concentration in low- to moderate-complexity commodities gives Papua New Guinea certain characteristics –namely, vulnerability to commodity price swings, but also strong export revenues when prices rise. However, this structure limits Papua New Guinea's ability to move up the value chain, develop export products with greater technology, branding, or processing content, and thus capture more value. Because many of its current comparative advantages are in goods that require fewer complementary capabilities (infrastructure, advanced skills, industrial inputs), moving into more complex products will require building those capabilities.

Diversification is similarly constrained by the geography of Papua New Guinea's product space: many products that are adjacent (i.e. "nearby" in the product space) to what Papua New Guinea currently exports are still commodities or low-processed goods (e.g. further processing of minerals

or agricultural goods). There appear to be few "low-hanging fruit" opportunities in high-tech or knowledge-intensive manufacturing already very close to Papua New Guinea's existing export basket. The product space suggests that Papua New Guinea must invest in capabilities (skilled labour, reliable infrastructure, regulatory capacity) to bridge larger "distances" in the product space for diversification. Innovation (in terms of product design, new export categories) will likely come slowly unless there is sustained policy effort toward human capital, research and development (R&D), value-added processing, and connectivity both domestically and with international value chains.

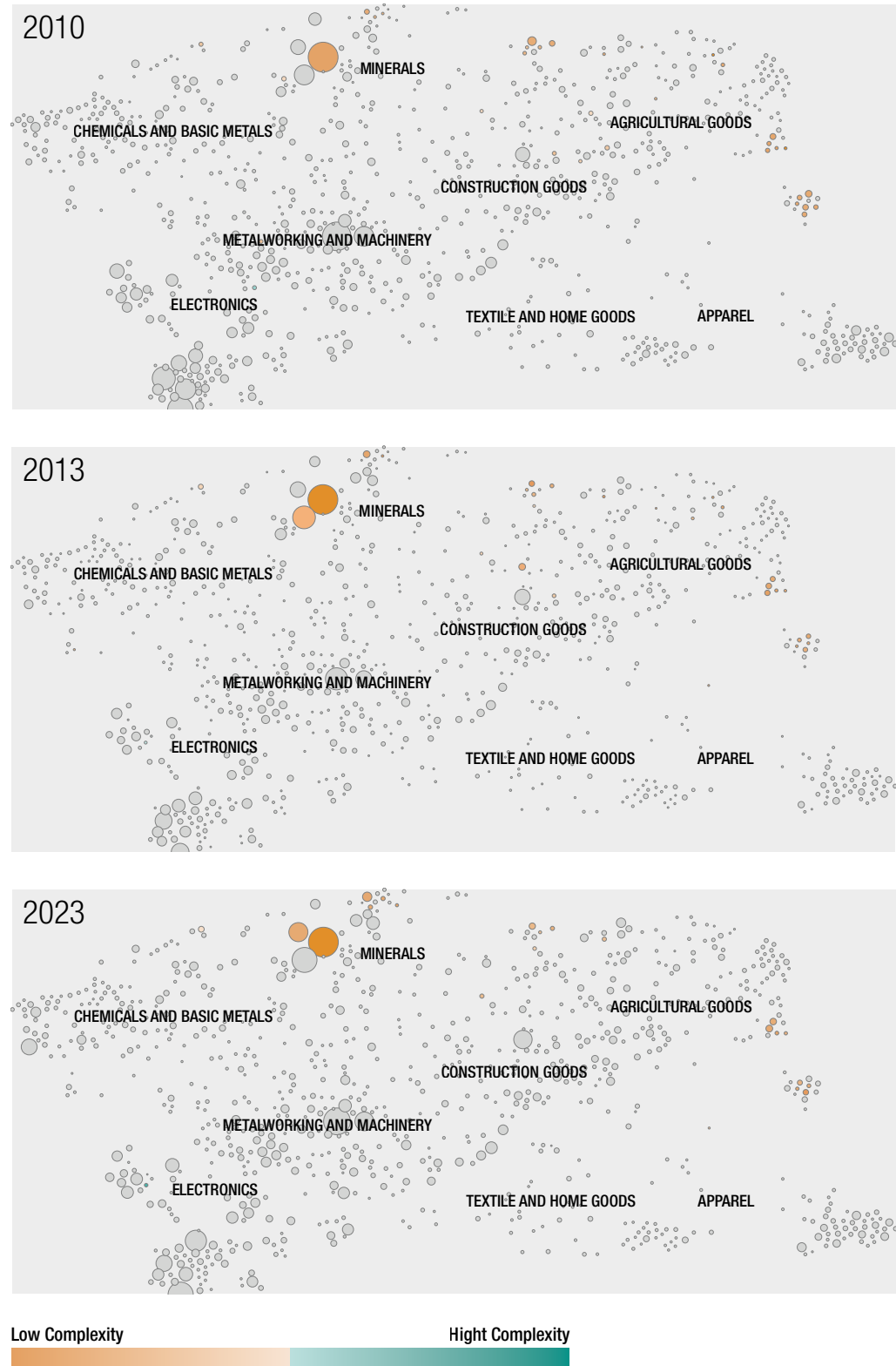
The evidence from 20 resource-rich economies suggests that resource-rich countries tend to exhibit a relatively constant level of economic complexity, with limited progress. Some explanations for this observation are that the findings of the study support the natural resource curse hypothesis, confirming that countries rich in natural resources tend to have lower economic complexity. Additionally, the study highlights the positive impact of corruption control on economic complexity, indicating that improving good governance can contribute to the improvement of economic complexity in resource-rich countries (Owjimehr & Jamshidi, 2024).



Figure 4.

Papua New Guinea's product space has evolved little over time

Changes in the country's export structure over time, 2000, 2010 and 2023



Source: Observatory of Economic Complexity. Accessed August 2025



B. Papua New Guinea's Tariff Reduction Programme: a chronology

1. Initiation of the Tariff Reduction Programme (late 1990s – early 2000s)

Papua New Guinea embarked on a significant Tariff Reduction Programme (TRP) in 1999 as part of a broader economic recovery and reform agenda.

The TRP was introduced in exchange for support from international financial institutions such as the International Monetary Fund (IMF) and World Bank, and was aimed at making Papua New Guinea's trade regime more open and competitive (Scollay, 2007). A new Value Added Tax (VAT) and Government Sales Tax (GST) were simultaneously implemented to offset the revenue losses from cutting tariffs. The government's 1997 Tariff Reform White Paper, inspired by a 1996 World Bank review, argued that high import duties were hindering efficiency, raising business costs, and protecting uncompetitive industries. Lowering tariffs was seen as a way to encourage a more productive private sector by exposing firms to international competition. At the same time, the new VAT would replace tariffs as a revenue source (Scollay, 2007). This policy shift came at a time of economic difficulty, following the Asian financial crisis and commodity price slumps of the late 1990s, but reflected Papua New Guinea's commitment to outward-oriented reforms under World Trade Organisation (WTO) membership and the Asia-Pacific Economic Cooperation (APEC) goals.

2. Phased tariff reductions (1999–2006)

Under the TRP, Papua New Guinea simplified its tariff structure and implemented gradual cuts between 1999–2006. Tariffs on goods produced locally, approximately three-quarters of all imports, were set to zero, making most products duty-free (Scollay, 2007). All remaining imported goods (those produced domestically or with production potential in Papua New Guinea) were classified into three tariff bands: “*intermediate*” (30 per cent), “*protective*” (40 per cent), and “*prohibitive*” (55 per cent) rates.

The TRP legislated stepwise reductions of 5 percentage points for each band at set intervals. In practice, tariffs fell in stages on January 1 of 1999, 2000, 2001, 2003, and finally 2006. By the end of 2006, the intermediate rate had been reduced to 15 per cent, the protective rate to 25 per cent, and the prohibitive rate to 40 per cent.² These cuts effectively halved Papua New Guinea's average applied tariff, with the unweighted average tariff dropping from about 20 per cent in the late 1990s to roughly 5 per cent by 2006. The new structure dramatically increased the share of duty-free lines from around 3 per cent before the cuts, to about 77 per cent of tariff lines by 2006. Only about 15 per cent of tariff lines (mainly for domestically produced foods and manufactures) still faced import duties at the protective 25 per cent or prohibitive 40 per cent rates by 2006 (UNCTAD, 2006). A few sensitive products, such as sugar, canned fish, salt, tobacco, alcohol, and plywood, had been granted special high protection due to prior government commitments or industry

Under the TRP, Papua New Guinea simplified its tariff structure and implemented gradual cuts between 1999–2006.

² The protective rate is intended to safeguard domestic producers of finished goods. The intermediate rate applies to production inputs that are produced, or could potentially be produced, in Papua New Guinea. The prohibitive rate is intended to strongly discourage imports of goods for which domestic production is considered particularly important (Scollay, 2007).

lobbying, and these were exempted from the standard TRP timetable.

Overall, the first phase of the TRP (1999–2006) achieved a more open trade regime with a simpler, lower tariff structure, aligning Papua New Guinea’s policies closer to global norms and its WTO bindings.

The rapid tariff reductions in the early 2000s generated controversy. On the one hand, the reform was praised for improving efficiency and lowering costs; on the other, domestic manufacturers grew concerned about increased competition (Fox et al., 2018; Laveil, 2020). By 2002–2003, evidence suggested that government revenue had fallen more than expected due to lower economic activities, particularly declining mineral revenue, and less revenue from tariffs (UNCTAD, 2006). These fiscal pressures and industry concerns led to calls to slow down or modify the TRP. In 2003, the Manufacturers’ Council of Papua New Guinea petitioned the government to suspend further tariff cuts, proposing to freeze rates at 45, 30, and 20 per cent (instead of dropping to 40, 25, 15 per cent, as scheduled) (Scollay, 2007). In response, the government commissioned a Tariff Review Taskforce in 2003 to evaluate the impacts of the TRP. This review found that the tariff reductions did not negatively affect the economy nor devastate the manufacturing sector (Laveil, 2020). Notably, industries with the highest protective tariffs (such as canned tuna/mackerel processing) were among the slowest-growing, suggesting that heavy protection was not yielding the intended industrial growth (Laveil, 2020). High import duties were also found to affect exporters by raising the cost of imported inputs.

As a result of these findings, the government rejected the moratorium proposal and decided to complete the TRP as planned (Fox et al., 2018). By 2006, the final step of the initial program was implemented, bringing tariffs to their new low bands of 40, 25 and 15 per cent, as mentioned above. The 2003 review even recommended extending the tariff

cuts beyond 2006, envisioning further 5 per cent reductions every few years such that all tariffs would eventually converge at 15 per cent (Scollay, 2007; UNCTAD, 2006). However, this suggestion was not adopted at the time, and trade policy liberalization essentially paused once the first TRP was completed in 2006 (WTO, 2010). The mid-2000s debate underscored the balance Papua New Guinea sought between fostering infant industries and pursuing an open, competitive economy. While the government remained committed to outward-looking reform, it became clear that complementary measures (improving infrastructure, good governance, and tax compliance) were needed to maximize the benefits of tariff reform (Scollay, 2007).

While an independent review of the tariff policy suggested that lower tariffs were not causing damage to national industries and could support broader economic growth, many stakeholders remained unconvinced about the overall benefits arising from the tariff reductions. As a result, further tariff reforms were suspended.

3. Continued tariff reductions (2007–2017)

After a brief pause in tariff changes, Papua New Guinea revisited its tariff policy to achieve further liberalisation in the late 2000s. The Tariff Review in 2007 (Scollay) assessed the post-TRP landscape and paved the way for a second phase of tariff reductions. The government’s development strategies, such as the Medium-Term Development Strategy and Vision 2050/Development Strategic Plan, emphasizing improving competitiveness and integrating Papua New Guinea’s economy with global and regional markets. In line with this, the 2007 review proposed additional gradual cuts to move Papua New Guinea toward a more uniform and lower tariff structure over the next decade (Fox et al., 2018). These recommendations were eventually codified in the Import and Export Customs Tariff Act of 2012, which legislated a schedule of tariff changes: some

reductions took effect immediately in 2012, followed by further 5 per cent cuts planned on January 1, 2015, and again planned on January 1, 2018. The ultimate goal was to reduce the remaining protective tariffs to around 10 per cent, flattening the tariff bands at a low rate across all products by 2018 (Laveil, 2020). For example, under this plan, the standard protective duty rate would drop from 25 per cent (post-2006) down to 15 per cent by 2015 and 10 per cent by 2018, while the highest “prohibitive” rate would fall from 40 to 30 per cent by 2015 and eventually 25 per cent by 2018. By the mid-2010s, Papua New Guinea

had implemented most of these scheduled cuts: the average of the three tariff bands declined from about 42 in 1999 to 18 per cent by 2015. Papua New Guinea’s trade policy in this era remained outward-oriented, influenced by international commitments. As a member of the WTO, Papua New Guinea had bound many of its tariffs at ceilings (often around 40-45 per cent), and its applied rates after the TRP were well below these bindings (UNCTAD, 2006). This gave Papua New Guinea flexibility to liberalise and policy space to raise tariffs up to the bound rates if deemed necessary.



Box 1. TRP summary (2007-2017)

After a 2007 Tariff Review, it was recommended that further liberalisation be implemented toward a uniform, lower tariff structure:

- Protective tariffs dropped from 25 to 15 and then 10 per cent;
- Prohibitive tariffs from 40 to 30 and then 25 per cent by 2018;
- Average tariff bands declined from 42 (1999) to 18 per cent (2015).

Reforms were supported in the Medium-Term Development Strategy and Vision 2050 goals for competitiveness and global integration.

In terms of international commitments, WTO bindings at 40-45 per cent provided policy flexibility; APEC Bogor Goals endorsed free trade by 2020, and Papua New Guinea engaged in regional agreements (see Box 2).

Reductions were resumed and subsequently halted during this phase of the program. Overall, tariffs fell considerably.



Box 2. Tariffs in the context of free trade agreements

Papua New Guinea was also mindful of its regional and bilateral trade agreements. It joined the Pacific Island Countries Trade Agreement (effective mid-2000s) and the Melanesian Spearhead Group (MSG) agreement, which progressively eliminated tariffs on trade with neighbouring Pacific states, though these had a limited impact on overall tariff policy since they apply only to regional partners. Under an interim Economic Partnership Agreement (EPA) with the European Union, signed in 2009 and ratified in 2011, Papua New Guinea committed to gradually fully liberalize 88 per cent of its imports from the European Union, with sensitive items excluded (European Commission, n.d.). Papua New Guinea chose to liberalise its trade with the European Union almost immediately, reflecting that many of those imports already faced low or zero most-favoured-nations (MFN) tariffs as a result of the TRP.

Meanwhile, as an APEC member, Papua New Guinea had endorsed the Bogor Goals of free and open trade by 2020 for developing economies.³ All these international frameworks reinforced Papua New Guinea's tariff reduction trajectory and the government's objective of using low, uniform tariffs to promote efficiency and investment. By 2015, tariffs were at historically low levels, and the country was more integrated into global trade networks, with few import restrictions and relatively low protective barriers.⁴

4. Reversal and shift towards protectionism (2017–present)

In the late 2010s, Papua New Guinea's tariff policy took a significant turn as a result of economic challenges.

Facing economic headwinds such as foreign exchange shortages, fiscal pressures, and a push to develop domestic industries, the government began reconsidering its open trade stance. A notable precursor was a brief import ban on select fruits and vegetables from Australia in late 2015, an attempt to protect local farmers (Fox et al., 2018).

In August 2017, the government formally suspended the TRP before the final 2018 cuts could be implemented.

Instead of reducing the remaining tariffs to 10 per cent as planned, Papua New Guinea moved to raise many tariffs in order to shelter domestic producers. In the 2018 Budget (enacted in late 2017), authorities announced increases on roughly 250 tariff lines and simultaneously scrapped or deferred over 600 scheduled tariff reductions. Average tariff rates on affected goods were raised by about seven percentage points in 2018. This policy reversal accelerated in the following year: by 2019, Papua New Guinea had introduced a total of 323

tariff line increases over two years, with an average hike of around 14 percentage points in the second round (Laveil, 2020).

Many of these changes targeted goods that compete with nascent or existing local industries, in line with the government's renewed emphasis on import substitution and industrialization.

For example, a 25 per cent import duty on milk was imposed (milk had previously been duty-free) to support Papua New Guinea's first domestic dairy venture, Ilimo Dairy Farm. Tariffs on items like frozen meats, packaged fruits, sugar and confectionery, flour and cereals, certain textiles and apparel, furniture, and household goods (soaps, detergents, etc.) were also raised significantly. The highest tariffs faced were for tobacco products with an ad valorem equivalent (AVE) of 155 per cent, followed by 65 per cent for beverages, 34 per cent for edible vegetables, 27 per cent for meat and meat products, 26 per cent for pearls and diamonds, 23 per cent for sugars, 20 per cent for furniture and 20 per cent for coffee (See Figure 5). The Customs Tariff Amendment Act of 2018 explicitly stated its intent was to "provide relief to local pioneer industries and existing local manufacturers from cheap imports" (Laveil, 2020). In practice, this meant reinstating protective barriers for a range of products to give local businesses a price advantage over imports.

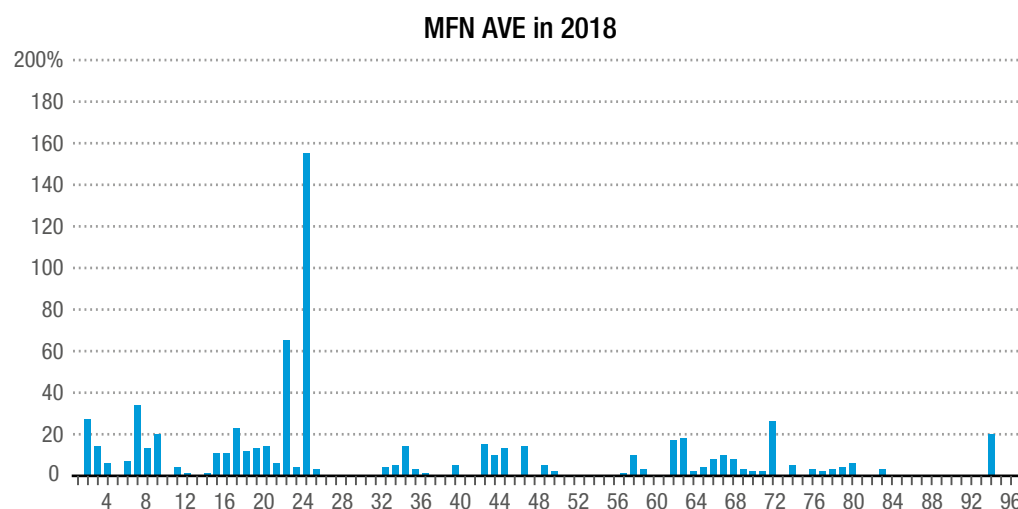
Average tariff rates on affected goods were raised by about seven percentage points in 2018.

³ The Bogor Goals, adopted at the 1994 APEC summit in Bogor, Indonesia, represent APEC's commitment to achieving free and open trade and investment in the Asia-Pacific region. Under these goals, developed member economies pledged to meet this objective by 2010, while developing economies, including Papua New Guinea, targeted 2020. The goals emphasize voluntary, non-binding cooperation and are pursued through Individual and Collective Action Plans that guide economies in reducing trade barriers, improving investment conditions, and enhancing economic integration. See APEC (1994). 1994 Leaders' Declaration. Bogor, Indonesia.

⁴ World Trade Organisation (2010). Trade Policy Review. Report by the Secretariat. Papua New Guinea.

**Figure 5.****Tariff levels vary across products in Papua New Guinea**

Tariffs at the harmonized system 2 level, 2018 (per cent).



Source: UNCTAD based on ITC Macmap. Accessed June 2025.

This reversal of past policies marked a new protectionist chapter in Papua New Guinea's trade policy. After nearly two decades of falling tariffs, Papua New Guinea's rates on some goods began climbing for the first time since the late 1990s. The tariff schedule became more complex, with new 12.5 per cent, 20 per cent, and 25 per cent bands added. The change sparked debate. Proponents argued this protection would nurture infant industries such as dairy, poultry, and textiles. Critics noted mixed past results from similar policies: Ramu Sugar struggled for decades despite high protection, while Halla Cement only became competitive after privatisation and lower tariffs (Fox et al., 2018). Critics of the 2018–19 tariff hikes pointed to these, arguing that prolonged protection can lead to inefficiency and high consumer prices without guaranteeing a globally competitive industry (Laveil, 2020). They also warned that reversing the TRP could harm Papua New Guinea's export sectors, as higher tariffs raise input costs for Papua New Guinea's exporters and risk retaliatory measures or losing credibility in trade negotiations.

By 2020, the government signalled a pause on further tariff changes. The

2020 budget contained no new increases, but the protective measures remained in place with no further tariff reduced. As of the early 2020s, Papua New Guinea's applied tariff rates remained higher than the lows achieved under the TRP, especially for products deemed important to "import-replacement" industries. Papua New Guinea stayed within its WTO bindings, avoiding breaches, but the reversal marked a clear shift from the liberalisation path followed since 1999.

This evolution from aggressive tariff cuts in the early 2000s to consolidation and commitment to free trade agreements and then to a resurgence of protectionist measures highlights how economic conditions, government objectives, and international influences have continually reshaped Papua New Guinea's tariff landscape over the past twenty years. Each phase of the Tariff Reduction Programme and decisions to partial reverse changes were driven by a mix of economic rationale and political debate, underscoring the challenge of balancing development goals with global trade commitments in Papua New Guinea.



Box 3.
TRP suspension summary (2017-2020)

In **August 2017**, the TRP was suspended before final reductions. In the **2018 Budget**, tariffs were raised on around 250 lines; more than 600 cuts were cancelled. In **2019**, they were 323 further increases, with an average hike of 14 points.

The highest duties in 2018 were: tobacco (155 per cent), beverages (65 per cent), vegetables (34 per cent), meat (27 per cent), furniture (20 per cent). New tariff bands (12.5 per cent, 20 per cent, 25 per cent) replaced the uniform tariff goal.

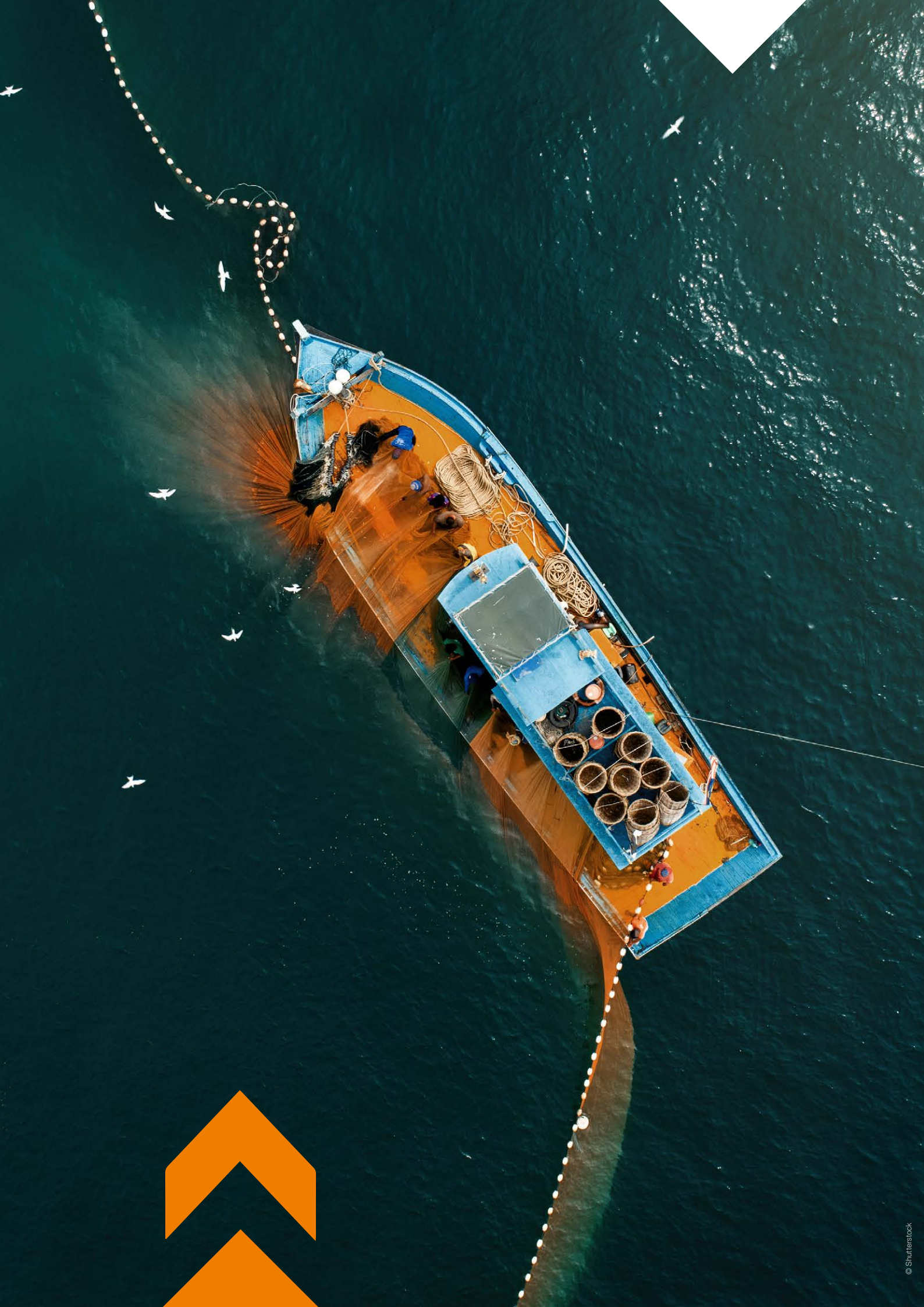
In 2020, further increases were paused, but higher tariffs remained.

Over the latest period of the tariff reforms, further reductions were implemented and subsequently halted. While progress was made, tariffs were above the lowest rates achieved earlier in the program.



Chapter 3

Government policies on trade and industry



Papua New Guinea implemented several sectoral and horizontal plans to support industry and trade. While tariff policies are typically decided by a Ministry of Finance (or Treasury in the case of Papua New Guinea), their objectives are often multiple and span several policy domains. These may include raising government revenue to ensure fiscal sustainability, restricting products harmful to the environment, protecting emerging industries, supporting industrial or agricultural development, and addressing external imbalances. Because of these diverse objectives, coordination across government institutions with different mandates is therefore essential to ensure alignment with the various policies of the Government.

Tariff policy aligned with the goals of the Medium-Term Development Plan (MTDP) IV (2023–2027) to promote diversification. By raising tariffs on processed foods, textiles, wood products and consumer goods, the government created a price advantage for domestic firms engaged in downstream processing, in line with the MTDP IV's focus on job creation, SME promotion, and adding value to Papua New Guinea's natural resources before export.

Sectoral development plans in forestry, fisheries, agriculture, and industry also reveal the way tariff policy was integrated

into the drive for diversification. High tariffs on wood and furniture, and the log export ban policy, supported local timber processing. In agriculture, tariffs on dairy, vegetables, and processed foods created space for new agribusiness ventures, while in fisheries, duties on canned and smoked fish reinforced incentives for local processing. While tariffs alone were not always sufficient to stimulate competitive industries, they formed part of a wider policy mix designed to encourage local value addition, support SMEs, and expand Papua New Guinea's industrial base.

A. National Trade Policy (2017–2032)

Papua New Guinea's National Trade Policy (NTP) 2017–2032 marked a strategic shift in trade policy. At its launch in 2017, the government suspended the TRP. Furthermore, the government declared that Papua New Guinea would review all trade agreements and even consider suspending trade with partners seen as unfair, underscoring a more assertive, Papua New Guinea-first stance. Reflecting this stance, Papua New Guinea declined to join the Pacific Agreement on Closer Economic Relations Plus (PACER-Plus) regional free trade agreement. Instead seeking arrangements with Australia and

New Zealand that yield tangible investment, export earnings and jobs for Papua New Guineans (Post Courier, 2017).

The subsequent tariff adjustments in 2018–2020 align with the NTP's protective orientation. According to the 2019 Customs Tariff Amendment Act, the rationale for recent tariff increases was “to provide relief to local pioneer industries and existing local manufacturers from cheap imports” (The National, 2020). In other words, the tariff policy has been reoriented to nurture domestic industries – a key goal of the NTP – by sheltering them from import competition.

B. Medium Term Development Plan IV (2023-2027)

Papua New Guinea's MTDP IV embodies objectives that relate to trade and tariffs. The MTDP IV targets broad economic growth and explicitly aims to “encourage downstream processing of our natural resources” (Ministry of Treasury, 2024). This priority reflects a desire to add value locally in sectors like agriculture, forestry, fisheries and mining, rather than exporting raw commodities. The tariff adjustments support this goal by making certain imported finished goods relatively more expensive and thus giving an edge to local producers of processed foods, wood products, textiles, and other manufactures. For example, tariffs were raised on a range of food products and consumer goods, including frozen meats, packaged fruits and vegetables, sugar, confectionery, flour, garments, wood and paper products, furniture and others, which is consistent with incentivizing domestic processing in those product

chains. The MTDP IV also emphasizes job creation (one million additional jobs) and promotion of Micro, Small and Medium Enterprises (Fox et al., 2018).

However, the MTDP IV simultaneously calls for a “conducive environment to attract Foreign Direct Investment”. This implies that tariff policy must strike a balance, providing reasonable protection to encourage local production, while not creating an overly prohibitive or unpredictable trade regime that could deter investors. Excessive or *ad hoc* tariff increases could conflict with the objective of a stable, investment-friendly climate. This tension is recognized by policymakers; indeed, observers noted that Papua New Guinea's abrupt deviation from the TRP in 2018 introduced new tariff rates (12.5, 20, 25 per cent) beyond the previous uniform structure and set a precedent for lobbying by industries for even higher tariffs (Fox et al., 2018).

C. Medium Term Revenue Strategy (MTRS) (2023–2027)

Papua New Guinea's MTRS 2023–2027 provides another lens through which to view tariff policy. The MTRS seeks to halt the decline in revenue-to-GDP and raise it to about 17.9 per cent by 2027 through improved compliance and a broader tax base (Ministry of Treasury, 2024). It emphasizes shifting toward consumption-based taxes in a fair, efficient manner while minimizing ad-hoc exemptions.

Tariffs play a dual role in this strategy. On one hand, they provide revenue, and the increases in 2018–2019 (averaging 7 per cent and 14 per cent) offered short-term relief to a budget under strain (The National, 2020). On the other hand, the MTRS cautions against measures that undermine efficiency. Higher tariffs

can discourage imports, reduce GST collections, or encourage smuggling if rates are excessive (The National, 2020).

The trade-off between revenue and efficiency is acknowledged by analysts – for example, an overvalued exchange rate was identified as a more direct cause of competitiveness issues, suggesting that devaluing the Kina would be a more efficient way to assist both import-competing and export sectors than raising tariffs (Fox et al., 2018).⁵ Thus, to remain consistent with the MTRS, tariff policy should be carefully calibrated: providing needed revenue and industry support, but complemented by broader tax reforms and macroeconomic adjustments to ensure it does not undermine productivity or compliance.

⁵ While potentially more efficient, exchange rate adjustment may be constrained by broader macroeconomic and regional factors.

D. Sectoral plans (fisheries, forestry, agriculture, industry)

Papua New Guinea's sector-specific strategies in industries like fisheries, forestry, agriculture and manufacturing also help contextualize the tariff adjustments. A common theme in these plans is to

promote downstream processing and value addition within Papua New Guinea, rather than merely exporting raw materials. In each case, tariffs were used to support downstream processing and value addition.

Table 1.
Tariff measures in support of sectoral strategies (2018–2020)

Sector	Policy goal	Tariff measure	Outcome
Forestry	Reduce raw log exports, promote domestic timber processing	40–50 per cent tariffs on timber, plywood, furniture; planned log export ban by 2020	Higher import prices supported local mills and carpentry firms
Fisheries	Expand onshore processing (e.g. tuna canning)	High tariffs on canned and smoked fish; temporary 2015 ban on fresh produce	Limited impact from tariffs; EU–Pacific iEPA gave duty-free access for tuna exports, encouraging investment in canneries
Agriculture	Strengthen food security and reduce imports	New tariffs on previously duty-free products such as milk (25 per cent) and vegetables	Supported ventures like dairy and vegetable farming; tariffs applied selectively to balance affordability and nutrition

Source: UNCTAD compilation based on World Trade Organization (2019); The National (2020).

Table 1 summarises the main goals, tariff interventions, and outcomes in each sector. Tariffs were used to encourage local processing in key industries. In forestry, high import duties and a log export ban supported timber processing. In fisheries, tariffs on canned and smoked fish had limited impact, but duty-free access to the European Union under the iEPA attracted investment in canneries. In agriculture, new tariffs on products such as milk and

vegetables supported domestic agribusiness ventures, though affordability and nutrition concerns required selective application and complementary support measures.

Overall, the 2018–2020 tariff adjustments can be seen as aligned with Papua New Guinea's sectoral ambitions to foster local industries in fishing, farming, and manufacturing, particularly by encouraging more local processing of fish, logs, cash crops and basic consumer goods.

E. Multilateral and regional trade agreements

Papua New Guinea's tariff policy operates within the bounds of several international trade agreements, and alignment with these commitments is crucial. A summary of these agreement is shown in Table 2. As a member of the WTO, Papua New Guinea has bound all its tariff lines in its schedule of commitments, generally capping the maximum tariff rates it can apply. The recent tariff increases were mostly within those bound rates, but there are a few potential

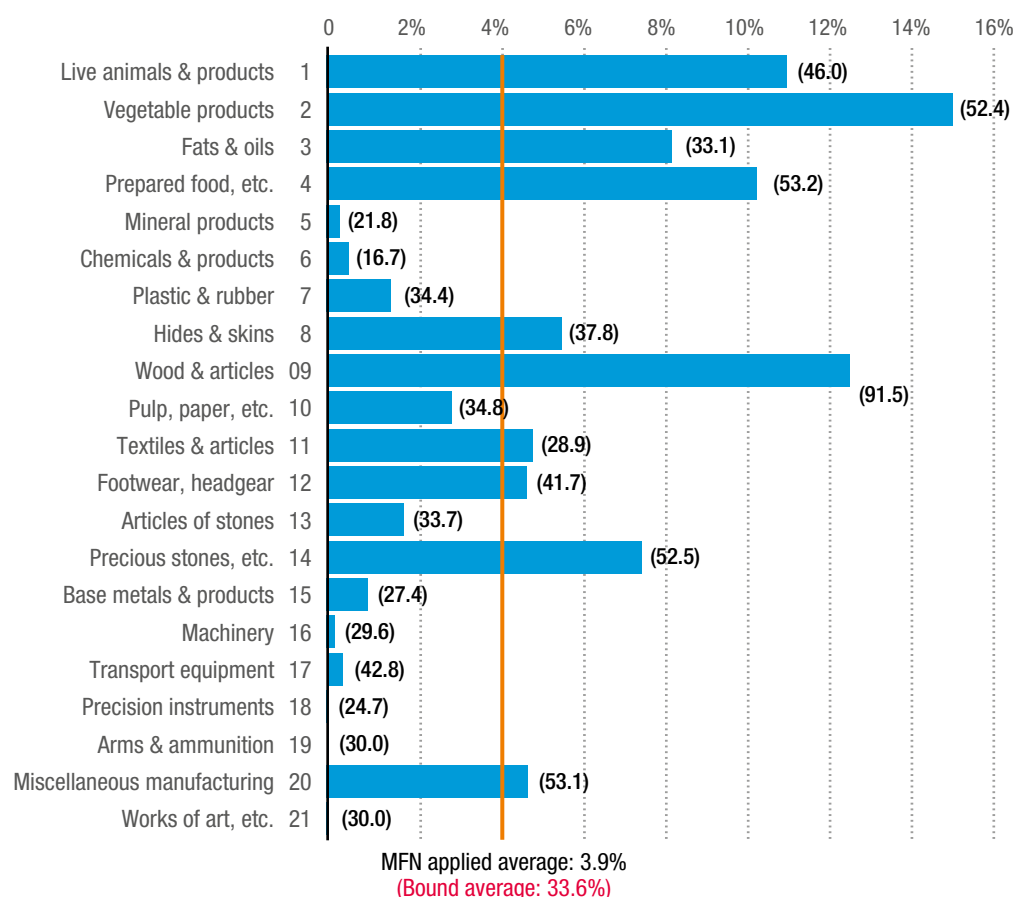
breaches: WTO data indicate that on 23 tariff lines, Papua New Guinea's applied tariffs now exceed the bound levels committed at the WTO (see Figure 6) (WTO, 2019).

At the regional level, Papua New Guinea is party to several trade agreements that aim to liberalize trade among specific partners. Under the Melanesian Spearhead Group Trade Agreement (MSGTA) and the Pacific Island Countries Trade Agreement



Figure 6.
Average applied MFN tariff rates differ across products in Papua New Guinea

Average applied MFN tariff rates by category, 2019 (per cent)



Source: WTO (2019). Papua New Guinea Trade Policy Review. Secretariat Report.



Table 2.
Papua New Guinea's network of FTAs

Agreement	Members	Year of enforcement
Australia - Papua New Guinea Agreement on Trade and Commercial Relations (PATCRA)	Australia; Papua New Guinea	1977
South Pacific Regional Trade, Economic and Commercial Agreement (SPARTECA)	Australia; Cook Islands; Fiji; Kiribati; Marshall Islands; Micronesia (Federated States of); Nauru; New Zealand; Niue; Papua New Guinea; Samoa; Solomon Islands; Tonga; Tuvalu; Vanuatu	1981
Melanesian Spearhead Group Trade Agreement (MSGTA)	Fiji; Papua New Guinea; Solomon Islands; Vanuatu	1994
Pacific Islands Countries Trade Agreement (PICTA)	Cook Islands; Fiji; Kiribati; Micronesia (Federated States of); Nauru; Niue; Marshall Islands; Palau; Papua New Guinea; Samoa; Solomon Islands; Tonga; Tuvalu; Vanuatu	2003
European Union – Pacific States iEPA	Fiji; Papua New Guinea; Samoa; Solomon Islands; European Union (EU)	2009
UK-Pacific Interim Economic Partnership Agreement (UK-Pacific iEPA)	Fiji; Papua New Guinea; Solomon Islands; United Kingdom	2021

Source: UNCTAD compilation.

(PICTA), for example, Papua New Guinea has pledged to reduce or eliminate tariffs on many goods traded with neighbouring Pacific countries. The protective tariffs imposed in 2018–2020 generally apply to MFN rates (i.e. to all suppliers globally), but Papua New Guinea must still grant preferential (lower or zero) rates to its MSG and PICTA partners on eligible products. In practice, this means the impact of some tariff hikes is blunted for imports from Fiji, Solomon Islands or other Pacific partners – which is in line with regional integration commitments, but also an important consideration for policy effectiveness (high MFN tariffs could be circumvented via regional partners if rules of origin are not strictly enforced). Papua New Guinea's stance on broader regional trade initiatives has been cautious: notably, it opted out of PACER Plus, as mentioned, due to concerns about losing policy space on tariffs. This reflects a deliberate alignment of tariff policy with what Papua New Guinea perceives as its development needs, even if it means standing apart from some regional liberalization efforts.

Under the European Union–Pacific EPA (and the United Kingdom–Pacific EPA), Papua New Guinea enjoys full duty-free, quota-free access for its exports to the European Union/United Kingdom, while committing to gradually liberalize 80–90 per cent of its own tariff lines over 15–25 years. The protective tariff measures taken in 2018–2020 do not necessarily violate the EPA, as the agreement allows asymmetric and gradual

liberalization with safeguards for food security (WTO, 2019). Many sensitive items can be excluded or liberalized at a slower pace. However, the spirit of the EPA and similar agreements is toward lower tariffs in the long run. Thus, Papua New Guinea will need to ensure that its interim tariff hikes on industrial and consumer goods are consistent with its schedule of reductions under these agreements. For instance, if milk or furniture imports from the European Union were slated for tariff cuts under the EPA schedule, Papua New Guinea might have to reconcile the temporary hike with its eventual phase-down commitments. WTO-consistency and regional commitments, therefore, set important outer boundaries for Papua New Guinea's tariff policy.

In summary, the 2018–2020 tariff adjustments were formulated to advance national policy goals (industrial development, revenue, value addition) and are broadly aligned with those domestic plans. But Papua New Guinea also faces the task of managing these tariffs in harmony with its international obligations, honouring its WTO-bound rates and progressive liberalization under trade agreements, to maintain credibility and avoid penalties. So far, Papua New Guinea's actions (such as suspending the TRP and raising certain duties) have been taken unilaterally in pursuit of national interest, but going forward, a careful alignment with trade agreements will be essential to sustain an open yet development-friendly trade regime.

Papua New Guinea's stance on broader regional trade initiatives has been cautious: notably, it opted out of PACER Plus.



Chapter 4

Assessment of TRP outcomes





The relationship between Papua New Guinea’s tariff policy evolution and broader economic performance is complex and requires a deeper dive to understand the underlying reasons for the outcomes obtained after tariff changes. For example, during the active TRP period, Papua New Guinea’s economy exhibited remarkable growth at an average of 6.3 per cent annually. This growth slowed down during TRP-suspension period at 2.0 per cent (IMF, 2025). Using a combination of quantitative and qualitative methods, this chapters dive into the linkages between tariffs and trade performance, and examines multiple contributing factors, such as global commodity cycles and major investment projects. Insights from the business sector is also incorporated to give a sense about the practical implications of the tariff policy and how it could be better designed in combination with the larger public policy framework.



Box 4.

Analytical approach employed for the analysis

To evaluate the outcomes and impacts of the TRP, the study relies on two complementary methodologies: an *ex-post* econometric analysis of trade flows, which assesses the results of the implementation of the TRP on specific economic variables, and an *ex-ante* partial equilibrium simulation, which assesses the expected changes to occur as a result of tariff changes. All analysis is based on data obtained from international data sets, namely from UN Comtrade, ITC TradeMap, ITC MacMap, and the World Bank WITS. This data was also compared to national data obtained from the Papua New Guinea Customs Service.

Partial equilibrium simulation (*ex-ante* projections): A partial equilibrium (PE) trade model was employed to simulate the impact of tariff changes on trade, welfare, and revenue. Two periods were examined: (1) TRP Implementation (2010–2018) – comparing outcomes with the tariff reductions against a counterfactual of no tariff change against the baseline (2010); and (2) TRP Suspension (2018–2023), comparing outcomes with the tariff increases against a counterfactual of retaining 2017 tariff rates, which were the lower tariffs. The model uses detailed import data by product and source country, along with import demand elasticities, to estimate how a tariff cut or hike would change import quantities, government tariff revenue, and welfare (measured by consumer surplus). For instance, for normal goods that exhibit a price elasticity, lowering a tariff should increase demand (and import volumes) but reduce per-unit tariff revenue. Raising a tariff would do the opposite, all else being equal. The results from the PE model quantify the effects on a product-by-product basis. Notably, the model captures the largest trading partners (like China and Australia) to see how their exports to Papua New Guinea were affected by tariff changes, and it computes the net welfare impact (consumer surplus change) for Papua New Guinea’s consumers from each policy change.

Gravity modelling (*ex-post* analysis): A standard bilateral gravity model of trade was employed to quantify how changes in tariffs affected Papua New Guinea’s import volumes. This model relates Papua New Guinea’s observed import values to various factors: (i) the tariff level on each product, (ii) the economic size of trading partners, and (iii) trade frictions. In formal terms, for each HS-6 product and partner country, the specification regresses the log of import value on the applied tariff (trade-weighted average) and controls. The regression includes partner GDP (to capture market size), distance between

Papua New Guinea and the partner (a proxy for transport cost), a dummy for whether the partner has a free trade agreement (FTA) with Papua New Guinea (e.g., Australia under PACER Plus, or other regional agreements), and a measure of the partner's regulatory quality (to proxy overall trade facilitation). Year fixed effects control for global shocks, and product-group fixed effects capture commodity-specific factors. This approach isolates the partial effect of tariff changes on imports, holding other factors constant. The gravity model estimation provides an elasticity of import demand with respect to tariff changes and indicates whether Papua New Guinea's recent tariff hikes or cuts significantly altered import behaviour.

A. Tariff outcomes

A tariff comparison between 2010 and 2023 indicates a shift in Papua New Guinea's trade protection policies over time. In 2010, after the first phase of the TRP, ad valorem equivalent (AVE) tariffs were relatively high and more dispersed across various product groups, showing peaks in sectors associated with agriculture, textiles, and manufacturing. During the TRP and up to mid-2018, tariff decreases were witnessed for several products, such as sugar and confectionery products, furniture, while witnessing a jump in tariffs for beverages and tobacco products (Figure 7). Several other tariffs have fallen during the TRP, though few tariff eliminations have been witnessed. The 2010 tariffs had a wider spread, indicating high protection levels for certain goods, particularly in sectors where local industries required shielding from international competition.

In contrast, the 2023 AVE tariffs show a more selective protectionist approach, with some tariffs significantly increased while others remain low. The tariff peaks in 2023 suggest renewed protection for specific industries, reflecting Papua New Guinea's post-2018 shift toward import substitution and infant industry protection (Figure 8). The higher tariffs in key sectors indicate targeted intervention rather than across-the-board tariff reductions. This change aligns with Papua New Guinea's suspension of TRP in 2018 and the subsequent introduction of higher tariffs on select products in the 2018–2020 budgets. The overall trend suggests that Papua New Guinea, after years of liberalization,

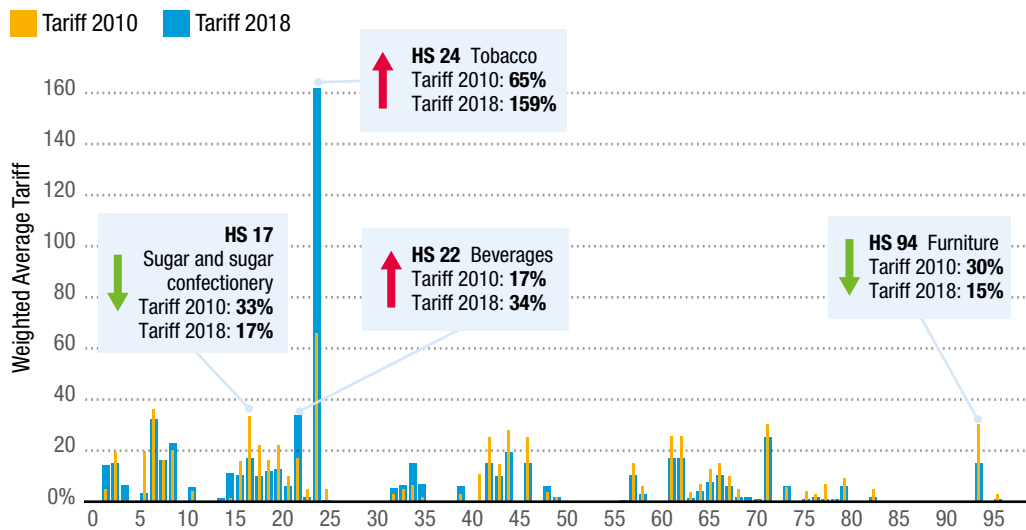
has partially reversed its approach, likely to support domestic industries and reduce dependence on imports, but with potential trade-offs such as higher consumer prices and limited international competitiveness in protected sectors.

Despite the rationalization of tariffs between 2010 and 2023, Papua New Guinea's protective measures continued to disproportionately affect more raw materials and continued to be applied to intermediate products. Grouping up the tariffs according to the product types, whether raw material, intermediary inputs to industry, consumer goods or capital goods, allows one to reflect on the levels of protection applied to strategic inputs into industry or to import competing industries. During the TRP (2012–2017), all tariff categories decreased. Raw materials dropped from 11.4 to 10.8 per cent, consumer goods from 6.4 to 5.2 per cent, intermediate goods from 2 to 1.5 per cent, and capital goods from 0.3 to 0.1 per cent. After the suspension of the TRP (2018–2023), most tariffs bounced back up, but only in a moderate fashion. Tariff protection on raw materials rose to 11.3 per cent, from 10.8 per cent in 2017, while tariffs on consumer goods rose from 4.9 (at their lowest point in 2018) to 5.3 per cent in 2023. Intermediate goods stayed steady at between 1.5–1.7 per cent on average, though the changes are still surprising from year to year. Finally, tariffs on capital goods remained very low, ranging between an average of 0.1 and 0.2 per cent (Figure 9).



Figure 7.
MFN AVE tariffs have changed since 2010

MFN AVE tariffs in Papua New Guinea, 2010 and 2018 (percentage)

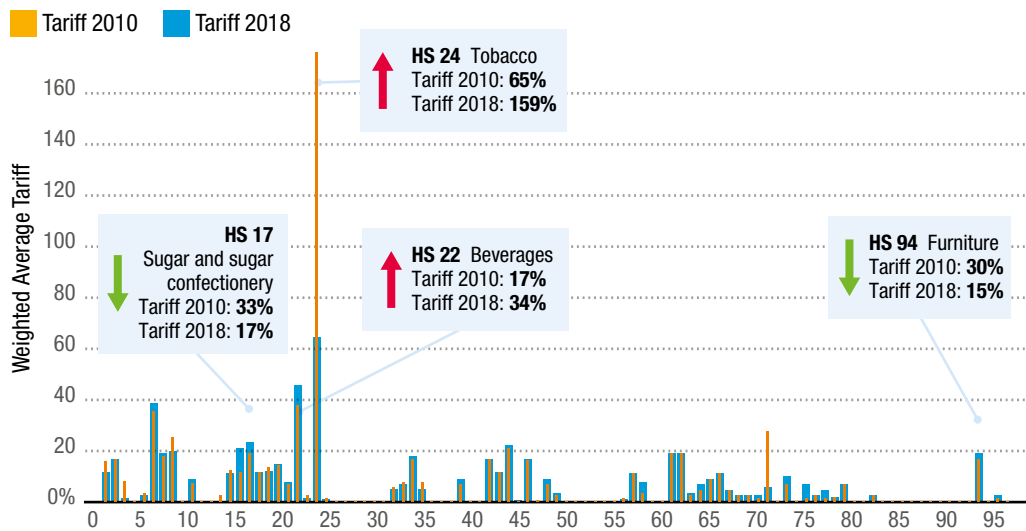


Source: UNCTAD based on ITC Macmap.



Figure 8.
MFN AVE tariffs in Papua New Guinea changed over the 2018–2023 period

MFN AVE tariffs in Papua New Guinea, 2018 and 2023 (percentage)

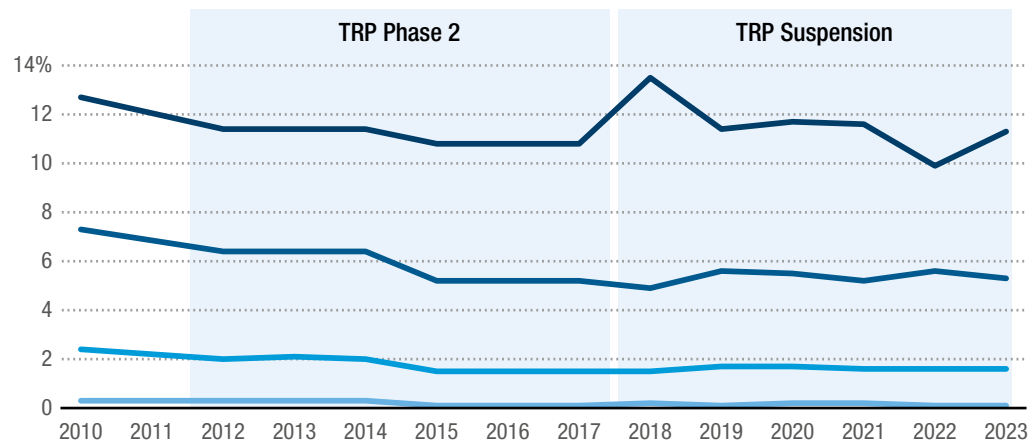


Source: UNCTAD based on ITC Macmap.

**Figure 9.****Simple average AVE tariffs of Papua New Guinea**

Simple average AVE tariffs by product category, 2010–2023 (percentage)

— Capital goods — Consumer goods — Intermediate goods — Raw materials



Source: Authors based on UNCTAD TRAINS.

The distribution of tariff levels across categories of products affects raw materials, consumer products and intermediary products most.

From the baseline in 2010 to the last year of the TRP's implementation in 2017, the distribution of tariffs has been rationalized, and average tariffs dropped. The fewer differences in rates of tariffs (indicated by blue dots in Figure 11) across the scale suggest a simplification of the tariff schedule. This is especially true for consumer products. When it comes to the TRP suspension period, it starts off in 2018 with a major increase in complexity and variability of rates across all types of products. The increase in intermediary products and raw materials is surprising from the perspective of simplifying and facilitating businesses engaged in trade. By 2023, the complex structure observed in 2018 had reversed, resulting once again in a more simplified and rationalized tariff schedule.

The distribution of tariff levels across Harmonized system (HS) codes has also varied and become less scattered over time during and after the suspension of the TRP.

In particular, in 2010, there was a high dispersion of tariffs along HS2 clusters, with many products facing a 20 to 40 per cent tariff rate. In some cases, there

were tariffs applied at a 60 per cent tariff in ad valorem terms. By 2018, the dispersion of tariffs along HS clusters fell, with the distance between tariff rates compressing. Most products also faced a far lower tariff rate of below 20 per cent. This reduction in dispersion is clearly visible in HS chapters such as HS17 and HS94, which show a much narrower range of tariff rates in 2018 compared to 2010. In particular, tariffs under HS94 exceeded 40 per cent in 2010 but fell below that level in subsequent years. By 2023, during the suspension of the TRP, some tariffs rose, but the pattern of dispersion along HS clusters was remarkably similar to that witnessed in 2018 (Figure 12).

The survey conducted for this study asked business respondents whether the nature of tariff revisions between 2018 and 2020 had any impact on their investment decisions. The results provide a clear indication that tariff uncertainty significantly influenced strategic planning for a majority of businesses. Out of the total respondents, 61 per cent reported that ad-hoc tariff changes directly affected their investment decisions. This indicates that most firms felt compelled to either delay, modify, or reconsider their investment plans in response to an unstable trade policy environment. Such findings highlight the disruptive role of

policy unpredictability in long-term financial commitments. Meanwhile, a quarter of firms stated that tariff revisions did not impact their investment choices. This suggests that either their operations were less exposed to tariff-sensitive sectors or they

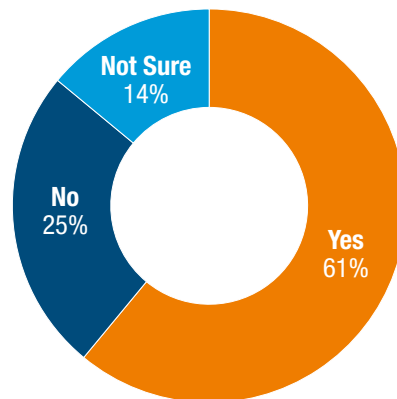
had sufficient flexibility to absorb potential risks without altering capital allocation. A smaller share, 14 per cent were unable to determine if the tariff uncertainty affected their business investments (Figure 10).



Figure 10.

Impact of tariff policy on investment decisions

Has the ad-hoc nature of tariff revisions (2018–2020) affected your investment decisions?



Source: Survey Results; n=28

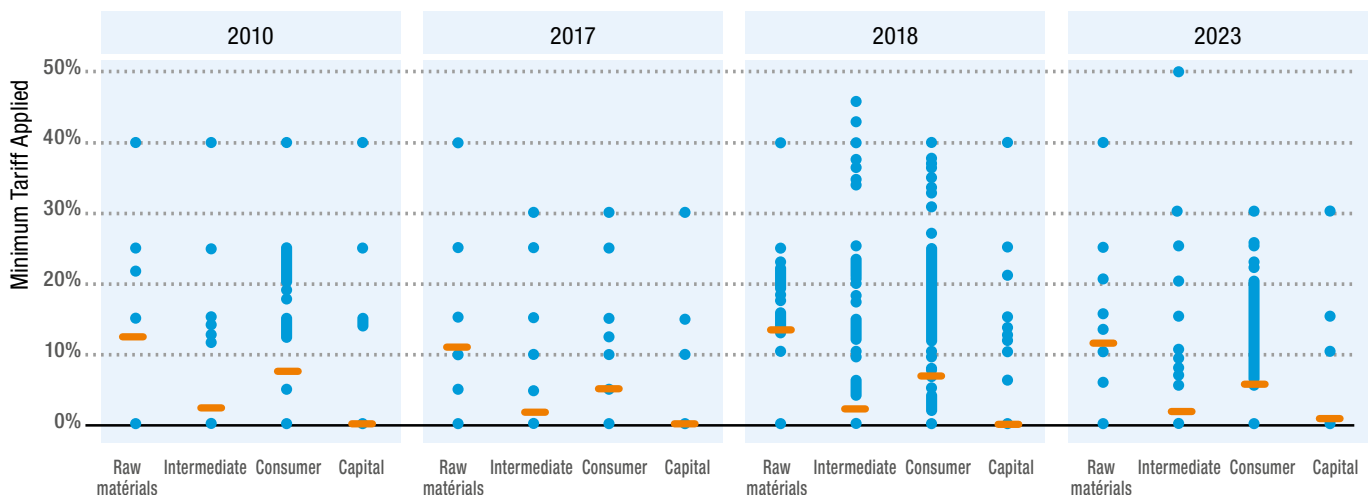


Figure 11.

Tariff levels by product category changed during and after the TRP

Minimum tariff applied by product category, 2010, 2017, 2018 and 2023 (percentage)

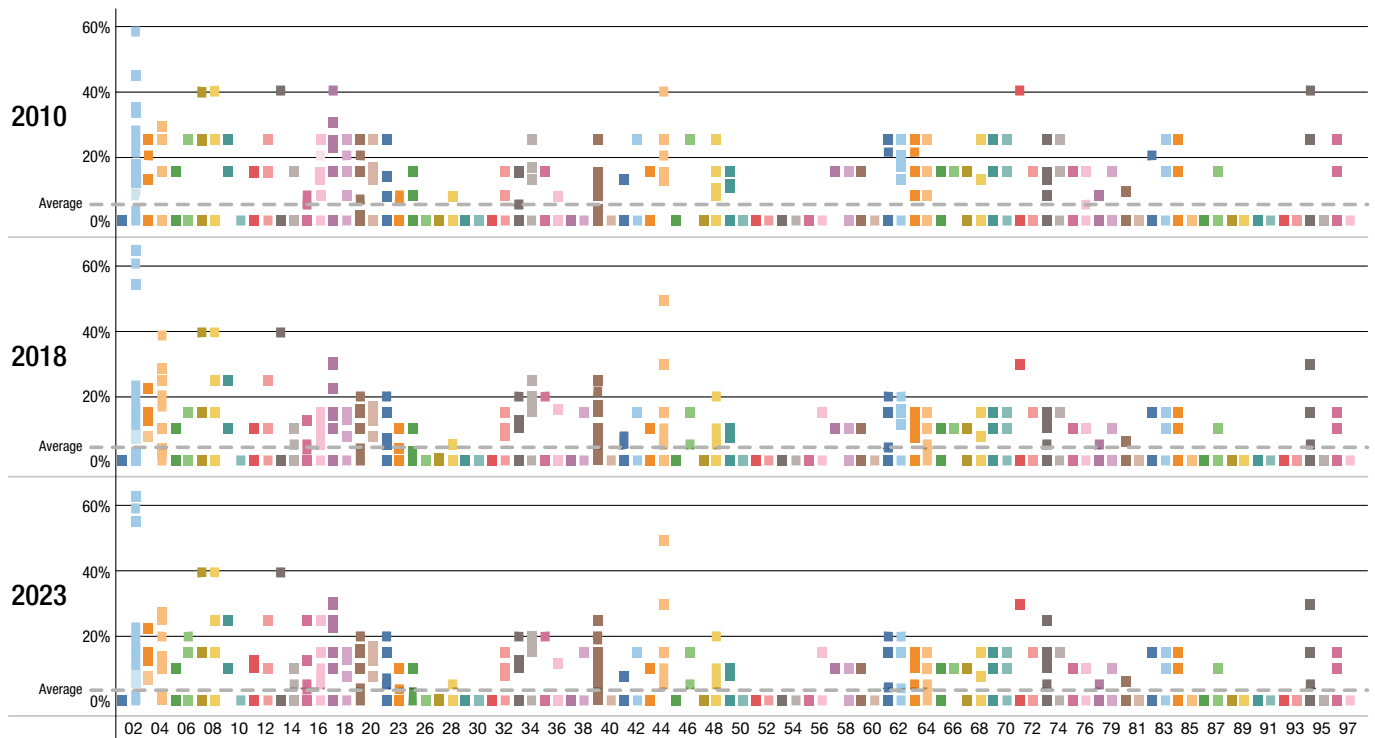
— Simple average tariff of product category • Products (HS6)



Source: Authors based on UNCTAD TRAINS.



Figure 12.
Tariff dispersions across HS 2 sectors
 Tariff applied, 2010, 2018 and 2023 (percentage)



Note: Each square represents an HS6 product within its corresponding HS2 sector. Sectors **HS 22 (Beverages)** and **HS 24 (Tobacco)** have been **excluded** from this visualization due to their extreme tariff values, which would distort the scale.

Source: Authors based on UNCTAD TRAINS.

B. Economic outcomes and sector performance

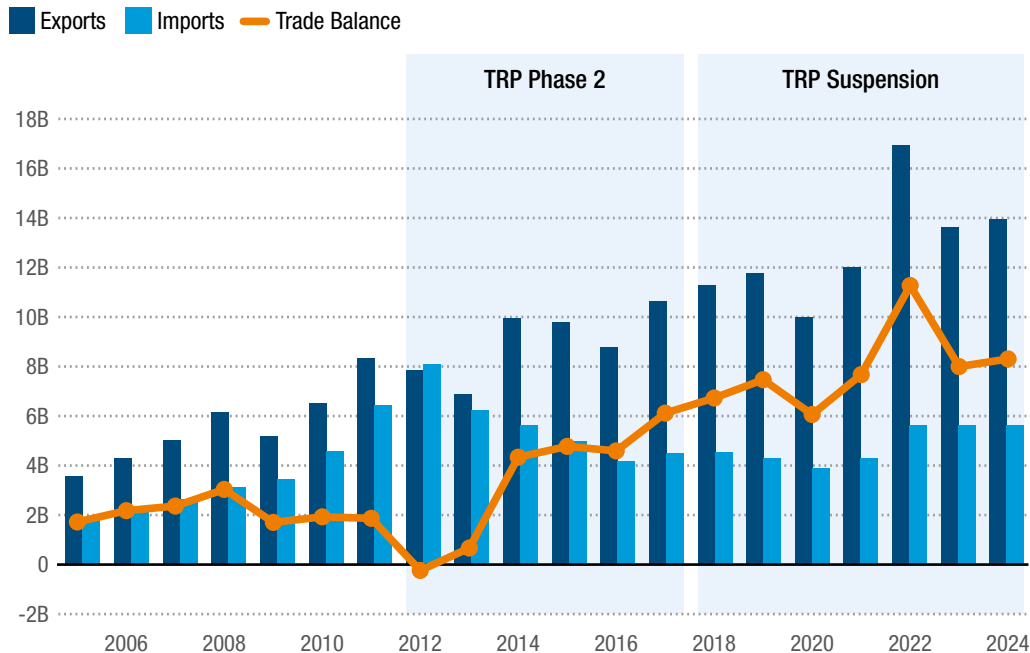
Overall observations

Trade trends in Papua New Guinea suggest a robust growth in exports and a decline in imports between 2010 to 2024. Imports declined significantly, from a high of \$8 billion in 2012, despite the gradual tariff reductions. At the same time, exports increased steadily, driven by strong performance in resource sectors like mining, oil, and gas. The trade balance for Papua New Guinea has overall remained positive over this period, with a slight dip in 2012, but improved sharply

afterwards as exports surged (Figure 13). The factors behind the improved performance in the trade balance reflect less the tariff policy and more other external factors, including the thriving mining sector. Following the suspension of TRP in 2018, import levels remained subdued, likely due to the introduction of higher protective tariffs and import substitution policies. Meanwhile, exports peaked in 2022, benefiting from high global commodity prices and the expansion of exports from extractive industries.

**Figure 13.****Exports in Papua New Guinea have expanded**

Merchandise trade in Papua New Guinea, 2005–2024 (US\$ billions)



Note: Due to the unavailability of trade data for certain years, mirror data has been used.

Source: UNCTAD based on ITC Trademap.

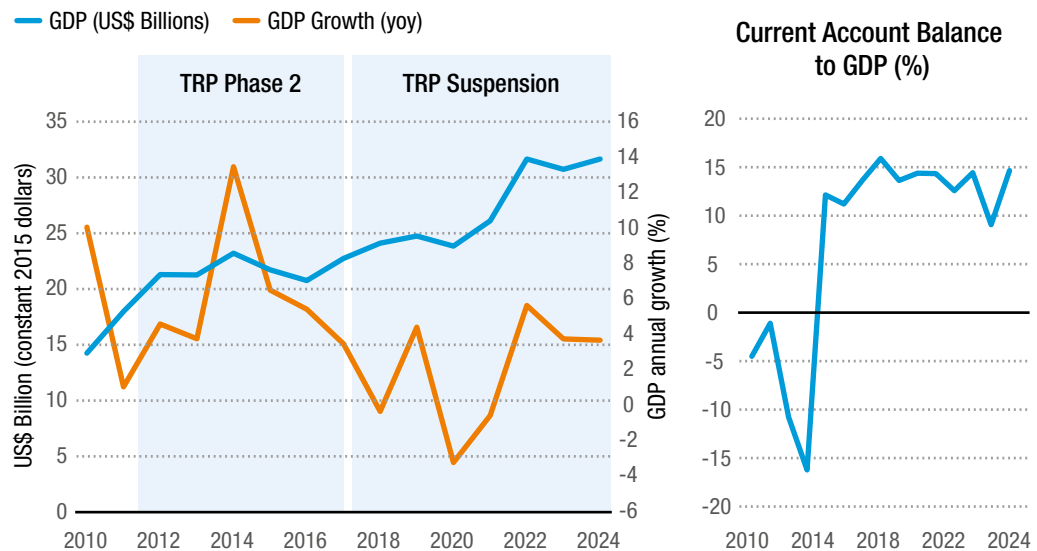
A number of factors explain why Papua New Guinea's imports fell during 2012–2017 despite tariff cuts.

One likely factor was broader economic conditions: Papua New Guinea was completing a large LNG investment project around 2014 and then faced a downturn in commodity prices, which constrained foreign exchange and import capacity. Thus, the macro environment dominated the tariff effect, masking any import-stimulating impact that lower tariffs would typically have. Additionally, Papua New Guinea maintained zero tariffs on a large share of essentials even before TRP, therefore cutting tariffs mainly affected other goods whose demand might be price-elastic but subject to income constraints.

During the active TRP period, Papua New Guinea's economy exhibited remarkable growth at an average 6.3 per cent annually. This performance stands in stark contrast to the post-

suspension period, which saw average growth rates decline to 2.0 per cent (Figure 14). While correlation does not equal causation, the timing and magnitude of economic changes during different policy phases provide important insights into the role of trade openness in small island economies. Furthermore, multiple factors contributed to the differential trends, including global commodity cycles and major resource project developments (for the TRP Phase under investigation), as well as the COVID-19 outbreak (for the TRP Suspension), necessitating careful interpretation of the correlation with trade policy phases. Papua New Guinea's external position, measured through the current account balance, has remained consistently strong. The balance has been positive since 2014 and has exceeded 10 per cent of GDP for the entire decade for which data are available.

Figure 14.
Growth has slowed after the TRP suspension
 GDP in Papua New Guinea, 2010–2024 (US\$ billion)

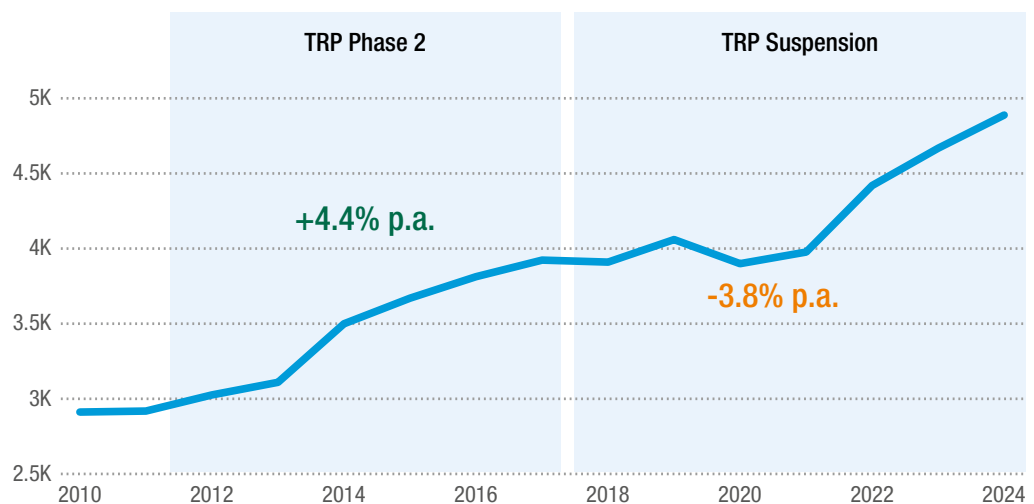


Source: World Bank WDI (2025).

Purchasing power showed a similar growth pattern over the same period. GDP per capita at Purchasing Power Parity (PPP) increased by 4.4 per cent annually between 2010 and 2017 and moderated to 3.8 per cent between 2018 and 2024 (Figure 15). This sustained improvement in

purchasing power often creates demand for both imported and domestically produced goods, thereby supporting increased economic activity across multiple sectors. Nevertheless, its purchasing power levels remained low compared to other island economies like Fiji or Sri Lanka (IMF, 2025).

Figure 15.
Overall, growth in Papua New Guinea has been strong
 Papua New Guinea, GDP in PPP, 2010–2024 (US\$ thousand)



Source: World Bank WDI (2025).

Impacts by product category

The gravity model analysis shows that tariffs do have a negative relationship with import volumes, but the elasticity is modest. Specifically, a 1 per cent increase in tariffs (ad valorem) is associated with a 2.1 per cent decrease in import value on average, holding other factors constant (see details in Annex 3). This implies that Papua New Guinea's import demand is somewhat price-sensitive, but this effect can be overshadowed by other forces. The results from the gravity model further suggest that tariff increases have a more significant effect in dampening the demand for importing raw materials and consumables. On the other hand, tariffs had little measured impact on intermediate goods imports, possibly because industries had no close substitutes for those inputs. For capital goods, the model found a positive correlation with tariffs, which is not causal but rather reflects that many capital goods imports in Papua New Guinea enjoy duty exemptions (for priority infrastructure or investment projects). This finding aligns with the practice of exempting crucial imports; for example, mining equipment or electricity generators might be exempted, meaning tariff hikes do not reduce those imports.

Import composition showed a strong shift towards consumer goods during the liberalization period. From 2012 to 2017, the share of consumer goods in total imports increased from 28 per cent to 41 per cent during the TRP, eventually reaching 46 per cent by 2023 (Figure 16). This structural change reflects both the direct effects of tariff reductions on consumer goods and the indirect effects of improved purchasing power driving consumption growth. The rising share of consumer goods imports could reflect increased demand for foreign consumer products as tariffs

fell and Papua New Guinea's purchasing power grew modestly. It might also indicate that local manufacturing did not expand to meet consumer needs, so imports filled the gap over time. Intermediate goods showed a gradual expansion from an average 12.5 per cent share during the TRP to 16 per cent following the suspension of the TRP, indicating continued dependence on imported inputs for domestic production processes. On the other hand, raw material imports experienced the opposite trend, with their share in total imports halving from 13 per cent to 7 per cent during the TRP and remaining at these low levels after suspending the TRP (Figure 16). The decline in raw material imports could be due to the expansion of Papua New Guinea's agriculture and resource sectors to supply more raw inputs locally, or due to the stagnating manufacturing (which uses raw materials) which required fewer imported inputs.

In value terms, despite liberalization measures, all import categories actually contracted during the TRP (2012-2017).

Raw materials declined at a rapid annual rate of 19.4 per cent, and capital goods fell 15.2 per cent annually. The post-suspension period from 2018 to 2023 witnessed recovery in three of four major import categories, with consumer goods leading strongly at an 8.1 per cent compound annual growth rate. Raw materials continued their decline at 4.1 per cent annually, possibly correlating with the increase in tariffs following the TRP suspension. On the other hand, both intermediate goods and capital goods returned to modest positive growth (Figure 17). These patterns may suggest that the earlier contraction reflected temporary factors such as major project completions and exchange rate adjustments rather than fundamental structural changes driven by tariff policy.

**Figure 16.****Consumer goods as a share of imports have grown steadily**

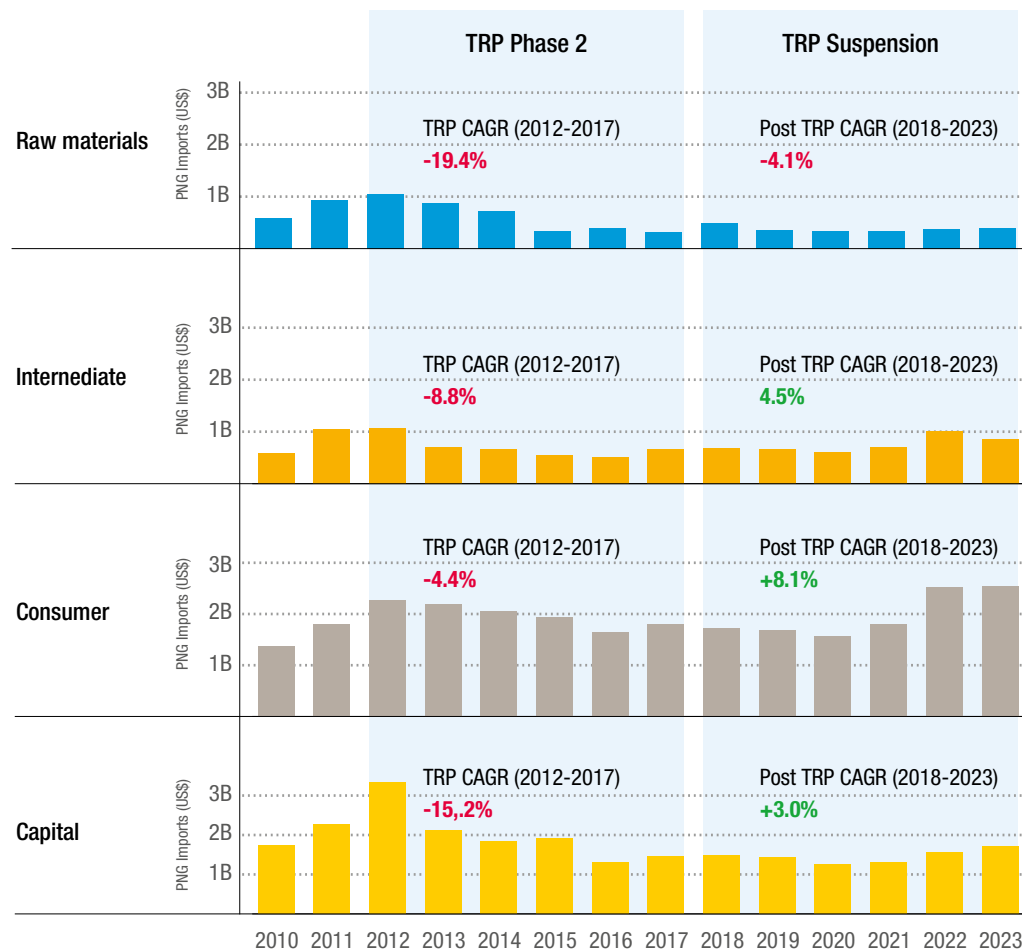
Structure of Papua New Guinea's imports during and after TRP phase 2, by type of goods (share, percentage)

	2010	2011	TRP Phase 2						TRP Suspension					
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Raw materials	13	15	13	14	13	7	10	7	11	8	9	8	7	7
Intermediate	13	16	13	12	12	11	12	15	15	16	15	17	18	15
Consumer	30	28	28	36	37	39	40	41	38	39	40	42	45	46
Capital	39	36	42	34	34	39	32	33	33	33	32	31	28	30

Source: UNCTAD based on UN Comtrade.

**Figure 17.****Intermediate, consumer and capital goods returned to modest positive growth**

Import value during and after TRP phase 2, by product category, 2010–2023 (US\$ billion)



Source: UNCTAD based on UN Comtrade.

Impact by sector

Examination of agriculture, mining, and manufacturing performance, using the gross value added (GVA) index over the past two decades, reveals that the tariff policy alone did not generate much change to the performance of major sectors of the economy. Agriculture exhibited steady, gradual growth throughout the entire period. Even when tariffs fell sharply under the TRP, Papua New Guinea's agriculture, forestry, and fishing sector maintained output expansion, though at a slow rate (Figure 18). This indicated that exposure to global competition did not cripple farmers; on the contrary, agricultural growth was "steady" and resilient. Mining and quarrying showed a different pattern: an initial dip around 2010–2012, followed by a significant surge from 2014 to 2019.

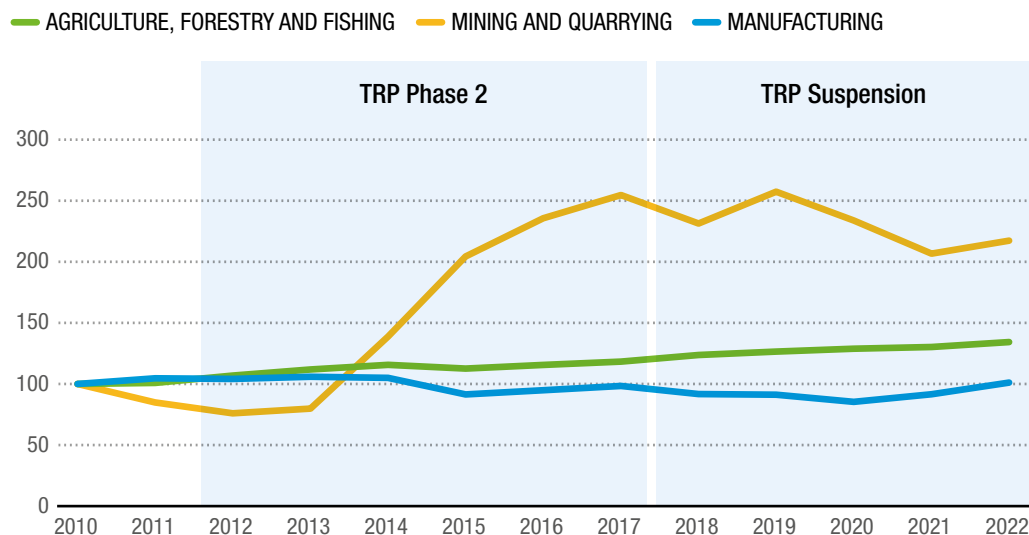
This boom was more likely to correspond to new resource projects (like the LNG project commencing in 2014) and high commodity prices rather than any tariff policy effect, as Papua New Guinea mining is largely an export sector with little reliance on import protection. Manufacturing, covering sectors such as agro-processing and light industry, clearly underperformed, with GVA revealing to be almost flat or declining throughout both the TRP and TRP suspension phases, making it the slowest-growing sector. Notably, even after 2017, when tariffs were raised, the manufacturing index was constantly below the 2010 level and only inched upward modestly by 2022, implying that protection gave limited relief. Furthermore, the stagnation reflects Papua New Guinea's ongoing struggle to develop its manufacturing base.

Even when tariffs fell sharply under the TRP, Papua New Guinea's agriculture, forestry, and fishing sector maintained output expansion.

Figure 18.

Manufacturing has stagnated

Index of gross value added by industry, 2010–2022 (2010 = 100)



Source: NSO, Papua New Guinea National Accounts Data (2023).

The above evidence suggests that tariff liberalization did not affect agriculture, as the sector continued to grow – possibly benefiting from cheaper imported farm inputs like machinery or fertilizer, and facing little direct competition since Papua New Guinea's agriculture is oriented around crops which were not heavily imported, such as coffee, cocoa, and palm oil. Manufacturing,

however, saw virtually no growth when tariffs were low, which may have reinforced the perceptions that it needed protection. However, it is important to note that Papua New Guinea's manufacturing faced small-scale and high-cost (energy, transport, and logistics, see Figure 27), which, even with tariffs, made it hard for local manufacturers to compete with lower-priced imports.

The sector's struggles during liberalization suggest that tariff policy alone cannot overcome fundamental structural constraints that limit manufacturing development in Papua New Guinea's economic context.

Impact by partner

The results from the partial equilibrium model estimation show limited effects of tariffs on Papua New Guinea's trade with partners. China and Australia are the most affected markets, due to the size of trade with these markets, although the change in tariffs (both during the TRP and after the TRP suspension) only caused a single-digit

million-dollar change in imports for both partners (Table 3). Between 2010-2018, the reduction in tariffs led to a simulated \$7 million more imports from China and \$2.7 million more from Australia. Between 2018-2023, the tariff rise led to a simulated \$6.8 million less imports from China. In contrast, Australia benefited from a tariff reduction during the Papua New Guinea TRP suspension, which is simulated to have resulted in an increase of \$4.5 million in imports from Australia. These small changes can be explained by the number and volume of imports affected by tariff changes during and after the TRP for Papua New Guinea's major trade partners (Table 4 and Table 5).



Table 3
Trade effects by partner

Scenarios	Export markets	Total trade effects (US dollars million)
Scenario 2010 - 2018	China	6.17
	Australia	2.70
	Thailand	2.40
	New Zealand	1.52
	Fiji	0.85
	United States of America	0.83
	Vanuatu	0.44
	United Kingdom	0.35
	Denmark	0.30
	France	0.28
	Others	-8.40
	China	-6.76
Scenario 2018 - 2023	Thailand	-2.44
	Philippines	-0.81
	Türkiye	-0.27
	Italy	-0.23
	United States of America	-0.12
	Poland	-0.12
	Rep. of Korea	-0.10
	Japan	-0.10
	Myanmar	-0.05
	Others	17.86

Source: Authors TradeSim based on data from ITC Macmap and UN Comtrade.

For example, for China, the TRP introduced changes to 370 product lines (at the HS-6-digit level), which affected \$97 million during the TRP, out of a total 1,614 product lines imported from China. The

TRP suspension introduced changes to even fewer products from China: just 65 product lines were impacted by the tariff change, out of 1,937 imported product lines, affecting total imports of \$59 million only.

Table 4.
Number of products affected by tariff change during and post TRP for China

	Scenario 2010 – 2018	Scenario 2018 – 2023
Number of products imported in baseline year	1,614 (total imports : \$352 million)	1,937 (total imports : \$719 million)
Number of products having the same tariff in the baseline year and the final year	1,244 (total imports : \$255 million)	1,872 (total imports : \$660 million)
Number of products having different tariffs in the baseline year and the final year	370 (total imports : \$97 million)	65 (total imports : \$59 million)

Source: UNCTAD based on ITC Macmap and UN Comtrade.

Table 5.
Number of products affected by tariff change during and post TRP for Australia

	Scenario 2010 – 2018	Scenario 2018 – 2023
Number of products imported in baseline year	1,985 (total imports : \$1.55 billion)	2,235 (total imports : \$1.30 billion)
Number of products having the same tariff in the baseline year and the final year	1,608 (total imports : \$1.44 billion)	2,181 (total imports : \$1.27 billion)
Number of products having different tariffs in the baseline year and the final year	377 (total imports : \$110 billion)	54 (total imports : \$30 billion)

Source: UNCTAD based on ITC Macmap and UN Comtrade.

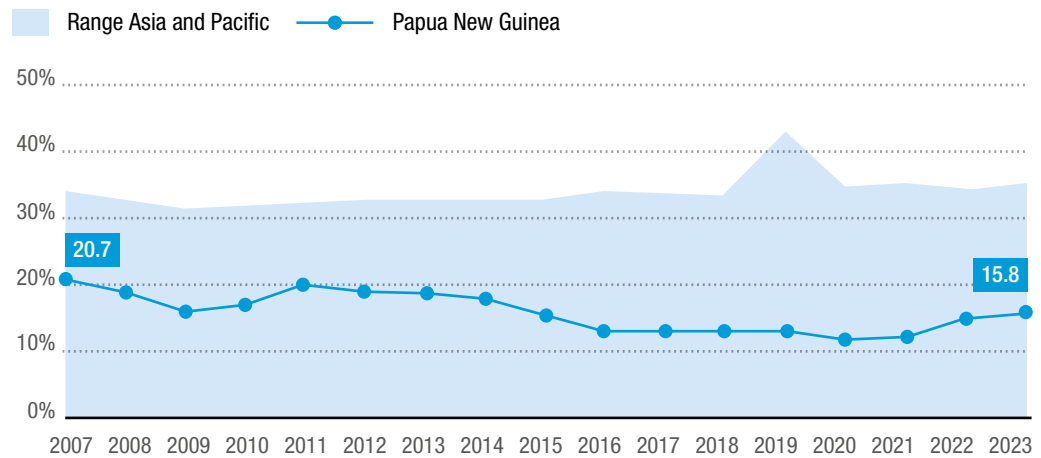
C. Tariff revenue

A critical aspect of tariff policy is its impact on government revenue. The tax-to-GDP ratio in Papua New Guinea has fallen between 2010 and 2023, reaching 15.8 per cent in 2023 (Figure 19). During the same period, tax revenue has increased from around 6.6 billion PGK in 2010 to 17.6 billion PGK in 2023. This apparent divergence reflects the faster expansion of nominal GDP relative to tax collections. Import duties have long been an important source of fiscal revenue for Papua New

Guinea, though their share in total tax collected has declined with the introduction of GST and tax reforms. Nevertheless, according to the OECD (2025), total tax collected by customs and import duties rose from 189 million PGK in 2010 to 403 million PGK in 2023 (Figure 20). While fiscal revenues increased in absolute terms, there was an observed marginal decline in relative terms during the TRP and an increase during the suspension, followed by a decline in 2023 to its lowest level.

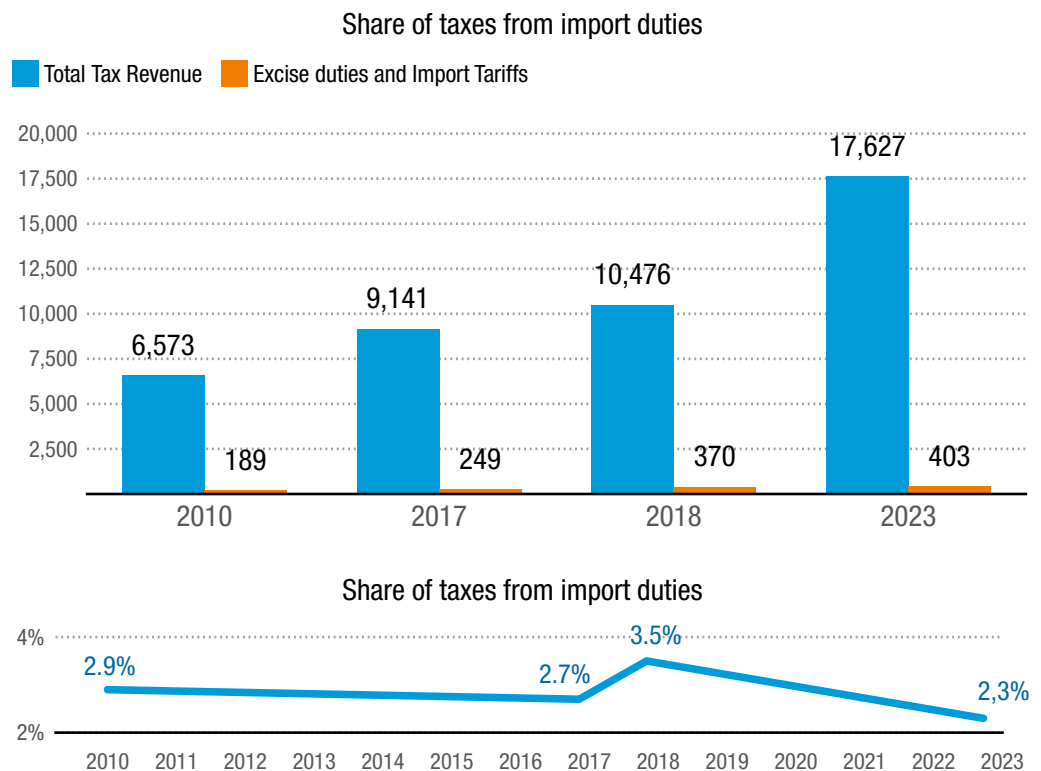
The tax-to-GDP ratio in Papua New Guinea has fallen between 2010 and 2023, reaching 15.8 per cent in 2023.

Figure 19.
Tax-to-GDP ratio has fallen in Papua New Guinea
 Fiscal revenue as a share of GDP (percentage), 2007-2023



Source: OECD (2025).

Figure 20.
Total tax revenue has increased in Papua New Guinea
 Excise duties and import tariff revenues, PGK million, 2010-2023



Source: OECD (2025).

The analysis of tariff revenue by product category revealed that consumer goods produced the bulk of Papua New Guinea's tariff revenue.

The fact that consumer products generate most revenue (Figure 21) is not surprising since consumer goods (e.g. fuel, vehicles, alcohol, tobacco, processed foods) have both high tariff rates and substantial import volumes. Intermediate goods and raw materials often have lower tariffs and/or low import quantities, contributing relatively little to fiscal revenue collection. The data showed that even though raw materials had high nominal tariffs in some cases, they generate minimal government revenue because import volumes are so low. In contrast, consumer goods – even with moderate tariffs – generated a lot of revenue due to the sheer volume of imports. For example, fuels (HS27) have AVE tariff rates of between 0 per cent to 0.5 per cent, yet the level of imports was \$751 million in 2018, generating large absolute values of revenue.

This suggests that there is little policy space for attempting to increase revenue by raising tariffs on other product categories, such as raw materials (due to the low volumes of imports), and raising tariffs further on consumer goods runs into the risk of lowering volumes of imports or incentivizing under-reporting, under-invoicing, or potentially attempting to evade duties. Any additional attempts to raise revenue may hit diminishing returns. Moreover, Papua New Guinea's Medium Term Revenue Strategy (2023–2027) emphasizes improving tax collection and diversifying revenue sources. The

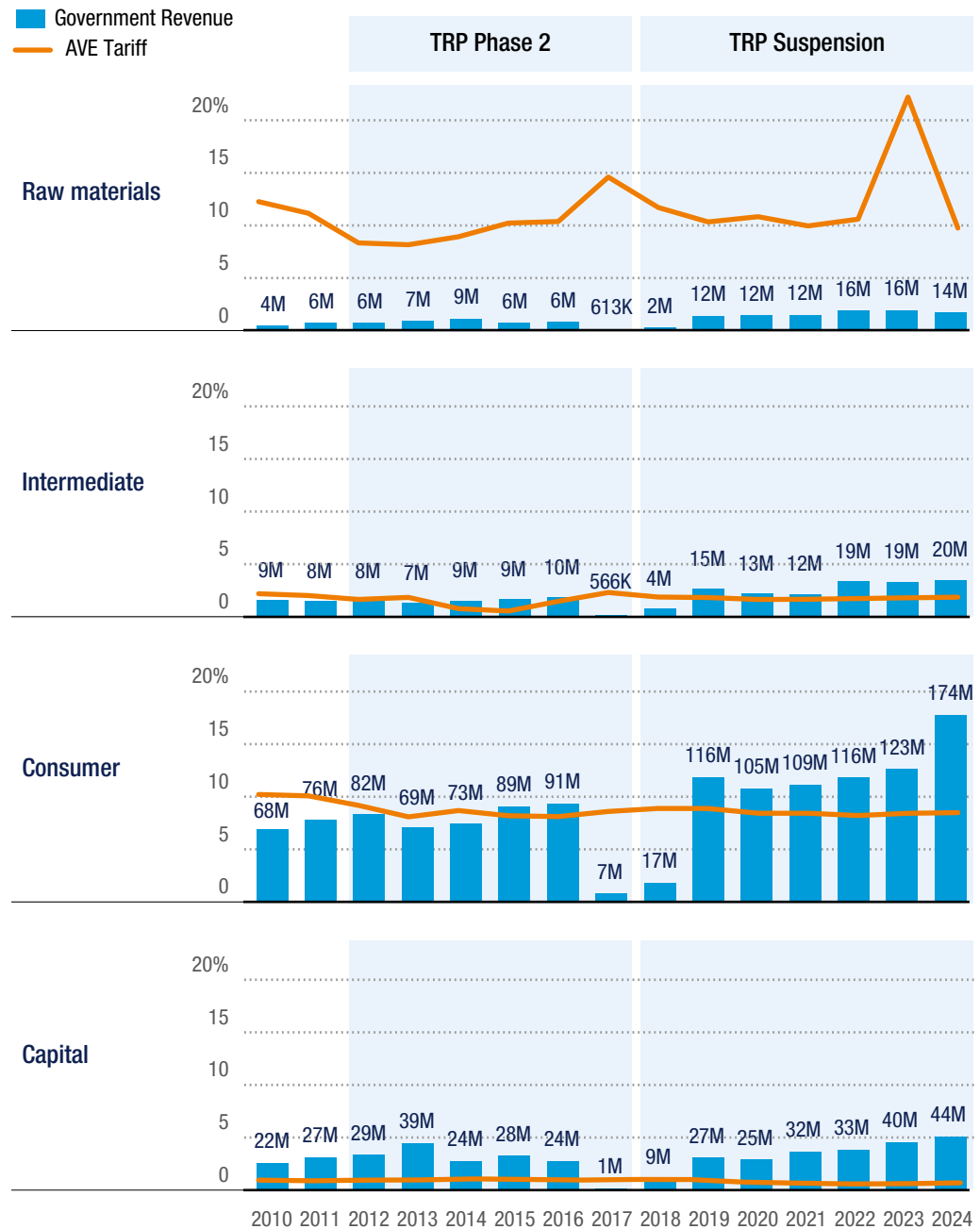
evidence suggests that trade taxes are a limited tool for revenue generation. Imports can contribute to fiscal revenue, but usually through the imposition of GST most directly, while expansion of the economy will lead to higher income tax, corporate tax and GST collection through the expansion of spending.

Lowering tariff rates may reduce tariff revenue collection or increase it, depending on the elasticities of demand.

Despite the evidence, holding all factors constant, it is interesting to simulate the expected changes in revenue from tariff changes. During the TRP, as tariff rates declined, revenue from import duties would be expected to drop proportionally. Using the partial equilibrium model, the modelled results estimate that tariff revenue in 2010–2018 would decline only marginally, holding flows of trade constant. Specifically, the partial equilibrium simulation estimated that the tariff reductions under TRP caused about a \$7.8 million annual revenue loss by 2018, relative to what revenue would have been without those cuts. To put this in context, this would present a shortfall in just over 1 per cent of tariff revenue collected, almost insignificant in terms of the overall fiscal revenue stream. This small impact is because many of Papua New Guinea's imports were either zero-rated already or not strongly affected by slight tariff changes (especially given import contraction in those years). Additionally, the GST in Papua New Guinea applies to imports, so as tariffs fell, GST collected on a possibly higher import base could offset some losses (Figure 22).

**Figure 21.****Tariff revenue increased since the TRP suspension**

Tariff revenue by product category, tariff rate (percentage) and revenue in Papua New Guinea Kina, 2010-2024



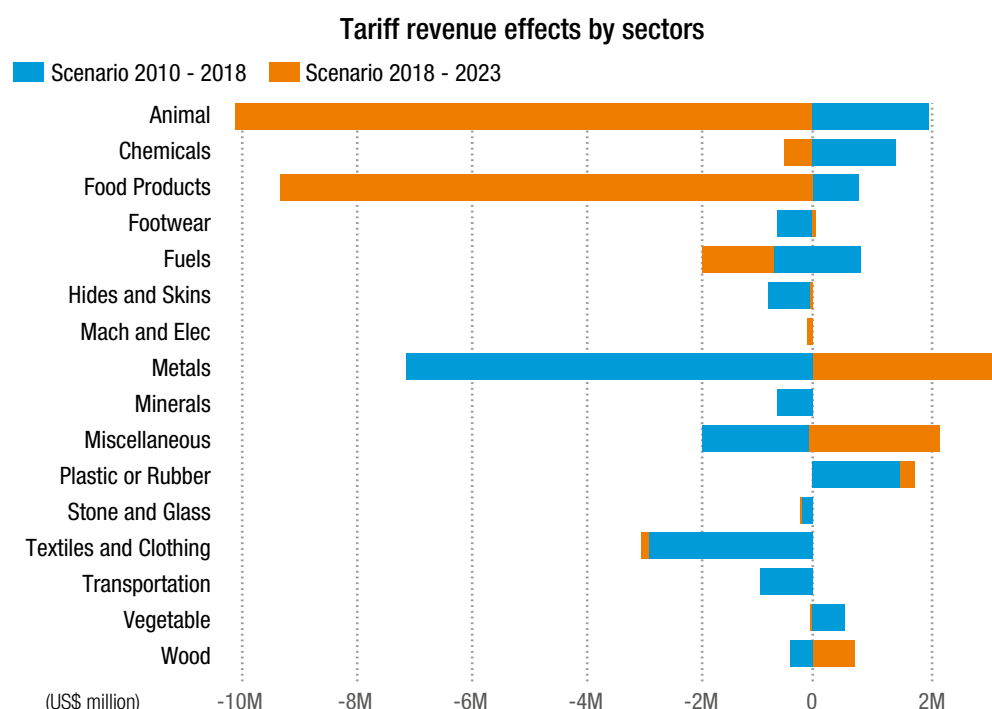
Source: UNCTAD analysis based on Papua New Guinea Customs Services data.



Figure 22.
Results from tariff revenue simulations

Change in tariff revenue during and after the TRP (Millions of dollars), overall and by sector

Scenario (US\$ million)	Current tariff revenue	New tariff revenue	Change in tariff revenue
During TRP (2010 - 2018)	615	607	-8
After TRP suspension (2018 - 2023)	696	680	-16



Source: International Economics TradeSim based on data from ITC Macmap and UN Comtrade.

After 2018, with the TRP suspension, the expectation was that fiscal revenue would increase when tariffs rise.

On a static basis, higher rates on the same import volume should yield more revenue.⁶ However, two dynamic effects played out: volume reductions (imports of some items fell because of higher prices) and policy reversals on specific items. The simulation for 2018–2023 shows a somewhat surprising result: tariff revenues were about \$15.6 million lower with the post-2018 tariff increases than they would have been if Papua New Guinea had maintained its baseline 2018 levels of tariffs. In other words, raising tariffs did not raise revenue. A deep dive into the results reveals the

primary reason for this outcome: Papua New Guinea slashed the tariff rate on tobacco soon after raising other tariffs. Tobacco is a top revenue-generating import (due to high volume and previously very high rates), so when its duty was reduced from 159 per cent to 58 per cent, the government forewent a huge amount of potential revenue on that item. Essentially, Papua New Guinea suspended the TRP “to gain revenue,” but then undermined that goal by cutting taxes on one of its most fiscal generating tariff lines (Figure 22).

Overall, tariff liberalization cost Papua New Guinea little in forgone revenue, while the attempt to reverse it did not yield the expected revenue gains. A

⁶ As the base flows of 2018 are maintained to project the tariff changes and make meaningful comparisons.

reason behind such unexpected results is collection efficiency and behaviour. It is possible that higher tariffs encouraged more evasion or smuggling, meaning the import volumes reported (and taxed) declined disproportionately. For instance, a 150 per cent tariff creates a strong incentive to under-declare, circumvent or smuggle. Papua New Guinea's large informal cross-

border trade (e.g. along the Indonesian border or via sea routes) could become a channel for tariff evasion when rates rise too steeply (Conroy, 2021). This suggests Papua New Guinea's optimal tariff (for revenue purposes) on many goods might be at a relatively low rate, where imports flow without significant distortions.

D. Industry perspectives and cross cutting observations

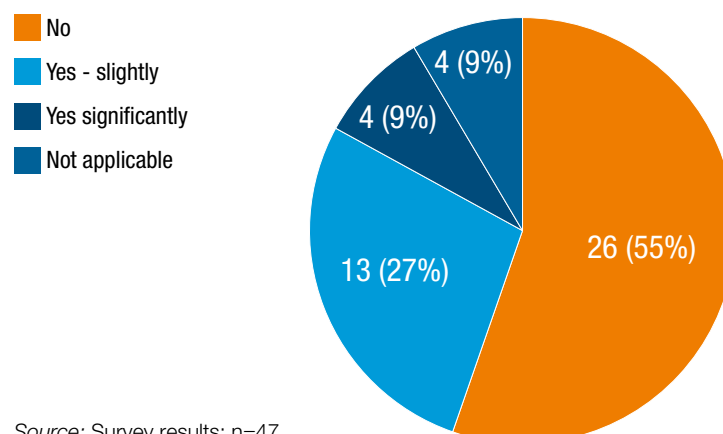
Industry views paint a complex picture of competitive challenges that are informative for trade policy considerations. During the liberalization period from 2012 to 2017, while 55 per cent of surveyed firms reported no benefits, 36 per cent of firms reported some benefits from tariff reductions, primarily through lower costs for imported inputs, capital goods, and equipment. These cost savings enabled some businesses to reduce sales prices,

expand operations, or invest in productivity improvements (Figure 23). Following the tariff increases since 2018, 53 per cent of surveyed businesses reported negative effects, while only 8 per cent reported positive impacts (Figure 24). This suggests that policy volatility creates potential winners and losers within the business community, and negative impacts are more easily felt than benefits, further necessitating scrutiny in designing trade policy measures.



Figure 23.
Perceived impacts of the TRP 2012-2017

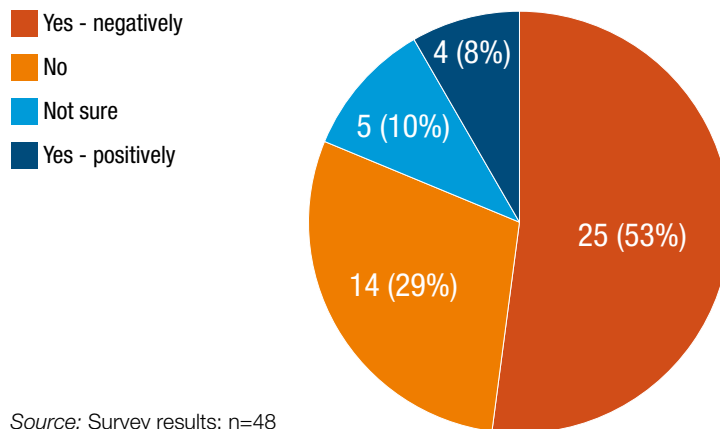
Did your business benefit from tariff reductions in the 2012–2017 period?



Source: Survey results; n=47

Figure 24.
Perceived impacts of the tariff increases post-2018

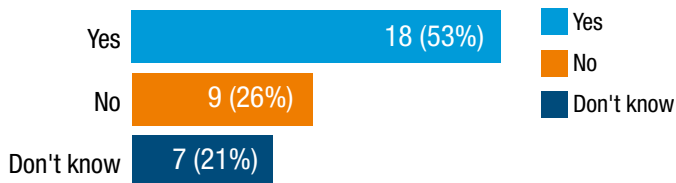
Has your firm been affected by tariff increases post-2018?



Source: Survey results; n=48

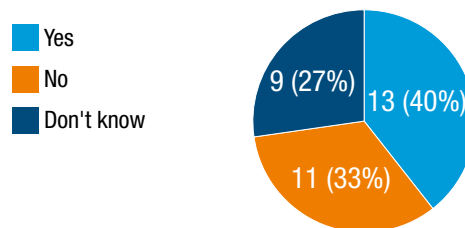
Figure 25.
Perceived level of competition linked to the TRP

Has your company experienced increased competition from imported products?



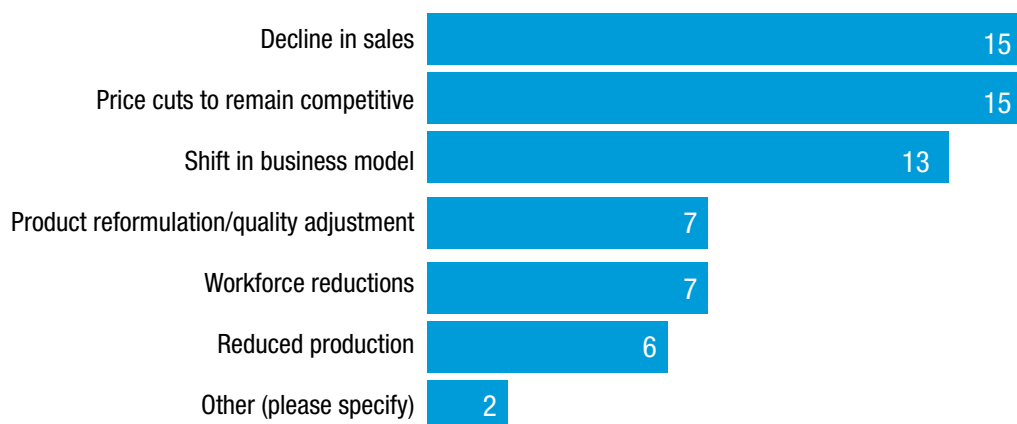
Source: Survey results; n=34

Did tariff reductions lead to perceived unfair competition from imports?



Source: Survey results; n=33

Specific competition impacts from imported products



Source: Survey results; n=21

Perceived competition was increasing, in line with the policy rationale of the TRP to expose domestic firms to international competition to encourage

private sector productivity. Indeed, 53 per cent of surveyed companies reported experiencing increased competition from imported products

Papua New Guinea businesses were concerned that they could not match import prices because their costs (transport, electricity, inputs) were too high.

since TRP implementation, with this competition affecting sales volumes, pricing strategies, and fundamental business models. Interestingly, 40 per cent of firms characterized this import competition as “unfair” (Figure 25). This perception could result from consumers’ price sensitivity: as reported by 55 per cent of businesses surveyed, customers would prefer cheaper imported goods over higher-priced local products. This consumer preference pattern suggests that successful import substitution requires achieving competitive domestic pricing rather than relying solely on tariff protection to create market space for local producers.

Price was considered a key issue: Papua New Guinea businesses were

concerned that they could not match import prices because their costs (transport, electricity, inputs) were too high. Next to price consideration, quality perception emerged as the second most important factor affecting competitiveness, per Papua New Guinea businesses’ perception (Figure 26). Even when local products are available at competitive prices, many consumers perceive imported goods as superior in quality, creating additional barriers for domestic producers. This perception challenge indicates that successful industrial development requires investments in quality improvement, branding, and consumer education that complement trade policy measures.



Figure 26.
Factors affecting competitiveness

Rank the following factors in order of their impact on your competitiveness



Source: Survey Results; n=27

Several structural challenges were shown as major hurdles for domestic production capacity and competitiveness. According to the survey results, transport and logistics costs emerged as overwhelming challenges for business operations, with 94 per cent of survey respondents rating these factors as high or very high barriers to competitiveness. This near-universal concern reflects Papua New Guinea’s fundamental geographic challenges, including its archipelagic geography, mountainous terrain, and underdeveloped infrastructure networks.

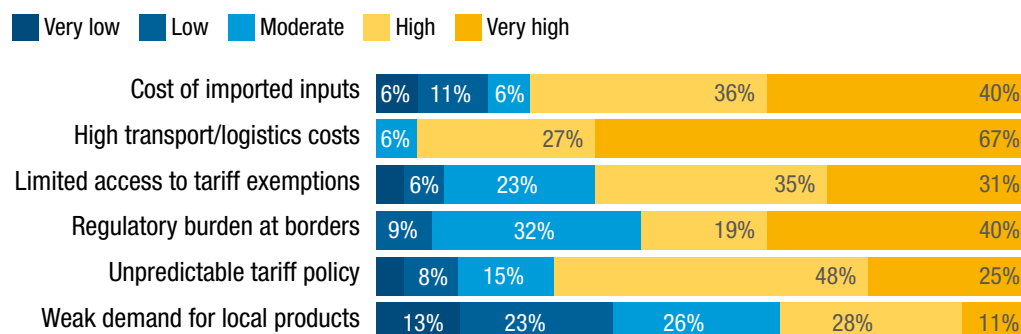
For many businesses, the cost of moving products within Papua New Guinea or connecting to international markets exceeds the impact of tariff policy changes by orders of magnitude. The survey further revealed what can be characterized as a “triple impact” affecting business operations beyond transport challenges, namely expensive imported inputs, unpredictable policies, and regulatory burdens. Specifically, 76 per cent of respondents identified expensive imported inputs as a significant operational barrier, while 73 per cent and 59 per cent cited policy

unpredictability and regulatory burdens as major concerns, respectively (Figure 27). These results suggest that Papua New Guinea's competitiveness challenges stem from fundamental structural issues beyond trade policy choices, and that

trade policy should be accompanied by an enabling, predictable regulatory environment and well-suited infrastructure to support businesses in productivity and competition capacity improvement.



Figure 27.
Main constraints to business operations



Source: Survey Results; n=49

E. Innovation

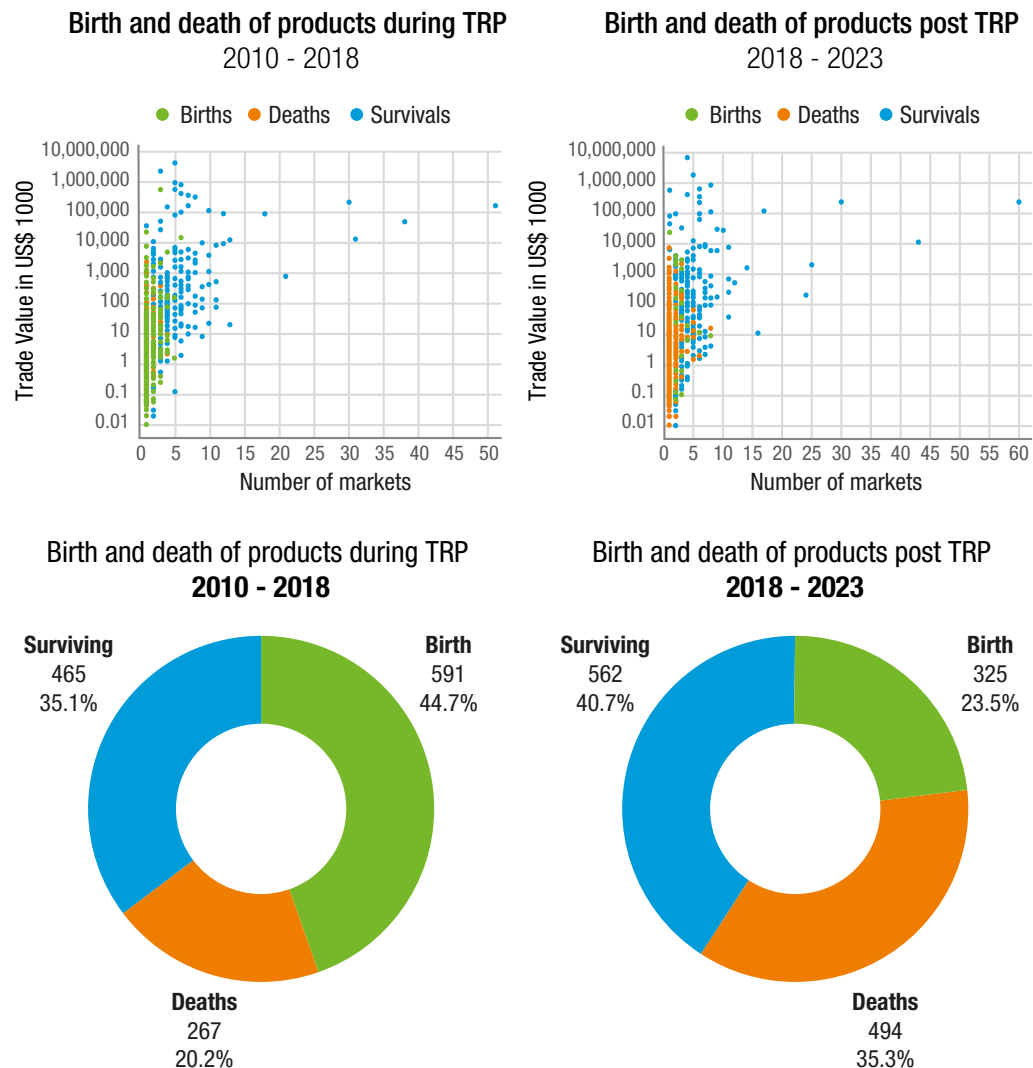
During the TRP years, Papua New Guinea's more open trade regime coincided with what can be described as an export product "boom". Data on the extensive margin of trade (the number of distinct products exported) show a clear uptick, suggesting that firms, enjoying lower input costs and easier access to imported equipment and materials, started venturing into new product lines for export. In the 2010-2018 period, 591 new products entered Papua New Guinea's export portfolio, representing 45 per cent of the total export basket (Figure 28). Most of the new products were of very little value (under \$100,000 in 2018). Only 44 products exceeding \$100,000 in exports in 2018 were new products in comparison to 2010 (see Annex 4 for the list of products). This increase in product innovation suggests that lower input costs enabled domestic firms to test international markets and explore new competitive opportunities. However, export sustainability proved challenging,

with only 465 products (35 per cent of the total) surviving the full TRP period. The figure indicates that cost advantages from input tariff reductions, while helpful for market entry, proved insufficient for sustained international competitiveness without addressing broader structural constraints affecting Papua New Guinea's productive capacity. On the other hand, the post-TRP period revealed the fragility of these export gains, with 494 products disappearing from Papua New Guinea's export portfolio (36 per cent of the total) while only 325 new products emerged (23 per cent). This pattern demonstrates how tariff increases can have unintended negative consequences for export competitiveness, particularly in sectors requiring imported intermediate goods or specialized equipment.

One key observation during the TRP was its association with export diversification and an entrepreneurial response by Papua New Guinea's private sector, followed by a setback once tariffs were raised.



Figure 28.
Innovation and decay in the export portfolio



Note: Blue dots represent product groups that existed in the export portfolio at the start and end of the period, while green dots represent product groups that exist at the end of the period but were not present in the country's portfolio at the start of the period. Data at the HS6 digit level.

Source: UNCTAD based on data from COMTRADE

The observed boom-and-bust pattern in export diversification is telling. Under the TRP, companies likely took advantage of cheaper imported inputs, intermediate goods, and capital machinery to venture into processing or manufacturing new products. Lower tariffs likely made imported inputs (materials, components) more affordable, thus reducing production costs for potential export industries. Furthermore, an open import regime can help exports because exporters often need imported machinery and raw materials to produce goods that

meet international standards. Additionally, the TRP signalled a commitment to open trade, possibly encouraging investment (domestic or foreign) in new ventures oriented towards export markets, knowing that inputs could be sourced at world prices. After 2018, the reversal in policy (i.e., higher tariffs) likely increased costs for those budding industries.

Firm-level behaviour and perceptions offer insights into how trade policy affected business decisions in the innovation area. Results from

the business survey reflect how firms perceived the impact of tariff changes on investment and innovation. Specifically, 46 per cent of surveyed firms reported launching a new product or export in the last five years. This suggests a strong entrepreneurial drive (Figure 29). But

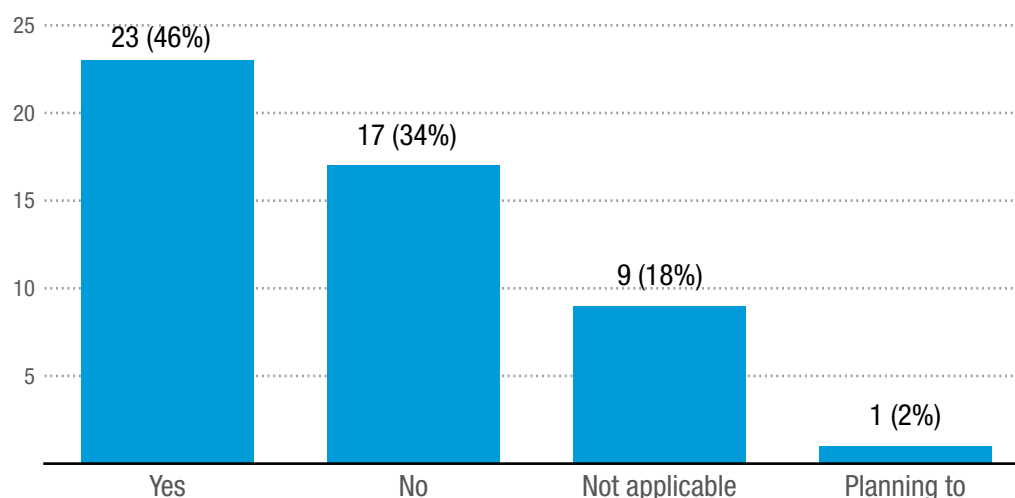
many also reported that higher input costs, equipment import restrictions, and constant policy changes made product development more difficult and expensive. This further emphasizes the roles of stable policy for investors' confidence and business innovation capacity.



Figure 29.

Reported innovation attempts: introducing new products

Has your company introduced new products or exports in the last 5 years?



Source: Survey results; n=50

F. Consumer welfare

Trade policy has direct consequences for consumer welfare through its impact on prices and product availability.

The welfare effects of the TRP can be assessed using consumer surplus, a measure of the benefit consumers obtain from being able to purchase goods at certain prices, derived from partial equilibrium simulations. In theory, consumer welfare outcomes depend on the balance between short-term price pressures caused by tariffs and the longer-term domestic industry development. In the short term, higher tariffs on finished products could lead to an increase in consumer prices for imported goods, thus reducing purchasing power, especially for lower income groups. Over the longer term, consumers may benefit from improved access to locally produced substitutes if protective tariffs succeed in stimulating domestic industries, they could

contribute to price stabilization, fostering import substitution, and strengthening local manufacturing capacity, thereby reducing dependence on expensive imports. However, there is also a risk that reduced competition from imports may limit product variety and quality in the domestic market, particularly if domestic industries do not simultaneously improve their production standards and capabilities.

The consumer welfare impact of the tariff policies was heavily impacted by the radical change in a few product groups. During the Papua New Guinea TRP implementation period (2010–2018), one would expect consumer welfare to improve because tariffs coming down should lower prices. Indeed, many categories of consumer goods did see tariff reductions (the average tariff on consumer goods

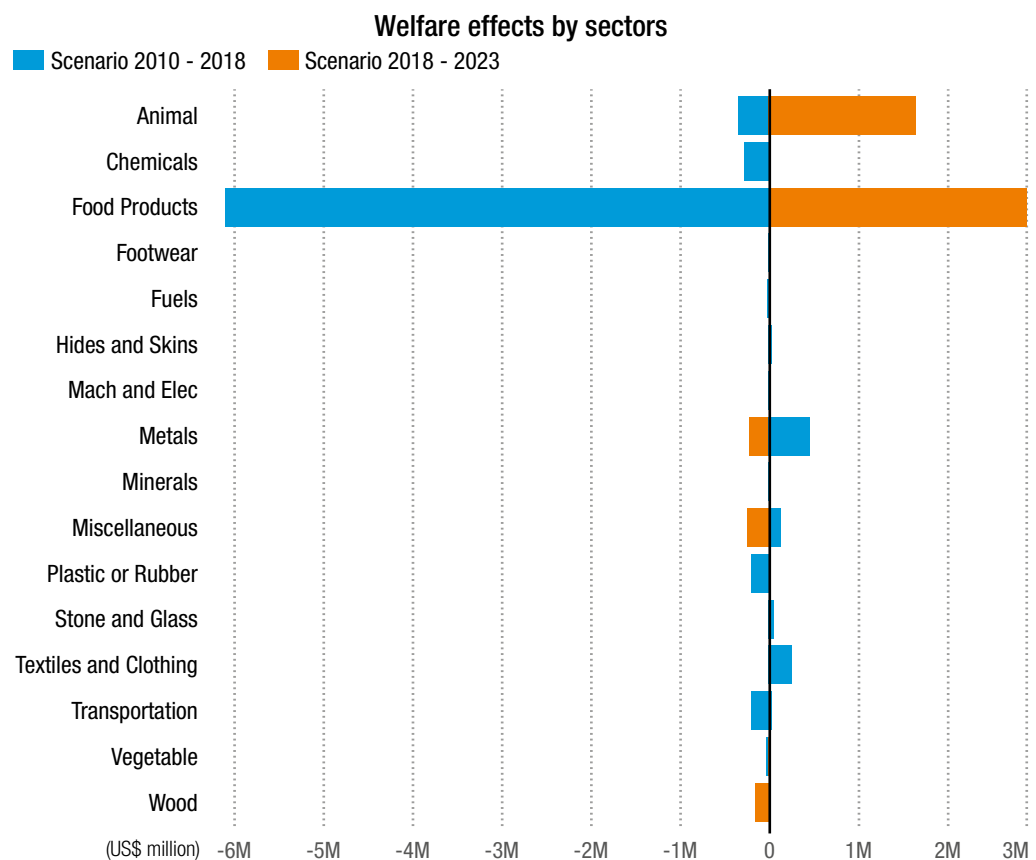
fell from ~23 per cent to ~14 per cent). However, the simulation results show a net welfare loss of around \$5.9 million for Papua New Guinea consumers in this period (Figure 30). This seemingly paradoxical outcome can be explained by the structure of the tariff changes. While hundreds of products had small tariff cuts, a few products retained very high tariffs, which resulted in large negative effects. Specifically, throughout the TRP, Papua New Guinea kept high tariffs on beverages (alcohol, soft drinks) and tobacco. The welfare analysis indicates that this single policy (high tariffs on alcohol

and tobacco) caused an estimated \$6.5 million loss in consumer surplus. Beyond beverages and tobacco products, Papua New Guinea increased tariffs on about \$390 million of other goods, lifting average rates from 5 per cent to 14 per cent, which contributed to a welfare loss of \$1 million. The total consumer welfare loss, therefore, outweighed the \$1.6 million gain consumers enjoyed from all the other tariff reductions. This finding is a reminder of the need for consideration of tariff impacts on items of common consumption or high expenditure with overall consumer welfare impact.

Figure 30.
Welfare implications of the TRP

Change in welfare during and after the TRP overall and by sector, (millions of dollars)

Scenario (US\$ million)	Welfare effects (Consumer surplus) per annum
During TRP (2010 - 2017)	-5.9
Pos TRP (2018 - 2023)	4.0



Source: International Economics TradeSim based on data from ITC Macmap and UN Comtrade

Conversely, after the TRP was suspended (2018-2023), the analysis shows a net welfare gain of roughly \$4 million in the post-2018 scenario. The cause of this shift is rooted again in the treatment of tobacco, with a drastic cut in the import tariff on tobacco from an average of ~159 per cent to 58 per cent. The effect of cutting such a high tariff was a significant drop in cigarette prices, which increased consumer surplus. That positive welfare effect outweighed the smaller, diffuse losses from the dozens of other tariff increases that remained in effect for items like dairy, meat, etc. In essence, Papua New Guinea's consumer welfare increased after the tariff reversal with the reduction of one extreme anomaly in the tariff structure (the punitive tobacco tax); had that not happened, the post-2018 welfare outcome would have been negative.

A critical observation is that conventional consumer welfare calculations from tariff reduction often assume that greater consumption, driven by lower prices, translates into higher welfare. However, this overlooks the negative externalities of harmful goods. For example, reduced tariffs on tobacco and alcoholic beverages may lower prices and increase consumption, which models interpret as a welfare gain. In reality, such increases can impose significant health costs on individuals and society,

such as higher medical expenditures, reduced productivity, and long-term public health burdens. These adverse effects are not captured in standard surplus-based measures, which focus narrowly on changes in prices and quantities consumed. As a result, the calculated welfare improvement may overstate the true benefits of tariff reductions when the products involved carry detrimental health or social consequences. This highlights the need to integrate broader social cost considerations alongside economic models when evaluating tariff policies.

Another dimension of welfare is the variety and quality of products available. Under a liberal trade regime, consumers enjoy greater choice – more brands, newer products – whereas high tariffs can limit choices to expensive local substitutes. While it is challenging to quantify the impact of increasing variety and choice, survey responses indicate that 55 per cent of businesses believe Papua New Guinea consumers prefer cheaper imported goods to higher-priced local ones (Figure 31), which further highlights the importance of price and variety for consumers. If tariffs force consumers to buy local products that are more expensive or of lower quality, that would result in a welfare loss not fully captured in the model.

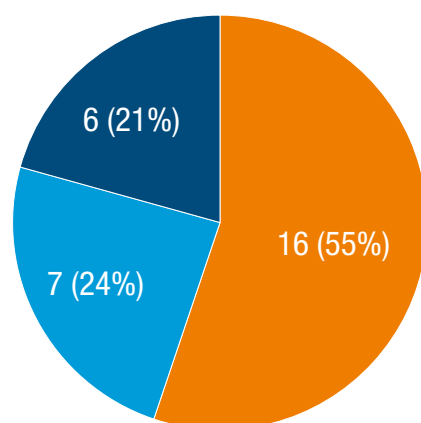


Figure 31.

Perceived price sensitivity of Papua New Guinea consumers

Do Papua New Guinea consumers appear willing to pay more for local products if protected from cheap imports?

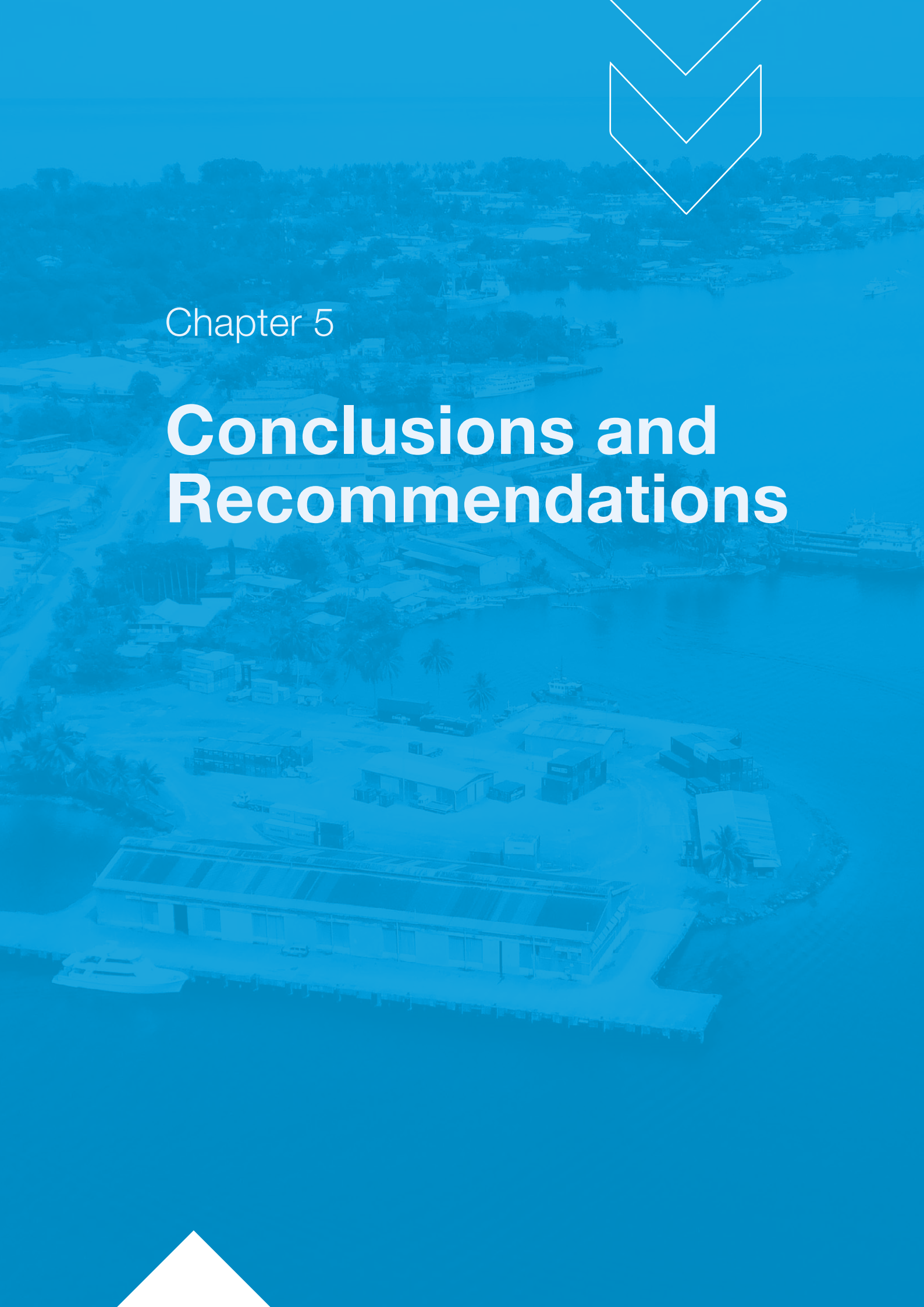
■ No
 ■ Yes
 ■ Not sure



Source: Survey results; n=29

Consumer welfare in Papua New Guinea suffered during the tariff reductions phase mainly due to selective high taxation on certain goods, and it improved slightly after 2018 solely because the tobacco tariff was cut. The lesson for policy is that extreme levels of tariff rates on popular consumption goods can cause loss to social and consumer welfare. A more balanced approach involving moderate tariffs across a broad base might achieve protection or revenue goals with less welfare damage.

Complementary measures such as excise taxes are more suitable for addressing public health concerns, for example in the case of tobacco. Stability in tariff policy is also important, as frequent shifts between high and low rates create uncertainty and uneven impacts on households. Continuous monitoring of tariff impacts across different segments of the economy is therefore essential to ensure that fiscal objectives or sectoral protection do not come at the expense of consumer welfare.

An aerial photograph of a coastal industrial area, likely a shipyard or port, with various buildings, docks, and ships. The image is overlaid with a blue gradient. In the top right corner, there is a white graphic of two stacked chevrons pointing downwards. In the bottom left corner, there is a white graphic of a triangle pointing upwards.

Chapter 5

Conclusions and Recommendations



Papua New Guinea embarked on the TRP as part of a broader post-crisis reform agenda, aiming to improve economic efficiency and integrate with global markets. The policy was intended to improve access to a larger variety of imported inputs and final products at lower cost, while exposing Papua New Guinea's domestic industries to international competition, thereby encouraging productivity gains and deliver consumer benefits.

The evidence from this review reveals that tariff policy alone cannot determine economic development outcomes in contexts where structural constraints fundamentally limit competitiveness. Papua New Guinea's businesses face high transport costs, energy expenses, and regulatory complexity that often dwarf the effects of tariff modifications, suggesting that successful development strategies must also address these underlying constraints to deliver more results from trade policy instruments like tariffs.

Nevertheless, trade policy remains an important component of the government policy toolbox for a comprehensive development strategy. Predictability emerges as perhaps the most important characteristic of effective trade policy, with businesses consistently emphasizing the importance of stable policy environments for long-term planning and investment. Whether tariff rates are high or low matters less than whether they are consistent and reliable over the time horizons required for building productive capacity -allowing businesses to predict and plan accordingly.

Tariff policy also operates withing wider regional and global frameworks. Free trade agreements with significant markets nearby, including Australia and New Zealand, and far away – with the European Union – reduce the policy space for tariff increases but provide predictability and the ability to tackle existing non-tariff barriers to trade.

Papua New Guinea's natural competitive advantages in agriculture, forestry, and mineral resources provide a foundation for economic growth and sustainable development. It is, therefore, important

that trade policy should support and enhance existing strengths rather than attempting to create artificial advantages in sectors lacking competitiveness.

Moving forward, Papua New Guinea should aim to formulate a tariff policy that balances development objectives with fiscal revenue needs, while taking account of the economy's reliance on imported inputs. The following recommendations are suggested:

1. Restore a path of gradual tariff rationalization

The Department of Treasury should consider resuming tariff reductions in a gradual, predictable manner. A phased approach over several years – for example, five to 10 years, similar to an FTA phasing out process – can help industries adjust, prepare for the changes and make necessary investments. The objective should be to establish a simpler, more uniform tariff structure, perhaps with a low single-digit uniform rate for most goods, removing nuisance tariffs, and a limited number of truly sensitive items at a slightly higher rate. This would reduce trade distortions and administrative complexity. Extremely high tariffs – e.g., above 40 per cent – could be counterproductive in both revenue and development terms (as they may encourage under-invoicing, smuggling, and may harm a variety of products for consumers). Generally, specific tariffs (covering nearly 7 per cent of imports for agricultural products and 2.2 per cent of imports for non-agricultural products in Papua New Guinea) are less transparent

Trade policy remains an important component of the government policy toolbox for a comprehensive development strategy.

than ad valorem tariffs, though they can be effective in keeping out low-value imports. A predictable and coherent tariff policy – possibly anchored in a revised National Trade Policy framework – would enhance certainty for businesses. An example to

consider is the Mauritius tariff profile, which has less uncertainty, less variation and fewer nuisance tariffs. Obviously, consultations and additional analysis are needed before changing the tariff profile to align with the country's objectives (Table 6 and Table 7).



Table 6
Mauritius tariff profile in 2025

Summary		Total	Ag	Non-Ag	WTO member since		1995		
Simple average final bound		86.6	119.4	17.0	Binding coverage:		Total	20.4	
MFN applied							Non-Ag	7.6	
Simple average	2024	0.8	1.6	0.7	Ag: Tariff quotas (in %)		0		
Trade weighted average	2024	1.5	4.8	0.7	Ag: Special safeguards (in %)		0		
Imports in billion US\$	2023	6.3	1.2	5.1					

Frequency distribution		Duty-free	0 <= 5	5 <= 10	10 <= 15	15 <= 25	25 <= 50	50 <= 100	>100	NAV in %
		Tariff lines and import values (in %)								
Agricultural products										
Final bound		0	0	0	0	0	2.5	1.3	96.1	0
MN applied	2024	92.7	0	3.0	2.6	0	1.1	0.5	0	0
Imports	2023	91.1	0	1.5	5.3	0	2.1	0	0	0
Non-agricultural products										
Final bound		5.7	0	0	0	0	0	1.8	0.1	0
MN applied	2024	96.3	0	0.1	2.8	0	0.9	0	0	0
Imports	2023	95.7	0	0.3	2.5	0	1.5	0	0	0



Table 7
Papua New Guinea tariff profile in 2025

Summary		Total	Ag	Non-Ag	WTO member since		1996		
Simple average final bound		32.8	43.2	31.1	Binding coverage:		Total	100	
MFN applied							Non-Ag	100	
Simple average	2024	4.1	12.2	2.8	Ag: Tariff quotas (in %)		0		
Trade weighted average	2024	2.4	7.6	1.4	Ag: Special safeguards (in %)		0		
Imports in billion US\$	2023	6.2	1.0	5.2					

Frequency distribution		Duty-free	0 <= 5	5 <= 10	10 <= 15	15 <= 25	25 <= 50	50 <= 100	>100	NAV in %
		Tariff lines and import values (in %)								
Agricultural products										
Final bound		0	1.7	0.3	17.8	6.3	38.2	35.3	0.3	6.9
MN applied	2024	46.7	1.5	9.1	20	10.6	10.6	0.4	1.1	6.2
Imports	2023	73.8	3.3	2.0	8.5	4.9	7.1	0	0.3	6.6

Frequency distribution		Duty-free	0 <= 5	5 <= 10	10 <= 15	15 <= 25	25 <= 50	50 <= 100	>100	NAV in %
		Tariff lines and import values (in %)								
Non-agricultural products										
Final bound		0	0.1	0	30.2	0	54.7	15.7	0	0.1
MN applied	2024	81.2	0.4	5.6	8.6	2.8	0.9	0	0	0.4
Imports	2023	89.8	2.0	2.5	2.2	2.7	0.5	0	0	2.2

Source: UNCTAD-WTO-ITC (2025) World Tariff Profiles 2025. Geneva July

2. Focus on fostering a competitive business environment

The government should pivot from protecting industries as a framework to enable them to compete. While adding additional layers of costs will undoubtedly provide some shielding from competition, this can have significant implications for consumers, businesses and ultimately growth. Papua New Guinea needs to focus on tackling the high-cost structure to do business on the island. Investments in infrastructure (roads, ports, electricity) are critical – e.g., improving transport could directly address the 94 per cent of firms citing logistics as their top constraint (Figure 32). Similarly, expanding a reliable and affordable power supply would benefit all manufacturers more than a tariff hike ever could. In fact, these are among the top policy recommendations to strengthen the industry's competitiveness as suggested by Papua New Guinea firms during the business surveys. Lowering input costs is

another quick win: for instance, eliminate or reduce tariffs on key intermediate inputs and capital goods (many of which are already low or zero). This would make it cheaper to produce “Made in Papua New Guinea” goods and services. With respect to product categorisation, Papua New Guinea should align its classification of goods for tariff purposes with the UNCTAD classification and justify any deviations. For example, products could be grouped by end use (raw materials, intermediary inputs, capital goods, or consumption goods) and further disaggregate into categories such as “import competing”, “environmental concerns”, or “fiscal revenue generating”. These categories should reflect stakeholders’ priorities and trade-offs from tariffs across.

In addition, simplifying customs procedures and reducing red tape can lower the non-tariff barriers that often exceed the tariff in impact. Improving governance, increasing multi-stakeholder engagement, and improving the efficiency of government lie at the heart of such trade facilitation measures.

Figure 32. Recommended areas of support for competitiveness

What are your top 3 policy recommendations to strengthen your industry's competitiveness?



Source: Survey results

- **Industry development programs:** Build foundation for competitiveness, such as infrastructure (roads, ports, power), workforce training, and technology upgrades.
- **Support schemes:** Direct financial help, such as access to affordable credit, subsidies for equipment/inputs, and tax breaks to reduce operating costs.
- **Trade policy measures:** Simpler exemption processes, lower input tariffs, and stable policy (stop changing rules).

3. Use targeted support for infant industries

If the government identifies certain sectors as priorities for diversification (e.g. agro-processing, light manufacturing, downstream processing of natural resources), it should employ targeted, time-bound support measures rather than blanket import protection. However, empirical evidence often highlights that governments frequently lack both the information and the ability to effectively target industries, leading to prolonged inefficiencies and entrenched protectionism rather than temporary support.⁷ In addition, a usual byproduct of tariffs is rent-seeking (or engaging in nonproductive efforts to extract resources and wealth) and other political economy considerations.⁸ Such political capture can entrench tariffs, reduce overall economic efficiency and complicate subsequent liberalization efforts.

A targeted support needs to be carefully identified and engineered. The types of support could include subsidies on inputs, tax holidays, credit facilities for capital investment, infrastructure investments, or technical assistance programs to improve quality and productivity. For example, instead of a 25 per cent tariff on dairy that raises consumer prices, a targeted program to provide dairy farmers with feed or veterinary services at subsidized cost would directly boost their productivity. Any protection that is provided should come with a clear sunset clause and performance benchmarks (as true infant industry support requires a path to competitiveness, not indefinite shelter). Engaging in public-private dialogues can help identify what specific support is most needed in each

sector, as suggested through the survey results, whereby many firms would prefer specific support over import tariffs.

4. Safeguard consumer welfare and food security

Tariff policy should be calibrated to avoid undue burden on consumers, especially for essential goods. Papua New Guinea has a relatively low purchasing power and a high poverty rate, and many households spend a large share of their income on food and essentials.⁹ High tariffs on staple foods or fuel are therefore regressive. It is recommended that essential goods (basic food items, medicine, fuel) be kept at low or zero tariffs to protect consumers' purchasing power. For goods like rice, flour, or tinned fish that are daily necessities, the government could explore buffer stocks or price stabilization schemes rather than high tariffs to support local producers. Where there are health or social reasons to tax a product (e.g. alcohol, tobacco), using excise taxes might achieve the goal without distorting trade since excise duties apply to both imports and local products and can be better targeted for health policy.

5. Enhance regional integration efforts

Lowering tariffs is only one aspect of openness – equally important is reducing non-tariff barriers and actively leveraging trade agreements. Papua New Guinea should continue to strengthen its customs service capacity to facilitate trade, for example, by adopting digital customs clearance, risk-based inspections, and

⁷ See for example Krugman, P. (1987). Is Free Trade Passé? in *The Journal of Economic Perspectives*, 1(2), American Economic Association and Lubik, T. A. (2025). When are tariffs optimal? in *Economic Brief*, 25(21). Federal Reserve Bank of Richmond

⁸ Grossman, G. M. & Helpman, E. (1994). Protection for Sale. in *American Economic Review*. 84(4). American Economic Association and Baron, P. (2002). Review of Grossman and Helpman's Special Interest Politics in *Journal of Economic Literature* 40(1)). American Economic Association

⁹ According to household surveys, 75 per cent of total household expenditure is dedicated to food (including the value of food grown and consumed on-farm). See Schmidt, E.; Fang, P.; Jemal, M.; Mahrt, K.; Mukerjee, R.; Rosenbach, G. & Yadav, S. (2024). 2023 PNG Rural Household Survey Report. International Food Policy Research: Washington, DC

working with neighbours to streamline border processes. This will reduce costs for importers and exporters alike. Papua New Guinea could strengthen its position by leveraging existing agreements such as the European Union interim EPA, MSG arrangements, and PACER Plus, while also considering participation in broader frameworks like the Association of Southeast Asian Nations Plus 1 (ASEAN+1) and the Regional Comprehensive Economic Partnership (RCEP), agreements. This would require deeper commitments from Papua New Guinea, but could yield significant market access. A final avenue to explore for Papua New Guinea is to capitalize on its success in implementing the WTO Trade Facilitation Agreement by extending efforts to other plurilateral agreements recently adopted in the framework of the WTO, such as the e-commerce joint statement initiative, the Domestic Regulations of Trade in Services, Investment Facilitation for Development, the older Information Technology Agreements I and II, and the Government Procurement Agreement. Implementing these agreements will help lower the cost of trade, improve the ability to source critical inputs for Papua New Guinea's industry, and raise efficiency for exploring new channels and opportunities in trade. Finally, Papua New Guinea should consider acceding to the Framework Agreement on the Facilitation of Cross-border Paperless Trade in Asia and the Pacific (CPTA) in order to facilitate the trade facilitation digitalization processes, which have been shown to have significant results in reducing trade costs.¹⁰

6. Institutionalize evidence-based review mechanisms

The tariff changes in 2018–2019 were made through budget announcements

without extensive public analysis. Going forward, Treasury would benefit from a more institutionalized review process for tariff policy. For instance, a Tariff Advisory Board could be established, including government, private sector, and independent experts, to evaluate requests for tariff changes and periodically assess outcomes. Regular impact assessments (such as the present report) should inform adjustments. Such a body could help gather evidence to inform policy formulation and ensure that any tariff protection is granted based on clear evidence of need and with a plan for phase-out. Transparency in tariff policymaking will also reassure investors that the environment will not shift unpredictably.

7. Invest in the capacity to monitor, investigate and deploy trade defense mechanisms

In the survey conducted under this review, the business community raised the view that unfair trade practices from some countries were posing a threat to their competitiveness. Papua New Guinea could adopt pending legislation on anti-dumping, countervailing, and safeguard measures, supported by a transparent legal framework. A specialised trade remedies unit with expertise in economics, law, and data analysis would strengthen monitoring and investigation capacity. Partnerships with regional and international organisations could provide training and technical support, helping ensure that Papua New Guinea can apply WTO-consistent measures when necessary. The last comprehensive call for this came in the UNCTAD Trade Policy Framework of 2006.¹¹ According to the Review, a draft Anti-dumping and Countervailing Measures Act and a draft Safeguards Measures Act were to be adopted, but this does not appear to have

¹⁰ ESCAP & ADB (2024). Asia-Pacific Trade Facilitation Report 2024: Promoting Sustainability and Resilience in Global Value Chains. Asian Development Bank: Manila.

¹¹ UNCTAD (2006). Papua New Guinea New Trade Policy Framework. UNCTAD: Geneva and New York. UNCTAD/DITC/TNCD/2003/10

happened.¹² By adopting the regulation and strengthening its institutional framework to undertake investigations, Papua New Guinea can protect vulnerable industries, mitigate economic shocks from external dumping, and navigate trade relations with greater confidence and authority.

Papua New Guinea stands at a crossroads in trade policy. The evidence indicates that while protective tariffs may offer short-term relief or revenue, they are not a sustainable strategy for long-term development.

A return to an outward-oriented trade policy, complemented by strong domestic investments in infrastructure, education, and governance, is likely to yield the best

outcomes in terms of growth, diversification, and consumer welfare. Papua New Guinea's own experience from 2010 to 2017 shows that it can successfully integrate with the world while bringing benefits home. The policy reversals of 2018 to 2023, meanwhile, serve as a caution that protectionism can be a costly detour. The recommended path forward is one of open but strategic trade engagement by keeping the economy open to gain efficiency, while using smart policies at home to build the competitiveness of industries so they can thrive in the global marketplace without needing high tariff walls. This balanced approach will support Papua New Guinea's aspirations for diversified, inclusive, and sustainable development.

¹² WTO (2019). Trade Policy Review Report by the Secretariat: Papua New Guinea. WTO: Geneva. WT/TPR/S/387

References

- ADB (2012). Papua New Guinea: Critical Development Constraints. Country Diagnostics Studies.
- Amiti, M., Redding, S.J., & Weinstein, D.E. (2019). The Impact of the 2018 Tariffs on Prices and Welfare. *The Journal of Economic Perspectives*, Fall 2019, Vol. 33, No. 4 (Fall 2019), pp. 187-210
- Anderson, J. E. & van Wincoop, E. (2003). Gravity with Gravititas: A Solution to the Border Puzzle. *American Economic Review*. Vol 93 (1). March
- APEC (1994). 1994 Leaders' Declaration. Bogor, Indonesia.
- Business Advantage Papua New Guinea (2018). Opinion: Papua New Guinea adjusting its tariff strategy in APEC year.
- Conroy, J. D. (2021). Economic informality in Pacific island countries. Griffith University. Griffith Asia Institute. 12 March
- ESCAP & ADB (2024). Asia-Pacific Trade Facilitation Report 2024: Promoting Sustainability and Resilience in Global Value Chains. Asian Development Bank: Manila.
- ESCAP (2023). Framework Agreement on Facilitation of Paperless Trade in Asia and the Pacific. ESCAP: Bangkok, May
- ESCAP (2023). Trade Statistics in Policy Making: A Guide on Using R for Calculating Trade Indices and Indicators: Weighted Average Tariff.
- European Commission (n.d.). The EU - Pacific States Interim Economic Partnership Agreement.
- Flaen, A. & Pierce, J. (2021). Disentangling the Effects of the 2018-2019 Tariffs on a Globally Connected U.S. Manufacturing Sector.
- Fox, R. (2018) Opinion: Papua New Guinea adjusting its tariff strategy in APEC year. Business Advantage PNG
- Fox, R.; Laveil, M.; Howes, S. (2018) PNG turns protectionist in its APEC year. Development Policy Centre
- Gilbert, J. (2017). Partial Equilibrium Analysis. UNESCAP: Bangkok.
- Grossman, G.M., Helpman, E. & Redding, S.J. (2024). When Tariffs Disrupt Global Supply Chains. *American Economic Review* 2024, 114(4): 988-1029.
- Head, K. & Mayer, T. (2014). Gravity Equations: Workhorse, Toolkit and Cookbook, in *Gopinath, Gita, Ehanan Helpman, and Ken Rogoff, eds Handbook of International Economics*, Vol 4. North Holland: Elsevier.
- IMF (2015). Papua New Guinea: Selected Issues. International Monetary Fund.
- IMF (2025). World Economic Outlook. March 2025. Washington, DC: International Monetary Fund.
- Laveil, M. (2020). Has trade protection assisted the manufacturing sector in PNG? Development Policy Centre
- Liu, B. Hudson, D. & Devine, J. (2022). A Gravity Model Analysis of Chinese Apparel Export Flows. Briefing Paper. ICAC – BP22 – 01. January
- Martin, W. (2020). Making Gravity Great Again. *Policy Research Working Paper*. WP9391. World Bank.
- Minister for Treasury (2024). 2025 National Budget. Volume 1 Economic and Development Policies. Securing Papua New Guinea in 2028 and Beyond.

- Nathan, Robert & Ram, Deepak & Balraj, Ann. (2017). Purchase Factors for Products with Inelastic Demand: A Study on Tobacco Sector. *Journal of Engineering and Applied Sciences*. 12. 1754 - 1761. 10.36478/jeas-ci.2017.1754.1761.
- National Statistics Office of Papua New Guinea (2025). Gross Domestic Products.
- OECD (2025). Revenue Statistics in Asia and the Pacific 2025: Personal Income Taxation in Asia and the Pacific. OECD Publishing: Paris.
- Owjimehr, S. & Jamshidi, N. (2024). Why do natural resource-rich economies resist improving the economic complexity?, in *Regional Science Policy & Practice*, Vol 16, Issue 7, Elsevier
- Papua New Guinea National Statistical Office (2025). PNG National Accounts (GDP).
- Post Courier (2017). Tariff Reduction Program suspended.
- Schmidt, E.; Fang, P.; Jemal, M.; Mahrt, K.; Mukerjee, R.; Rosenbach, G. & Yadav, S. (2024). 2023 PNG Rural Household Survey Report. International Food Policy Research: Washington, DC
- Scollay, R. (2007). Review of the Tariff Reduction Program. Prepared for Treasury PNG.
- The National (2020). Trade protection, did it help?
- Thiagarajan, R., Brender, J. & Metcalfe, M. (2025). Understanding the Economic Tradeoffs of Tariffs. State Bank. February 2025
- UNCTAD (2006). Papua New Guinea New Trade Policy Framework. UNCTAD: Geneva and New York. UNCTAD/DITC/TNCD/2003/10
- United Nations and World Trade Organisation (2012). A Practical Guide to Trade Policy Analysis. Chapter 2: Quantifying trade policy.
- United Nations Conference on Trade and Development (UNCTAD) (2006). Papua New Guinea Trade Policy Framework. UN: New York and Geneva.
- World Trade Organisation (2006). World Tariff Profile 2006.
- World Trade Organisation (2010). Trade Policy Review. Report by the Secretariat. Papua New Guinea.
- World Trade Organisation (2019). Trade Policy Review Report by the Secretariat: Papua New Guinea. WTO: Geneva. WT/TPR/S/387

Annexes

Annex 1. Project terms of reference

Review of the Papua New Guinea Tariff Reduction Program

A. BACKGROUND

In 1986, the Bogan Review covered Papua New Guinea Tariff Policy and Administration in Papua New Guinea, which resulted in the release of the 1997 White Paper on the Tariff Reform Program. Subsequently, the first Tariff Reduction Program (TRP) began in 1999. The TRP was aimed at encouraging a more efficient and productive private sector through greater exposure to international competition. As part of the reform, duty rates for most goods (three-quarters) were set at zero, with the exception of goods that are produced in Papua New Guinea or could potentially be produced in Papua New Guinea. The duty rates for goods that are produced or could potentially be produced in Papua New Guinea, the duty rates were combined into three categories: intermediate, protective, and prohibitive rates. The three categories of duty rates were reduced in three scheduled steps at a rate of 5 per cent.

The introduction of the Value Added Tax (now the Goods and Services Tax) was designed and introduced to ensure that there would be no concern over revenue losses arising from the TRP. The revenue collection mechanism for GST became a more modern and efficient approach to the collection of indirect taxes.

In the lead-up to the 2003 National Budget, the Manufacturers' Council of Papua New Guinea sought a temporary suspension of the TRP. The government responded by commissioning two reviews of the Tariff Reduction Program, conducted separately in 2003 and 2007, to establish the future direction of tariff reduction policy for Papua New Guinea.

Both reviews confirmed that the TRP had achieved its intended purposes and recommended that the government implement further tariff reduction, aimed at achieving more uniform as well as lower tariff rates for a more uniform effective rate of protection. Following these reviews, the government implemented a scheduled or phased tariff reduction in three-year intervals, which commenced in 2012 and ended in 2017. The table below shows the tariff reduction rates and schedule prior to 2018.

Tariff Reduction Program announced in the 2011 Budget.				
				Current Rates
Category rate	2011	2012 - 2014	2015 - 2017	2018
Intermediate	15%	15%	15%	15%
Protective	25%	20%	15%	15%
Prohibitive	40%	30%	30%	30%
Tariff Reduction Program with Extension announced in the 2012 Budget.				
Intermediate	15%	12.5%	10%	
Protective	25%	20%	15%	
Prohibitive	40%	35%	30%	

According to the Tax Review Committee's Final Report 2013 - 2015, it mentions that several local manufacturing businesses argued that the reduced import rates through the TRP program were unfair competition resulting from lower import duties on products. The Report further stated that some businesses will be affected or even forced to close if TRP does not address their concern.

Prior to 2018, a proposal was made to carry out a review into the last phase of TRP implementation and to recommend the next step going forward. However, the proposal was unsuccessful due to funding issues. Therefore, in the 2018 Budget, the government suspended the TRP and maintained the 2017 TRP rates in 2018 pending the review.

Whilst the TRP is suspended, the government introduced several changes to specific tariff items, mainly to protect infant local manufacturers from cheap imported products. The amendments were made to the *Customs Tariff Act 1990* in the 2018, 2019 and 2020 National Budgets. The introduction of Tariff policies on an ad-hoc basis provided uncertainty to the wider business community.

TRP Review and its relevance to Papua New Guinea National Development and Trade Strategies

The review of the Tariff Reduction Program is highly relevant to the Papua New Guinea National Trade Policy (NTP) 2017 – 2032, Medium Term Development Plan (MTDP IV) 2023-2027, Medium Term Revenue Strategy 2023 - 2027 and other sector-specific plans such as fisheries, forestry and others. Identifying the priority products by sectors that will contribute towards a stronger and resilient economy and grow these strategic sectors through investment, value adding and providing information for exporters and importers and related policy making.

Papua New Guinea is a small open economy producing a diverse range of raw products from agriculture, forestry, fisheries, mining and petroleum sectors. Papua New Guinea does not produce

many of the products it consumes and relies on international trade for imports of physical capital, intermediate products and final consumer goods and services.

The TRP review will set a new policy direction for the future tariff to promote sustainable economic growth and a smooth trading system for Papua New Guinea over the medium to long term.

B. Terms of reference

Review of Import Tariff

The Government of Papua New Guinea through the Department of Treasury, is engaging a consultant to undertake an impact assessment of the TRP to ascertain the costs and benefits of the program.

The consultant, with the assistance of the Department of Treasury, Papua New Guinea Customs Service and the National Trade Office, will consult widely with stakeholders in preparing the Impact Assessment Study Report.

Objective of the Engagement

The objective of the engagement is:

- 1. To undertake a comprehensive assessment of the costs and benefits owing to the implementation of the Tariff Reduction Program that is temporary suspended.**
 - 1.1.** Determine whether the TRP has achieved its intended purpose to expose Papua New Guinea companies (manufacturing, agriculture and etc.) to international competition through reduced tariff rates. Use case studies to verify.
 - 1.2.** Determine whether the implementation of the tariff rate reductions encouraged a more efficient and productive private sector. Selected companies from relevant sectors of the economy will be interviewed to ascertain the outcome.

- 1.3. Determine the benefit, costs and overall effect of the TRP to Papua New Guinea, including consumers (reduced retail price), employees, businesses (producers), as well as importers and exporters. Use empirical analysis to explain.
 - 1.4. Determine whether the TRP's contribution toward the country's overarching goal of encouraging downstream processing and value adding was achieved.
 - 1.5. Determine the benefits and costs arising to Papua New Guinea from international trade reforms under Papua New Guinea's various trade agreements, such as the Interim Economic Partnership Agreement (IEPA) with the European Union, the Melanesian Spearhead Group and other Regional Agreements that Papua New Guinea is part of.
 - 1.6. Determine whether the high cost of utilities within the structure of the Papua New Guinea Economy has undermined TRP not achieving its intended objective.
- 2. Undertake a diagnostic overview (including international benchmarking and comparative analysis) of Papua New Guinea's current import tariff policy based on the above findings and highlight issues for future import tariff policy and possible reform options.**
- 2.1. Establish the effective rates of protection provided through import tariffs to support local companies against imports, boost downstream processing thus economic growth. Advise whether this is an appropriate measure for PNG to take to achieve economies of scale. Provide examples of countries that are using such impositions. If not, propose options.
 - 2.2. Establish the advantages and disadvantages of protecting certain sectors based on economic contributions and long-term sustainability of the industry. For example, is providing incentives to value-added industries in PNG, particularly the downstream processing of agricultural goods (including forestry and fisheries), minerals and hydrocarbons appropriate?
 - 2.3. Establish the advantages and disadvantages of PNG's commitments from international treaties. Suggest options on how PNG could improve or take advantage of the benefits through its international trade relations, consistent with PNG's Trade Policy.
 - 2.4. PNG's trade policy is aimed in principle at the "domestication" of value adding and to promote import substitution and the diversification of the economy from the finite mineral and petroleum sector. Advise whether the TRP has achieved this objective. If not, further advise what specific direction the future tariff policy should take; and
 - 2.5. Establish what possible direction the future tariff policy should take, given the high cost of doing business (utilities) in PNG.
- 3. Undertake a detail assessment on whether the ad hoc changes on Import tariff rates of selected items in the 2018, 2019 and 2020 National Budgets achieve their intended aim to protect infant local industries from similar cheap imported products. Provide a case study of selected infant manufacturing companies that benefit from the reform.**
4. Any other findings that will add value to the discussion to improve Papua New Guinea tariff policy direction into the future, consistent with the Government development policies, such as the Medium-Term Revenue Strategy 2023 – 2027, the Medium-Term Development Plan III, the National Trade Policy 2017 – 2032 and other sector-specific policies.

Expected outcome of the engagement

1. A series of presentations and workshops to be undertaken to discuss the findings with the Department of Treasury, PNG Customs Service and National Trade Office officials and other relevant stakeholders.
2. To build capacity and transfer knowledge to Government officers assisting in the review.
3. Submit a comprehensive and detailed Final Report to the Department of Treasury upon completion.

Methodology

The engagement will involve the following phases:

1. **Inception Phase:** Hold an inception meeting with the Department of Treasury and the PNG Customs Service. The inception meeting will validate the Terms of Reference, identify necessary stakeholders, information and documents required to assist in the engagement and agree on a work plan.
2. **Interface with key stakeholders:** Develop a methodical process to engage and consult key organisations, not limited to PNG Customs Service, Department of Commerce, Trade & Industry and the Manufacturers Council.
3. **Comprehensive Final Report:** Prepare and submit a Final Report to the Department of the Treasury.

Projected Timeline

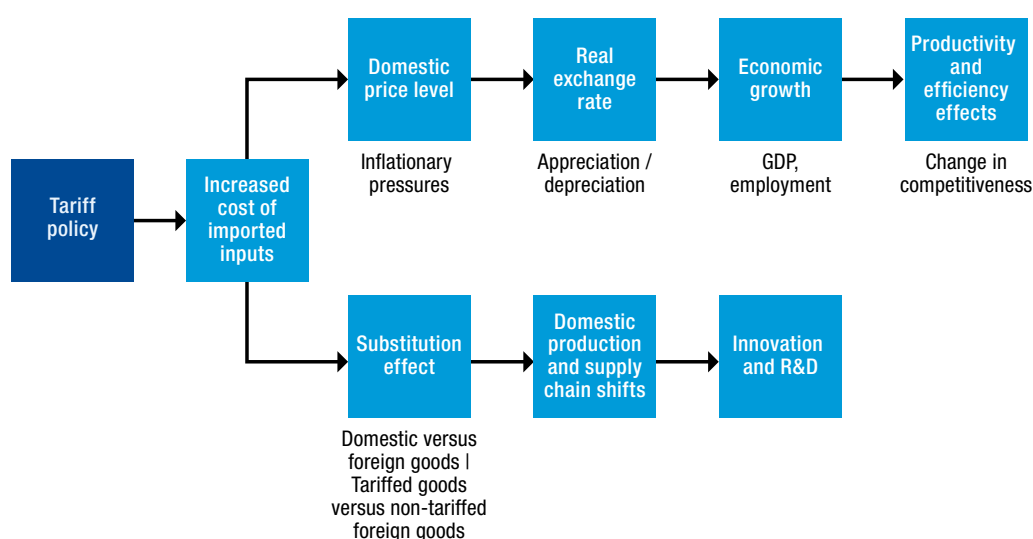
Phase	Timeline (2025)
i. Inception	2 weeks
ii. Interface with key stakeholders	16 weeks (4 months)
iii. Preparation of Final Report	4 weeks (1 month)

Annex 2. Methodology

Background

There are several channels through which tariff changes may alter consumers' consumption (and by default welfare), government spending and revenue, and enterprises' production and investment decisions. Most directly, tariff change can lead to increased costs for intermediary inputs or machinery, leaving firms to either postpone investment decisions or retain operational levels, and either absorbing some of the additional costs by reducing margins, or passing the additional costs onto consumers. A simplified and illustrative suggested pathway for tariff policies on firm decisions is illustrated in Figure 33.

Figure 33.
Tariffs and transmission of firm decisions



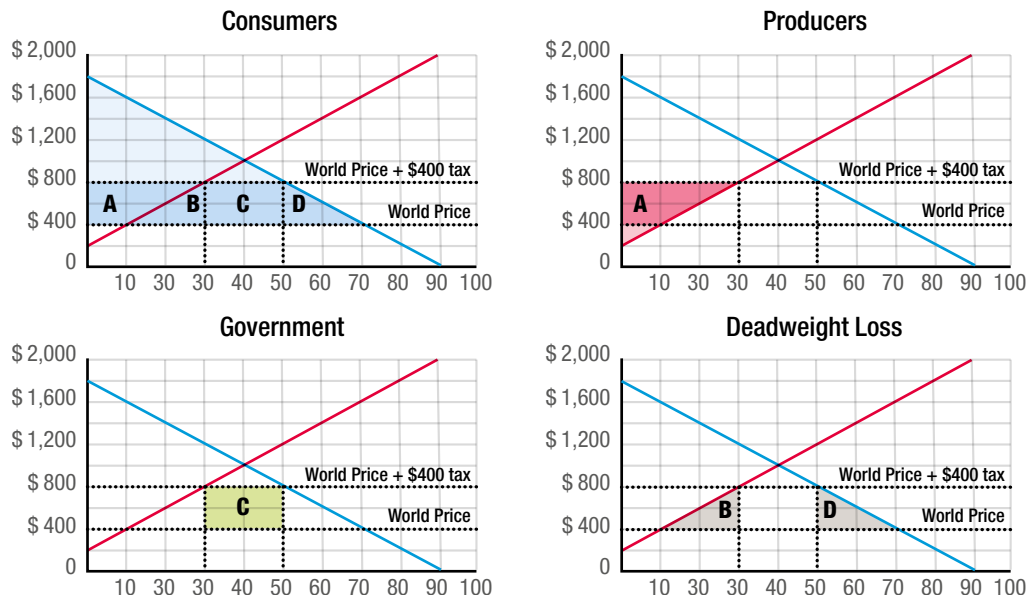
Source: Thiagarajan et al. (2025)¹³

In reality, the effects are more complicated owing to the nonlinear nature of effects, the lags in response (also affected by accumulated inventories) and a series of unknowns such as the propensity for consumers to pay a higher price (modelled through price elasticities of demand).

The simple method for calculating the impact of tariffs on consumers, government revenue and businesses is to adopt a partial equilibrium model. In such a model, the increase in a tariff adds a premium on price over the world price of a commodity.

This leads to a reduction in demand from consumers, which is illustrated in Figure 34, as increase price increases from \$400 to \$800 (a 100 per cent tariff is applied). Producers will be willing to supply more products at that price and will make up for some of the shortfall in Imports (shown as the addition of “A” in the producer quadrant). The government will collect the tariff on imports, equal to “C” in the Government quadrant. Due to inefficiencies in domestic industries and the absolute loss of consumption, triangles B and D are ‘lost’ (called deadweight losses).

¹³ Thiagarajan, R., Brender, J. & Metcalfe, M. (2025). Understanding the Economic Tradeoffs of Tariffs. State Bank. February 2025

**Figure 34.****Impact of a tariff on consumers, producers, government and welfare**

Source: Hutchinson, E. (2017). Principles of Microeconomics. University of Victoria

In identifying the impact of the United States' 2018 tariffs on prices and welfare by running a regression on the HS10 imports by month, Amiti et al. (2019) found negative impacts of tariffs on domestic prices, availability of imported varieties, and overall welfare. Specifically, cumulative deadweight welfare cost (reduction in real income) from the United States tariffs was around \$8.2 billion during 2018, with an additional cost of \$14 billion to domestic consumers and importers in the form of tariff revenues transferred to the government.¹⁴ Regarding the effect of the same United States tariff on the United States manufacturing sector, Flaaen and Pierce (2021) examined the industry's exposure in terms of manufacturing employment, production, and prices under three channels, namely import protection, rising input costs, and export retaliation. It is found that tariff increases enacted in early 2018 were associated with relative reductions in United States manufacturing employment and

relative increases in producer prices. For manufacturing employment, rising input costs and export retaliation accounted for the negative relationship, and the contribution from these channels more than offset a small positive effect from import protection.¹⁵ In analysing the impact of an unexpectedly higher import tariff from the United States on Chinese imports to companies involved in global value chains, Grossman et al. (2024) found that higher import tariff leads to increasing cost for searching and switching suppliers, which amounted to a welfare loss of 0.12 per cent of Gross Domestic Product (GDP).¹⁶

Protection measures

A first step consists of preparing the simple average and weighted average tariffs by sector over time. Dispersion rates, maximum and minimum rates will also be presented by sector to illustrate the level of protection, using the published

¹⁴ Amiti, M., Redding, S.J. & Weinstein D.E. (2019). The Impact of the 2018 Tariffs on Prices and Welfare. The Journal of Economic Perspectives, Fall 2019, Vol. 33, No. 4 (Fall 2019), pp. 187-210

¹⁵ Flaaen, A. & Pierce, J. (2021). Disentangling the Effects of the 2018-2019 Tariffs on a Globally Connected U.S. Manufacturing Sector.

¹⁶ Grossman, G.M., Helpman, E. & Redding, S.J. (2024), «When Tariffs Disrupt Global Supply Chains», American Economic Review 2024, 114(4): 988-1029.

tariff rates at the HS-6-digit level (or a more detailed level, depending on the availability of data provided by Papua New Guinea)¹⁷ to avoid underestimation of the level of tariff protection.

The simple average tariff is calculated by adding the tariffs on all lines and dividing by the number of those tariff lines. In other words, each tariff line has the same weight irrespective of its economic importance. The simple average tariff provides a simple indication of how much protection is applied by an economy or region, on average. Higher values indicate a more protected economy, lower values a less protected economy.

$$\bar{\tau} = \frac{\sum_i \tau_i}{n}$$

Where τ indicates the tariff applied to specific product i , and n is the number of products.

The weighted average tariff is calculated as the average rate of duty per imported value unit. Like the simple average, the weighted average tariff tells us how much protection is applied by an economy or region, on average. Higher values indicate a more protected economy, lower values a less protected economy. The difference is that the weighted average tariff considers the volume of imports in each product category.¹⁸ The weighted average tariff is normally lower than the simple average tariff because products with high tariffs are normally imported less, while products with low tariffs are imported more. In other words, high tariffs tend to deflect imports and thus lower the tariff average.¹⁹

$$\bar{\tau} = \sum w_k \tau_k$$

where k indexes imported goods and is the weight given to tariff k in the average, calculated as the share of goods k in the country's overall imports.

Dispersion. Average tariffs (both simple and weighted) can be considered as not truly reflecting the level of protection.²⁰ As an alternative, the dispersion of tariffs provides the level of distortion of tariffs to trade: the higher the dispersion, the greater the distortion. The dispersion of tariffs can be captured by using the standard deviation (see below) to show how tariffs are widely spread out (compared to the average tariff).²¹ The standard deviation (σ) is calculated as below:

$$\sigma = \sqrt{\frac{1}{N} \sum_{k=1}^N (\tau_k - \bar{\tau})^2}$$

Categories of products

The study classifies products using the UNCTAD classification. It classifies products into four broad categories: raw materials (primary products used in production), intermediate products (semi-finished goods used to make other products), consumer products (for final consumption), and capital goods (manufacturing goods like machinery for production). This system, based on the Broad Economic Categories (BEC) classification and the Harmonized System (HS), helps to analyse trade by the stage of processing and use of the goods. The study uses this classification system throughout for the analysis of tariffs and trade flows, as well as revenue and consumer welfare effects. The classification used by UNCTAD has been compared to the categories of products used in the TRP. The analysis revealed the following:

¹⁷ This could be done at the national tariff line level if data is provided by Papua New Guinea.

¹⁸ ESCAP (2023). Trade Statistics in Policy Making: A Guide on Using R for Calculating Trade Indices and Indicators: Weighted Average Tariff.

¹⁹ World Trade Organisation (2006). World Tariff Profile 2006.

²⁰ World Trade Organisation (2006). *ibid*

²¹ United Nations and World Trade Organisation (2012). A Practical Guide to Trade Policy Analysis. Chapter 2: Quantifying trade policy.

Table 8.
UNCTAD and TRP classification of goods

UNCTAD classification	Papua New Guinea TRP classification of products				
	Duty free	Intermediate rate	Prohibitive rate	Protective rate	Grand total
Capital goods	631	10	1	2	644
Consumer goods	618	164	12	303	1097
Intermediate goods	973	74	7	40	1094
Raw materials	133	18	38	42	231
Grand total	2355	266	58	387	3066

Note: Numbers represent tariff lines at HS6 level

Source: UNCTAD

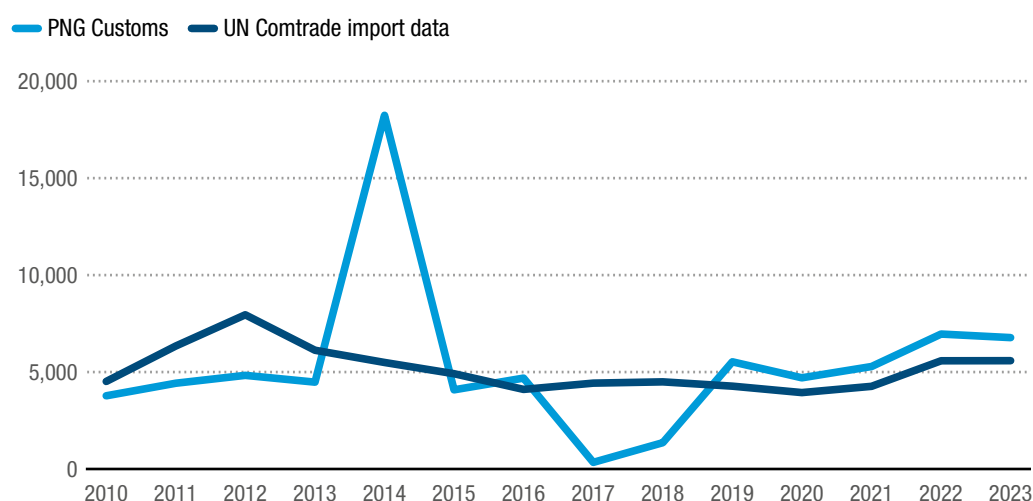
Data choices

Official import data from Papua New Guinea Customs Services were cross-checked against international sources, and several discrepancies were identified. For example, 2014 showed an anomalously large import value in Papua New Guinea's customs data (due to a one-off inclusion of around \$11.4 billion for petroleum imports) which did not appear in mirror statistics (Figure 35 and 6). Given the international scope of UNSD statistics, those data were used as

the primary reference. All monetary values are in United States dollars at current prices unless stated otherwise. Where averages or changes in "tariffs" are mentioned, these refer to trade-weighted ad valorem tariffs, providing a sense of effective protection. Tariff rates also different between the revenue declared to be collected by PNGCS and the tariff rates compiled from reputable sources (UNCTAD and ITC). Internationally comparable methodologies for calculating tariff rates were used once more.

Figure 35.
PNGCS and UNDS data for Papua New Guinea imports

Yearly imports in US\$ millions, 2010–2023.



Source: UNCTAD based on PNGCS and UNSD Data



Table 9.
PNGCS and UNDS import data for Papua New Guinea

Year	PNGCS import data (US\$ million)	UN Comtrade import data (US\$ million)	Discrepancy (per cent)
2010	3 775	4 515	16
2011	4 423	6 342	30
2012	4 829	7 953	39
2013	4 479	6 124	27
2014	18 240	5 494	-232
2015	4 083	4 914	17
2016	4 687	4 105	-14
2017	342	4 428	92
2018	1 357	4 494	70
2019	5 528	4 272	-29
2020	4 703	3 941	-19
2021	5 285	4 263	-24
2022	6 959	5 581	-25
2023	6 776	5 586	-21

Source: UNCTAD based on PNGCS and UNSD data



Table 10.
Tariff rates estimated for Papua New Guinea

Year	PNGCS Ratio of revenue collection on import values (per cent) (A)	World Bank WITS/ UNCTAD Trains Tariff Data (per cent) (B)	Discrepancy of values (per cent) (C)=(B)-(A)
2010	4.4	5.6	1.2
2012	3.9	4.5	0.6
2013	3.9	4.5	0.6
2014	3.9	4.8	0.9
2015	3.2	4.6	1.3
2016	3.2	4.5	1.3
2017	3.2	4.8	1.5
2018	3.2	4.8	1.6
2019	3.4	4.6	1.2
2020	3.3	5.3	2.0
2021	3.2	5.5	2.3
2022	3.5	5.4	1.9
2023	3.2	5.7	2.5

Note: PNGCS average tariff rate is calculated as the sum of tariff revenue collected divided by the value of imports. UNCTAD average tariff rate is an estimation (unweighted) of the ad valorem equivalent tariffs obtained from WITS.

Source: Author based on PNGCS data and UNCTAD TRAINS through WITS

Impact analysis

For the **ex-post analysis**, a standard bilateral gravity model of trade was applied to quantify how changes in Papua New

Guinea's tariffs affected its import volumes. This model relates Papua New Guinea's import values to various factors: (i) the tariff level on each product, (ii) the economic size

of trading partners, and (iii) trade frictions. In formal terms, for each HS-6 product and partner country, the model regresses the log of import value on the applied tariff (trade-weighted average) and relevant controls. The regression includes partner GDP (to capture market size), distance between Papua New Guinea and the partner (a proxy for transport cost), a dummy for whether the partner has a free trade agreement (FTA) with Papua New Guinea (e.g. Australia under PACER Plus, or other regional agreements), and a measure of the partner's regulatory quality (to proxy overall trade facilitation). Year fixed effects control for global shocks, and product-group fixed effects capture commodity-specific factors. This approach isolates the partial effect of tariff changes on imports, holding other factors constant. The gravity model estimation provides an elasticity of import demand with respect to tariff changes and indicates whether Papua New Guinea's recent tariff hikes or cuts significantly altered import behaviour.

For the **ex-ante analysis**, a Partial Equilibrium Simulation Model was applied to complement the econometric approach. The model is similar to the World Bank's SMART model, which is used to simulate the impact of tariff changes on trade, welfare, and revenue. Two scenarios were examined: (1) TRP Implementation (2010–2017) – comparing outcomes with the tariff reductions against a counterfactual of no tariff change; and (2) TRP Suspension (2018–2023) – comparing outcomes with the tariff increases against a counterfactual of retaining the lower tariffs. The model uses detailed import data by product and source country, along with import demand elasticities, to estimate how a tariff cut or hike would change import quantities, government tariff revenue, and consumer surplus. For instance, lowering a tariff should increase import volume (and consumer surplus) but reduce per-unit revenue, while raising a tariff would do the opposite, all else equal. These simulations draw on data from ITC MacMap for tariff rates and UN Comtrade for trade values. They allow us to quantify effects on a product-by-product

basis and aggregate to sector or economy-wide totals. Notably, the model captures the largest trading partners (like China and Australia) to see how their exports to Papua New Guinea were affected, and it computes the net welfare impact (consumer surplus change) for Papua New Guinea's consumers from each policy change.

For the underlying data, Papua New Guinea's official import data was cross-checked against international sources, and some discrepancies were noted. For example, 2014 showed an anomalously large import value in Papua New Guinea's customs data (due to a one-off inclusion of ~\$11.4 billion in petroleum imports), which did not appear in international data sets, nor in mirror statistics. Adjustments were made in the analyses to account for such anomalies, ensuring they did not skew the results. All monetary values are in US dollars at current prices unless stated otherwise. Where averages or changes in "tariffs" are mentioned, these refer to trade-weighted ad valorem tariffs, providing a sense of effective protection.

By combining these methods – regression analysis for observed historical impact and simulation for scenario assessment – a robust picture is obtained of how the TRP influenced Papua New Guinea's trade outcomes and what the effects of reversing it have been. The following sections present the findings on trade flows, sectoral performance, welfare, revenue, and business sentiment, each supported by evidence from the data and the models.

Methodology for measures of protection and trade flow analysis

For Papua New Guinea, the level of processing of Papua New Guinea exports is notably skewed towards raw materials and semi-finished intermediary products for further processing. Around 1 per cent of all exports are final goods (consumer goods or capital goods), 76

per cent are intermediary products, and 23 per cent are primary products.

In the case of imports, a more complex picture arises. Final goods represent around 33 per cent of the share of imports, while intermediates account for around 63 per cent, and raw materials around 4 per cent. The analysis on the structure of tariffs based on stages of processing will help to determine the potential impact that such tariffs may have on manufacturing, dependent on those inputs.

Due consideration is given to the network of FTAs that Papua New Guinea enjoys.

Economic modelling strategy

During the inception phase, the team was able to review data provided by the Treasury and submitted by other ministries and from the Papua New Guinea Customs Service (PNGCS). The data on trade was very rich and is provided at a shipment level by day for the years 2010-2024 and includes the value and volumes of imports and exports, source of origin and destination, and duties collected (tariff and excise taxes).

The modelling is inevitably influenced by the availability and quality of data provided. As such, the modelling strategy at the inception phase proposes several avenues for investigating the relationships between tariff changes and relevant economic aggregates, with some approaches requiring less ambition underlying data. The analysis begins with less data-intensive models and increases toward more granular results that require more refined data sets.

Partial equilibrium model

The partial equilibrium (PE) model framework is a useful modelling tool since it demands a minimal amount of data and is relatively easy to understand and use. In this case,

the policy instrument is treated as a tariff, and the underlying assumption is that the country is a small country that has no influence on world prices (influence on world prices would be expected from a large supplier). In this model, where the country is a price taker, the objective will be to estimate the effects of introducing higher tariffs on prices, demand, and supply for the product directly affected by the measure.

In contrast to a general equilibrium model, which considers the whole economy, including potential interactions across all markets, the advantage of partial equilibrium modelling is the level of detail that can be examined. Negotiation of trade agreements rarely encompasses product group level blanket tariff cuts, such found in the general equilibrium context, which takes an average tariff cut for an entire sector, thereby leading to varying degrees of aggregation bias, with potentially generalised conclusions. These biases are minimised in the PE model. It is usually, therefore, helpful to use a PE model to get a better understanding of the product line (and sector trade flow effects), but also to quantify the effects on macroeconomic aggregates and capture cross-sectoral effects arising from liberalisation.

To split imports across different exporters, the well-known Armington equation is used, where the shift parameter, import shares of each country, and the degree of substitution between imports from different countries are the key variables. From the model, it is straightforward to determine the changes in imports, tariff revenue, and welfare following differential changes in bilateral tariffs.

A basic PE model essentially has three equations, specifying demand, supply, and a market-clearing equation that demand and supply must equate (equations (1) to (3)). Additionally, the price linkage equation (4) shows how import prices relate to export prices.

$$M = \alpha_M PM^\varepsilon \quad (1)$$

$$X = \alpha_X PX^\mu \quad (2)$$

$$X = M \quad (3)$$

$$PM = PX \times (1 + t) \quad (4)$$

$$TR = PX \times t \times M \quad (5)$$

$$\Delta WM = \int_{PM_1}^{PM_0} \alpha_M PM^\varepsilon dPM = \frac{\alpha_M}{\varepsilon + 1} (PM_0^{\varepsilon+1} - PM_1^{\varepsilon+1}) \quad (6)$$

$$\Delta WX = \int_{PX_0}^{PX_1} \alpha_X PX^\mu dPX = \frac{\alpha_X}{\mu + 1} (PX_1^{\mu+1} - PX_0^{\mu+1}) \quad (7)$$

Where M is imports, X exports, PM price of imports, PX price of exports, $\alpha_M > 0$, and $\alpha_X > 0$ are shift parameters, $\varepsilon < 0$ and $\mu > 0$ are the elasticities of import demand and export supply, and t is the ad valorem tariff rate. TR is tariff revenue. ΔWM and ΔWX are the changes in the gains for importing and exporting countries, respectively.

Given the parameter values ε , μ , α_M , α_X and the policy variable t, equations (1) to (4) are a system of four equations in four unknowns, PM, PX, M, and X. Once the equilibrium prices and quantities are determined, one can determine the components of economic surplus — equations (5), (6) and (7).

This model is a helpful ex-ante check on the direction and estimated trade effects arising from tariff changes. The modelling does not involve significant data challenges and is expected to be possible. A ready-built tool partial equilibrium model, based on John Gilbert (2017) is proposed for us,²² which is structurally similar to the SMART model in the World Bank's WITS. The model is quite flexible and can be extended to include imports from different sources as well as allow the tariffs to vary by source of imports using Armington elasticities. Extensions to the model allow for the inclusion of several parameters, e.g. inclusion of Non-Tariff Measures (NTMs) AVE in the simulations.

Gravity model

A common approach for estimating the impact of tariff changes on export performance is to use a gravity model of trade. Gravity models enjoy a robust theoretical foundation²³ and remains to this day one of the most popular approaches to estimate the flow of international trade between countries, based on a host of variables which attract or repel countries to trade with each other, such as the size of economies, their geographical distance, common languages and legal systems, etc.

Building on the gravity model framework developed by Anderson & Van Wincoop (2003),²⁴ which provides a theoretically sound foundation for analysing trade flows. Extending this by incorporating multilateral resistance terms (MRTs) to correct for third-country effects and extending it by incorporating sectoral-level tariff effects, the following approach used by Liu (2020) is considered,²⁵ enabling a more disaggregated analysis of tariff impacts on specific industries.

According to Anderson & Van Wincoop (2003), a gravity model accounts for bilateral trade costs and the global resistance factor:

²² Gilbert (2017). Partial Equilibrium Analysis. UNESCAP: Bangkok.

²³ Head, K. & Mayer, T. (2014). Gravity Equations: Workhorse, Toolkit and Cookbook, in *Gopinath, Gita, Ehanan Helpman, and Ken Rogoff, eds Handbook of International Economics*, Vol 4. North Holland: Elsevier.

²⁴ Anderson, J. E. & van Wincoop, E. (2003). Gravity with Gravititas: A Solution to the Border Puzzle. *American Economic Review*. Vol 93 (1). March

²⁵ Liu, B. Hudson, D. & Devine, J. (2022). A Gravity Model Analysis of Chinese Apparel Export Flows. Briefing Paper. ICAC – BP22 – 01. January

$$X_{ij} = \frac{Y_i Y_j}{Y_w} \left(\frac{t_{ij}}{\pi_i P_j} \right)^{1-\sigma}$$

Where, X_{ij} , Y_i , Y_j are GDP of exporting and importing countries Y_w is the world GDP t_{ij} are trade (including tariffs), π_i , P_j are Multilateral Resistance terms (MRTs) and σ is the elasticity of substitution.

Following Liu (2020), the gravity model is extended to a sectoral level, estimating the impact of tariffs on different industries. This extension is presented in the equation below:

$$\ln X_{s,ij} = \beta_0 + \beta_1 \ln GDP_i + \beta_2 \ln GDP_j + \beta_3 \ln DIST_{ij} + \beta_4 \ln Tariff_{ij} + \beta_5 \ln RTA_{ij} + \beta_6 \ln Contiguity_{ij} + \beta_7 \ln Common\ language_{ij} + \beta_8 \ln Population_{jt} + \beta_9 \ln Population_{it} + \beta_{10} \ln Population_{jt} + \beta_{11} \ln controls_{ij} + \delta_i + \delta_j + \epsilon_{s,ij}$$

$X_{s,ij}$ is Exports of sector s from country i to j , GDP_i and GDP_j are the GDPs of exporting and importing countries, bilateral Distance between countries and tariff rate applied to sector s in trade between i and j . $\ln Population_{jt}$ is Natural logarithm of an exporter's or importer's population $\ln Contiguity_{ij}$ is a dummy if countries share a boarder and $\ln Common\ language_{ij}$ dummy if countries share a common language

Anderson emphasises the importance of controlling for third-country effects, which is implemented by including δ_i and δ_j which represent country-specific trade barriers, which proxy for unobservable MRTs.

Such a model can be used to measure the effects of tariff change on driving bilateral trade between Papua New Guinea and its trading partners, with the sign of the elasticity of β_4 expected to be negative, as a tariff increase (reduction) will lead to a

decrease (increase) in exports between a trading partner and Papua New Guinea.

The gravity model is helpful to explain changes in trade flows ex-post, but it does not help to identify output, employment or productivity changes from tariffs, which is of fundamental importance for this study. As such, this model is not proposed to be prioritised over others, unless data availability proves to be overly challenging to use other modelling efforts.

Stakeholder engagement

To supplement the quantitative analysis of the TRP impact assessment, stakeholder consultations were conducted and facilitated by the Department of Treasury with national government agencies, business associations, and companies relevant to the covered subject matters.

Annex 3.

Results from the Gravity Model

Methodology and data

This section presents the detailed statistics and estimates of the impacts of tariff changes on Papua New Guinea imports.

A log-linear regression model is employed to examine the impact of tariffs on Papua New Guinea imports. The specification is as below:

$$\ln(\text{Imports}) = \alpha_0 + \alpha_1 \text{Tariff} + \sum_j \beta_j \text{Tariff} * \text{SoP}_k + \alpha_2 + \alpha_1 \ln(\text{GDP}) + \alpha_3 \ln(\text{distance}) + \alpha_4 \text{FTA} + \alpha_5 \text{RQI} + \mu_t + \varphi_k + \epsilon$$

Where

- “Imports” denotes annual import value into Papua New Guinea at the Harmonised System (HS) 6-digit level;
- Tariff denotes trade-weighted tariff at the HS 6-digit level;
- SoP denotes commodity group;
- GDP denotes the GDP of Papua New Guinea’s partner countries;
- Distance is the geographic distance (in kilometres) between Papua New Guinea and its partner countries;
- FTA is a dummy variable indicating the existence of a regional trade agreement between Papua New Guinea and its partner countries;
- RQI is the Regulatory Quality Index of Papua New Guinea’s partner countries, capturing the ability of a government to formulate and implement sound policies and regulations that permit and promote private sector development;
- μ_t denotes year fixed effects and φ_k denotes commodity group fixed effects. They are incorporated into the model as different dummy variables.

The model is estimated using annual trade data at the HS 6-digit level over multiple years (details to be discussed later). For simplicity, subscripts for partner country, HS6 product, and year are suppressed in the notation above.

A log-linear model is used for this estimation. The main reasons for this model selection include:

- A log-linear specification is preferred over Poisson Pseudo-Maximum Likelihood (PPML) because the dataset includes only imports into Papua New Guinea, rather than bilateral trade flows as in a typical gravity model. Since zero import values are not widespread, the log-linear model is appropriate and directly provides the elasticity of trade with respect to tariffs. By contrast, PPML is generally advantageous in settings with a high prevalence of zero trade flows and heteroskedasticity, as it accommodates zeros without transformation and yields consistent estimates under heteroskedastic errors.
- The log-linear model is estimated with clustered standard errors at the partner level, allowing for arbitrary correlation of residuals across product categories and years within each partner country. This adjustment accounts for unobserved partner-specific shocks (e.g., macroeconomic or policy changes) that may simultaneously affect multiple trade flows, thereby ensuring consistent inference.

- SoP dummies are included to capture differences in both tariff responsiveness and import values across commodity groups.
- Time dummies control for year-specific shocks affecting all trade flows.
- Given the data availability, partner-specific variables such as GDP, distance, FTA, and RQI are incorporated as they are key determinants of imports.

Data sources

This study draws on data compiled from 3 main sources:

- World Bank World Integrated Trade Solution (WITS) provides Papua New Guinea import values and associated trade-weighted tariffs at the HS6 level for the years 2010, 2012, 2015, and 2018–2023. WITS also classifies HS6-level commodities into four groups: (i) raw materials (SoP1), (ii) intermediate goods (SoP2), (iii) consumer goods (SoP3), and (iv) capital goods (SoP4).

- The World Bank publishes the Regulatory Quality Index (RQI) and GDP data, available at the country level.
- The CEPII Gravity database supplies various country-specific variables.

Table 119 reports total import values and by category over time. Although expressed in current prices, a significant decline in Papua New Guinea imports is evident from 2015 to 2019, followed by a gradual recovery. Capital goods constitute the largest share of imports, followed by consumer goods.

Table 11.
Import values in \$ thousands, current prices

Year	Total imports	Raw materials	Intermediate goods	Consumer goods	Capital goods
2010	5 839 478	323 345	992 690	1 968 087	2 555 356
2012	8 104 091	1 120 831	1 084 514	2 407 693	3 491 053
2015	4 286 046	376 379	756 706	1 457 844	1 695 116
2018	2 944 827	469 883	370 337	908 720	1 195 888
2019	2 800 029	348 081	399 686	829 501	1 222 760
2020	3 340 988	337 249	556 932	1 197 689	1 249 119
2021	3 392 458	337 659	563 721	1 228 971	1 262 106
2022	3 909 660	385 200	687 221	1 376 489	1 460 750
2023	4 398 581	381 361	743 762	1 617 897	1 655 562

Source: Comtrade via WITS

10 reports descriptive statistics for the variables included in the estimation, while Appendix 1 reports the summary statistics by year.

Table 12.
Summary statistics

Variables	Mean	Sd
Imports value (1'000s \$, 2015 price)	199.80	2 120.05
Imports value (1'000 \$, current price)	209.19	2 267.88
Tariff (weighted average in %)	3.81	7.69
GDP (million \$, 2015 price)	3 090 064.58	5 290 844.02
GDP (million \$, current price)	3 238 905.68	5 645 719.91

Variables	Mean	Sd
Geodesic distance (km)	7 106	4,180
FTA (1=yes)	55 484	
RQI (Regulatory Quality Index)	0.99	0.91
SoP1-Raw materials (1=yes)	6 867	
SoP2-Intermediate goods (1=yes)	39 123	
SoP3-Consumer goods (1=yes)	83 129	
SoP4-Capital goods (1=yes)	57 391	
Number of observations	186 510	

Results from the Gravity model

Table 11 reports the results from the model estimation. Results indicate that a 1 per cent increase in tariffs leads to a 2.1 per cent decline in imports, holding other factors constant.

The interaction terms with commodity group show how the tariff effect varies relative to SoP1 (raw materials). For SoP2 (intermediate goods), the elasticity is -0.005 ($-0.021 + 0.016$), implying a near-zero effect. Consumables (SoP3) also exhibit little deviation from the average effect (-0.021), but the coefficient is not statistically significant, indicating a similar impact as raw materials. By contrast, capital goods (SoP4) show a positive elasticity of 4.6 per cent ($-0.021 + 0.067 = 0.046$). This result suggests that the observed increase in imports of capital goods alongside higher tariffs after 2018 may reflect

unobserved factors, such as exemptions for priority projects, concessional schemes, tariff adjustments in response to import surges, or misclassification.

As expected, partner GDP is positively associated with imports, while distance has a negative effect. The existence of a trade agreement with partners promotes imports increase by around 40 per cent. Large declines in import values are observed in 2015 and 2022, with the latter plausibly attributable to the impact of COVID-19. The positive coefficient on regulatory quality, although statistically insignificant, is consistent with the expectation that stronger institutional and regulatory frameworks facilitate trade. Enhanced regulatory quality mitigates uncertainty, reduces transaction costs, and establishes a stable and predictable environment for international transactions, thereby creating conditions conducive to greater trade flows.

Table 13.
Estimation results

Variable	Estimate	Std.error	Significance
Tariff (weighted average in %)	-0.021	0.008	***
Tariff*SoP2	0.016	0.006	***
Tariff*SoP3	0.006	0.011	
Tariff*SoP4	0.067	0.013	***
GDP in logarithm form	0.282	0.049	***
Distance in logarithm form	-1.042	0.127	***
FTA (1=yes)	0.403	0.126	***
RQI	0.086	0.083	
SoP2	0.065	0.195	
SoP3	0.159	0.312	

Variable	Estimate	Std.error	Significance
SoP4	0.621	0.317	*
Year-2012	0.080	0.038	**
Year-2015	-0.277	0.059	***
Year-2018	-0.011	0.148	
Year-2019	-0.063	0.133	
Year-2020	0.035	0.137	
Year-2021	-0.011	0.132	
Year-2022	-0.466	0.074	***
Year-2023	-0.020	0.150	
Cluster	Yes		
Observations	186,510		
R ²	0.12		

Note: *p<0.1; ** p<0.05; ***p<0.001

In summary, tariffs are found to negatively affect imports of raw materials and consumables, with little impact on intermediate goods. For capital goods, the positive association may reflect

exemptions for priority projects, concessional tariff schemes, or reverse causality, whereby higher imports lead to tariff increases.

Appendix to Annex 3. Summary statistics by year

Variables	2010		2012		2015	
	Mean	Sd	Mean	Sd	Mean	Sd
Imports value (1'000 \$,2015 price)	214.73	2 266.20	242.50	3 394.80	151.20	1 433.08
Imports value (1'000 \$, current price)	210.74	2 221.83	276.42	3 919.56	151.20	1 433.08
Tariff (weighted average in %)	4.66	9.29	4.11	7.94	3.44	6.69
GDP (million \$, 2015 price)	2 502 725.69	4 170 872.59	2 703 021.40	4 520 899.78	3 139 695.14	5 105 593.09
GDP (million \$, current price)	2 353 091.88	3 780 342.38	2 765 069.56	4 339 614.74	3 139 695.14	5 105 593.09
Geodesic distance (km)	7 039	4 182	7 041	4 193	7 311	4 290
FTA (1=yes)	7 372		7 509		8 096	
RQI (Regulatory Quality Index)	0.92	0.88	0.95	0.88	0.96	0.95
SoP1-Raw materials (1=yes)	907		1 116		1 157	
SoP2-Intermediate goods (1=yes)	5 886		6 373		6 267	
SoP3-Consumer goods (1=yes)	12 618		13 010		12 672	
SoP4-Capital goods (1=yes)	8 298		8 819		8 251	
Number of observations	27 709		29 318		28 347	

Variables	2018		2019		2020	
	Mean	Sd	Mean	Sd	Mean	Sd
Imports value (1'000 \$,2015 price)	210.01	2 131.41	205.18	1 995.86	236.42	1 814.78
Imports value (1'000\$, current price)	214.92	2 133.46	204.77	1 956.36	232.05	1 748.90
Tariff (weighted average in %)	3.46	7.20	3.60	7.39	3.51	7.37

Review of the Papua New Guinea Tariff Reduction Programme

	2018		2019		2020	
Variables	Mean	Sd	Mean	Sd	Mean	Sd
GDP (million \$, 2015 price)	1 715 301.70	3 493 820.33	1 793 775.31	3 578 493.21	3 617 743.19	5 587 389.76
GDP (million \$, current price)	1 789 055.72	3 679 042.62	1 862 683.70	3 829 381.02	3 689 030.76	5 758 580.93
Geodesic distance (km)	7 034	4 239	7 310	4 358	7 109	4 086
FTA (1=yes)	5 173		4 901		4 603	
RQI (Regulatory Quality Index)	1.20	0.89	1.17	0.84	1.00	0.88
SoP1-Raw materials (1=yes)	545		480		486	
SoP2-Intermediate goods (1=yes)	2 663		2 626		2 876	
SoP3-Consumer goods (1=yes)	6 023		5 953		6 198	
SoP4-Capital goods (1=yes)	4 471		4 615		4 838	
Number of observations	13 702		13 674		14 398	

	2021		2022		2023	
Variables	Mean	Sd	Mean	Sd	Mean	Sd
Imports value (1'000 \$,2015 price)	204.29	1 573.17	128.04	1 200.63	255.79	1 883.96
Imports value (1'000\$, current price)	222.33	1 702.30	140.07	1 306.46	271.69	1 970.43
Tariff (weighted average in %)	3.48	7.34	3.87	7.53	3.44	7.24
GDP (million \$, 2015 price)	3 671 582.54	5 923 068.62	4 221 832.77	6 773 593.68	3 998 848.28	6 442 224.32
GDP (million \$, current price)	4 102 655.88	6 682 039.87	4 760 697.62	7 848 338.57	4 338 036.16	7 082 695.96
Geodesic distance (km)	7 000	4 054	7 224	4 223	6 758	3 834
FTA (1=yes)	4 989		7 820		5 021	
RQI (Regulatory Quality Index)	0.94	0.91	0.94	0.92	0.97	0.95
SoP1-Raw materials (1=yes)	495		1 089		592	
SoP2-Intermediate goods (1=yes)	3 018		6 113		3 301	
SoP3-Consumer goods (1=yes)	6 684		12 939		7 032	
SoP4-Capital goods (1=yes)	5 062		7 772		5 265	
Number of observations	15 259		27 913		16 190	

Annex 4.

Births of products

Table 14.

List of products that were exported in 2018 but not exported in 2010

HS6 Code	HS6 Product Description	Value \$ million
020736	Meat and edible offal; of the poultry of heading no. 0105, of ducks, geese or guinea fowls, frozen	0.13
030621	Crustaceans; not frozen, rock lobsters and other sea crawfish (Palinurus spp., Panulirus spp., Jasus spp.), in shell or not, smoked, cooked or not before or during smoking; in shell, cooked by steaming or boiling in water; edible flours, meals, pellets	0.75
030551	Fish; cod (gadus morhua, gadus ogac, gadus macrocephalus), dried (whether or not salted but not smoked)	0.59
030549	Fish; smoked (whether or not cooked before or during the smoking process), n.e.c. in item no. 0305.4 (including fillets)	0.21
030741	Molluscs; cuttle fish and squid, live, fresh or chilled (whether in shell or not)	0.15
100110	Cereals; durum wheat	0.44
110630	Flour, meal and powder; of the products of chapter 8	0.30
140120	Rattans used primarily for plaiting	0.15
151329	Vegetable oils; palm kernel or babassu oil and their fractions, other than crude, whether or not refined, but not chemically modified	0.32
160510	Crustacean preparations; crab, prepared or preserved	4.63
160590	Molluscs and other aquatic invertebrates; prepared or preserved (excluding crustaceans)	13.10
220720	Ethyl alcohol and other spirits; denatured, of any strength	0.28
220290	Non-alcoholic beverages; n.e.c. in item no. 2202.10, not including fruit or vegetable juices of heading no. 2009	0.74
230690	Oil-cake and other solid residues; whether or not ground or in the form of pellets, resulting from the extraction of oils, n.e.c. in heading no. 2306	1.15
240110	Tobacco, (not stemmed or stripped)	1.49
252329	Cement; portland, other than white, whether or not artificially coloured	0.73
261000	Chromium ores and concentrates	21.13
271320	Petroleum bitumen; obtained from bituminous minerals	0.23
271113	Petroleum gases and other gaseous hydrocarbons; liquefied, butanes	2.09
280440	Oxygen	0.19
290121	Acyclic hydrocarbons; unsaturated, ethylene	7.12
290129	Acyclic hydrocarbons; unsaturated, n.e.c. in heading no. 2901	0.12
290122	Acyclic hydrocarbons; unsaturated, propene (propylene)	2.96
310230	Fertilizers, mineral or chemical; nitrogenous, ammonium nitrate, whether or not in aqueous solution	0.46
360300	Fuses and detonators; safety or detonating fuses, percussion or detonating caps, igniters, electric detonators	0.73
401194	Rubber; new pneumatic tyres having other than a herring-bone or similar tread, of a kind used on construction or industrial handling vehicles and machines and having a rim size exceeding 61cm	0.21
440726	Wood, tropical; white lauan, white meranti, white seraya, yellow meranti and alan, sawn or chipped lengthwise, sliced or peeled, whether or not planed, sanded or end-jointed, thicker than 6mm	0.17
440392	Wood; beech, in the rough, whether or not stripped of bark or sapwood, or roughly squared, untreated	1.66
440320	Wood; coniferous, in the rough, whether or not stripped of bark or sapwood, or roughly squared, untreated	0.26

Review of the Papua New Guinea Tariff Reduction Programme

731100	Containers for compressed or liquefied gas, of iron or steel	0.15
750120	Nickel; oxide sinters and other intermediate products of nickel metallurgy	522.47
830120	Locks; of a kind used for motor vehicles (key, combination or electrically operated), of base metal	1.15
830990	Stoppers; caps and lids, of base metal	0.16
841231	Engines; pneumatic power engines and motors, linear acting (cylinders)	0.49
841210	Engines; reaction engines, other than turbo-jets	1.01
841950	Heat exchange units; not used for domestic purposes	0.17
847090	Machines incorporating a calculating device; n.e.c. in heading no. 8470	0.12
841350	Pumps; reciprocating positive displacement pumps, n.e.c. in heading no. 8413, for liquids	0.14
841121	Turbo-propellers; of a power not exceeding 1100kW	1.56
850211	Electric generating sets; with compression-ignition internal combustion piston engines (diesel or semi-diesel engines), of an output not exceeding 75kVA	0.22
851629	Heating apparatus; electric soil heating apparatus and space heating apparatus (excluding storage heating radiators)	0.45
870510	Vehicles; crane lorries	0.48
940600	Buildings; prefabricated	0.14
950691	Athletics and gymnastics equipment	0.16

Note: Reported export lines where the value of exports was above \$100,000 in 2018; Source: UNCTAD based on data from COMTRADE

